



Fundamentals of neurobiology

ch3 : synapse

R.C. Lambert

Neuronal networks and physiopathological rhythms

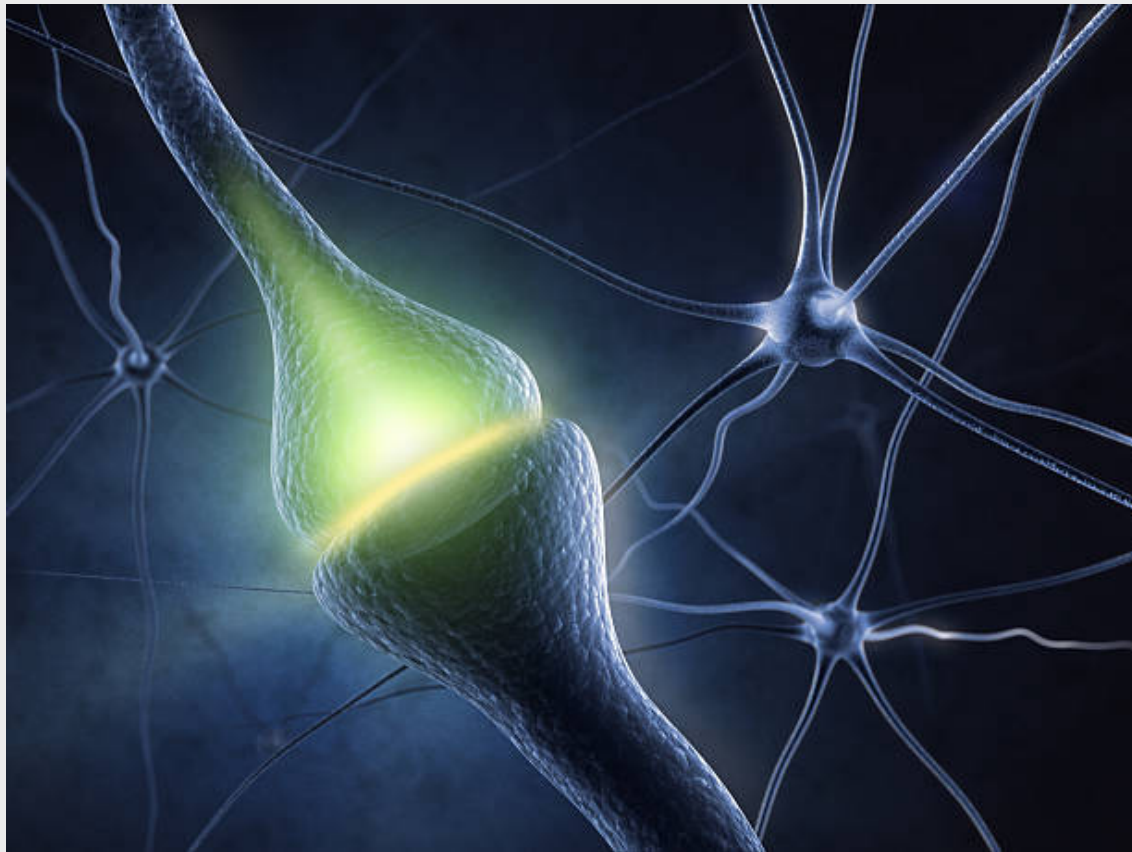
<http://www.ibps.upmc.fr/fr/Recherche/umr-8246/reseaux-de-neurones-et-rythmes-physiopathologiques>

Sorbonne Université- Neuroscience Paris Seine (NPS)

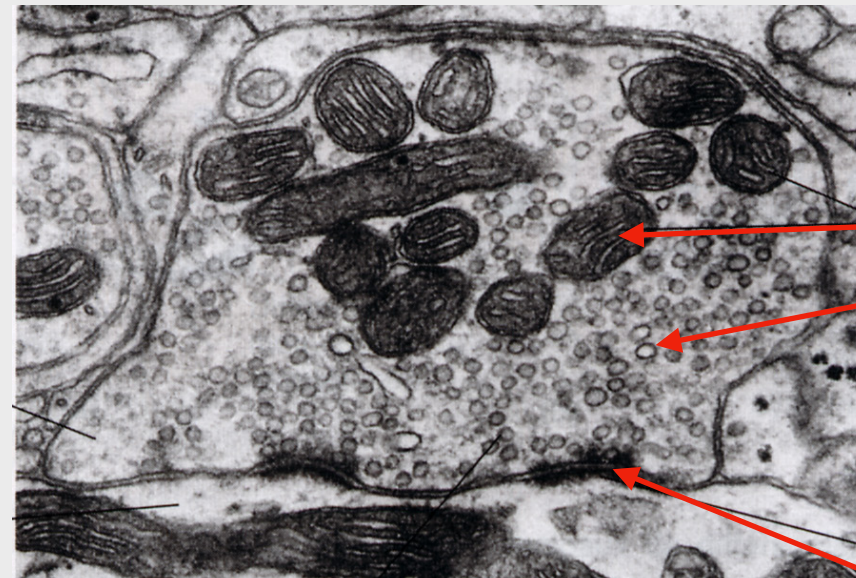
CNRS UMR8246, INSERM U1130, UPMC UMI19



structure of the synapses



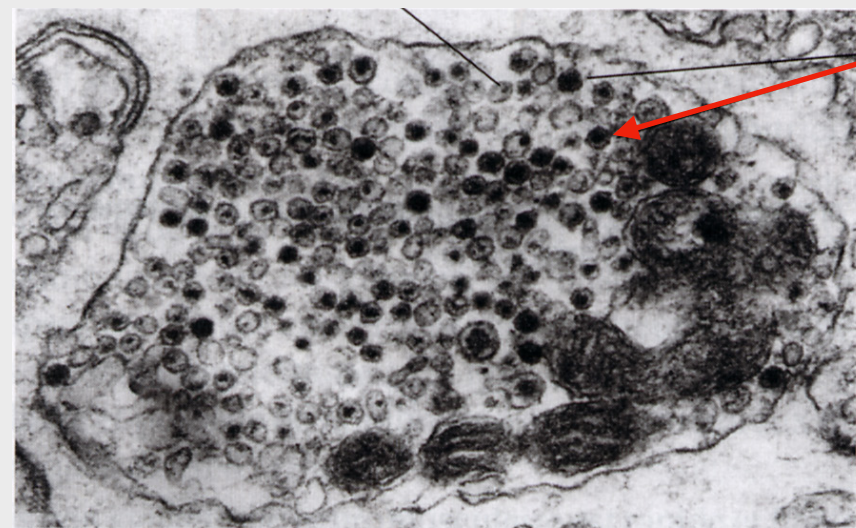
electron microscopy image



mitochondria

vesicles

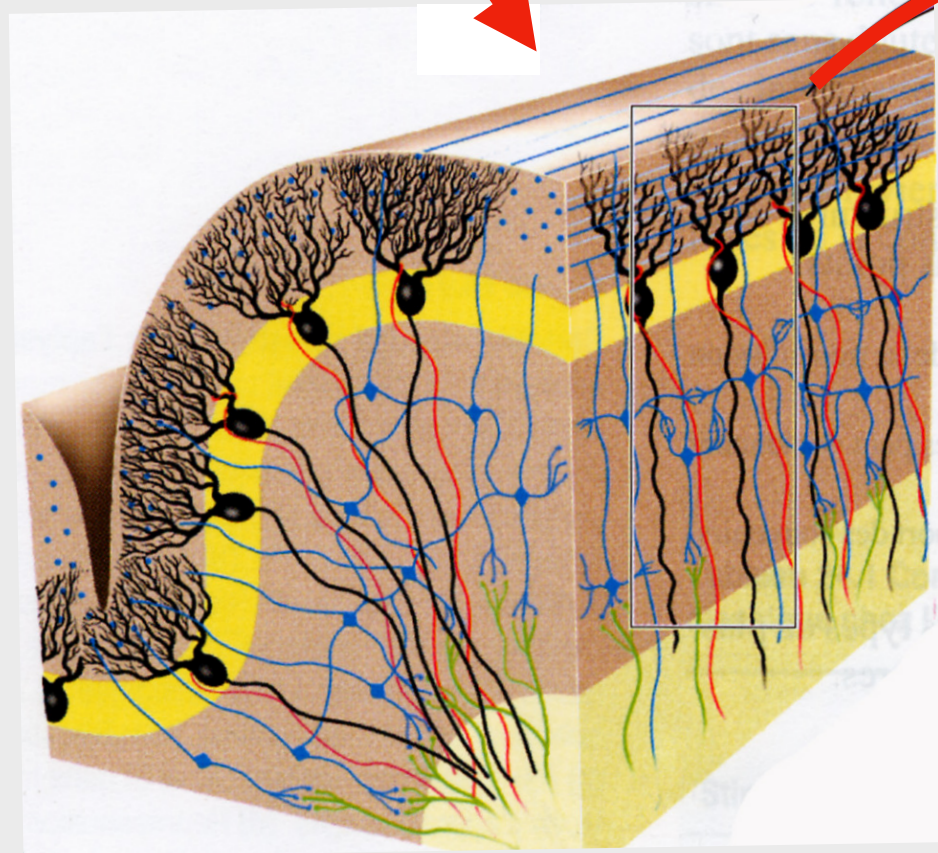
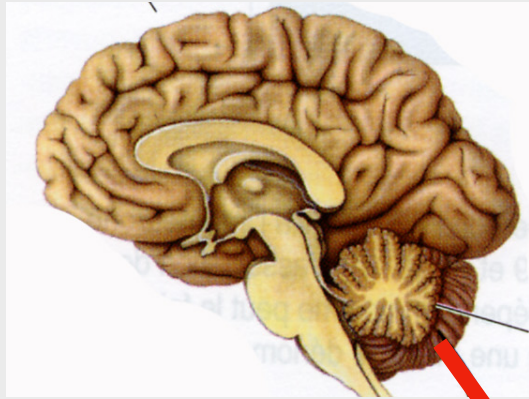
active zone



« dense core »
vesicles

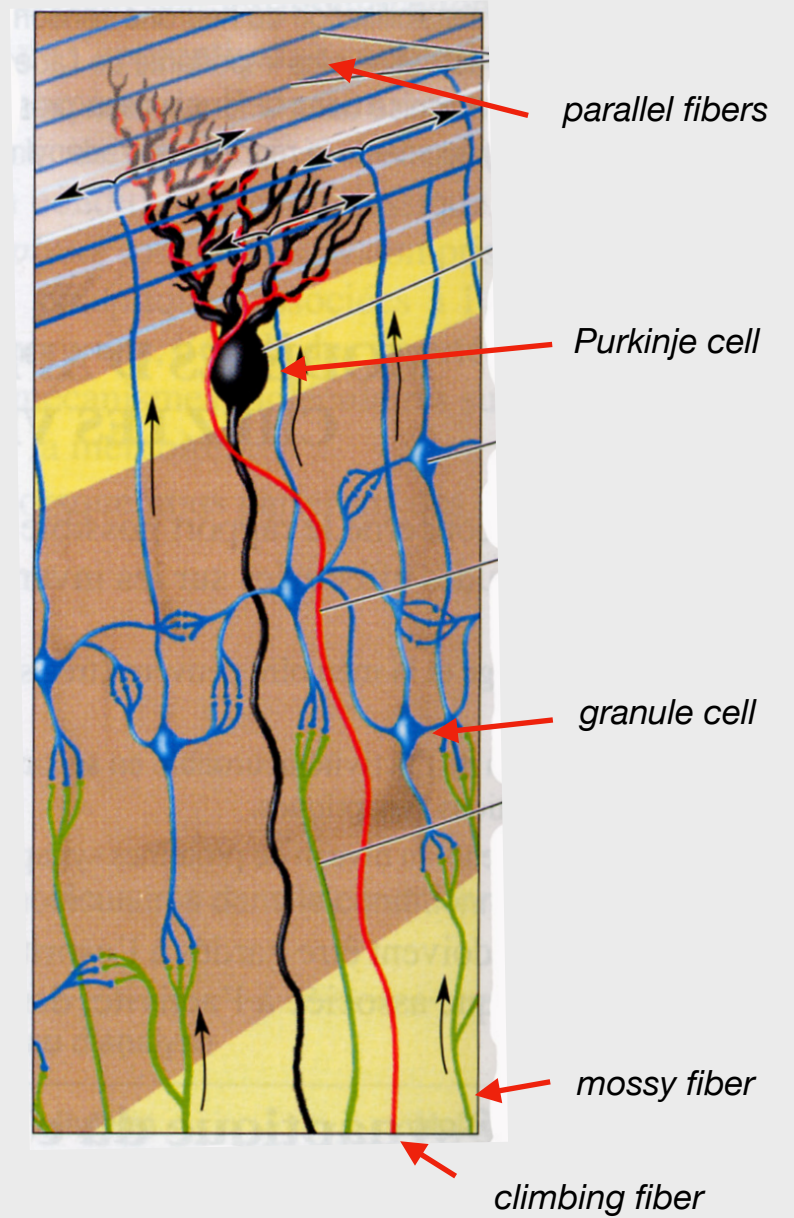
excitatory synapse of the PNS

the cerebellum



molecular layer
Purkinje cell layer

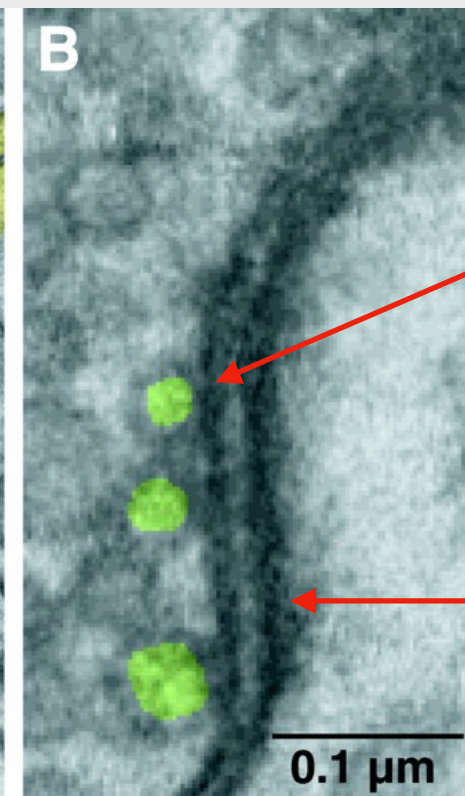
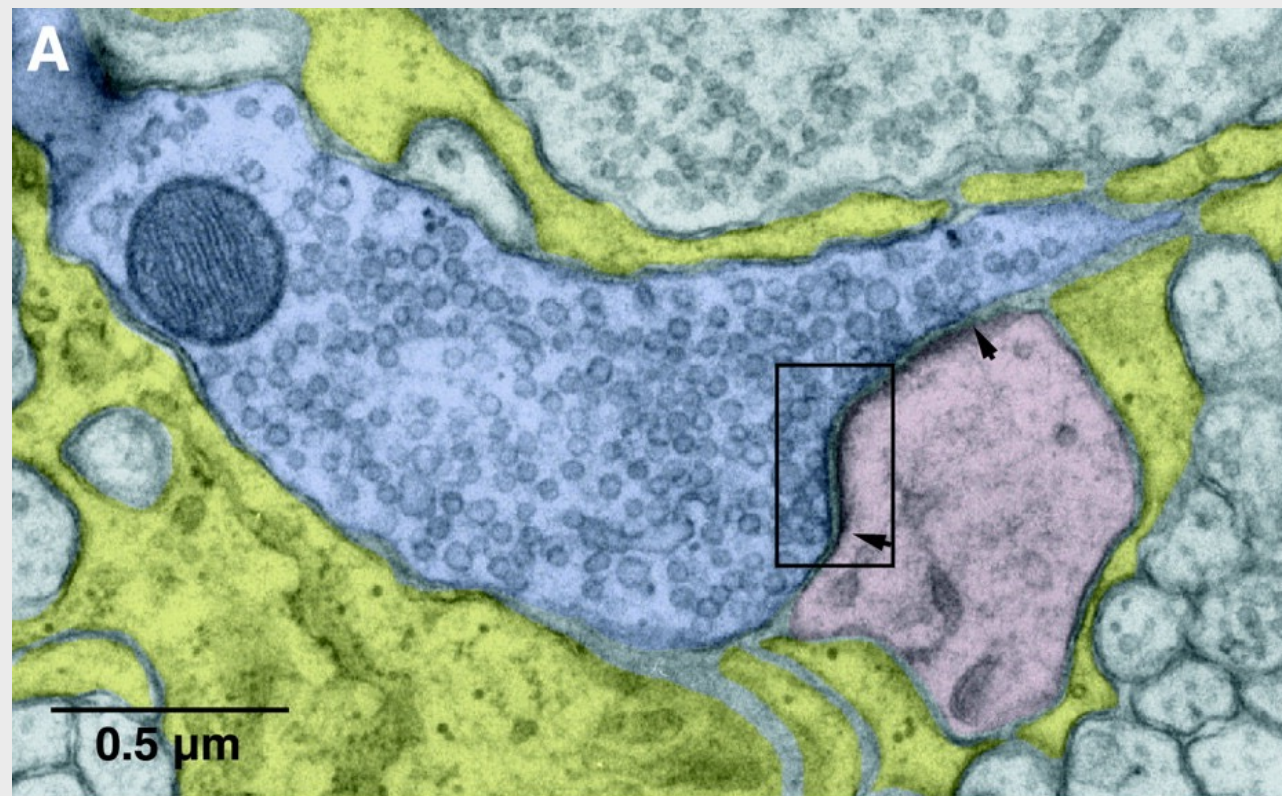
granule cell layer



structure of the synapses

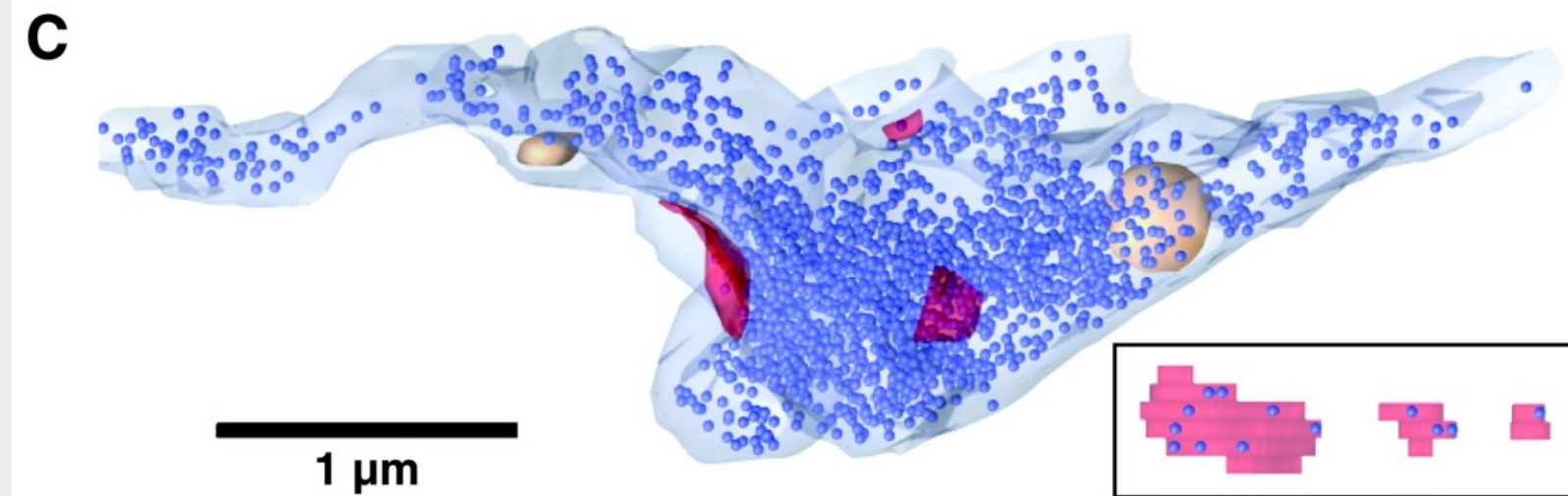
reconstruction of the climbing fiber synapse from EM images

- blue : climbing fiber
- pink : Purkinje cell
- yellow : glial cell



vesicle docked
at the active zone

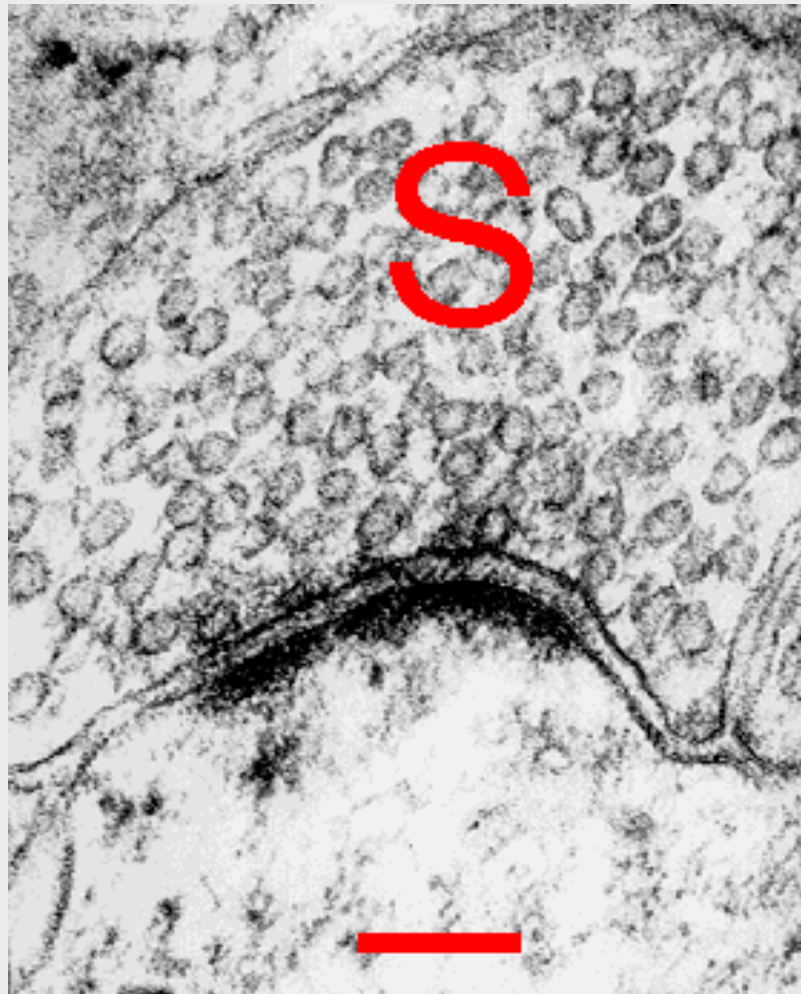
post-synaptic density



- blue : vesicles
- brown : mitochondria
- pink : active zones

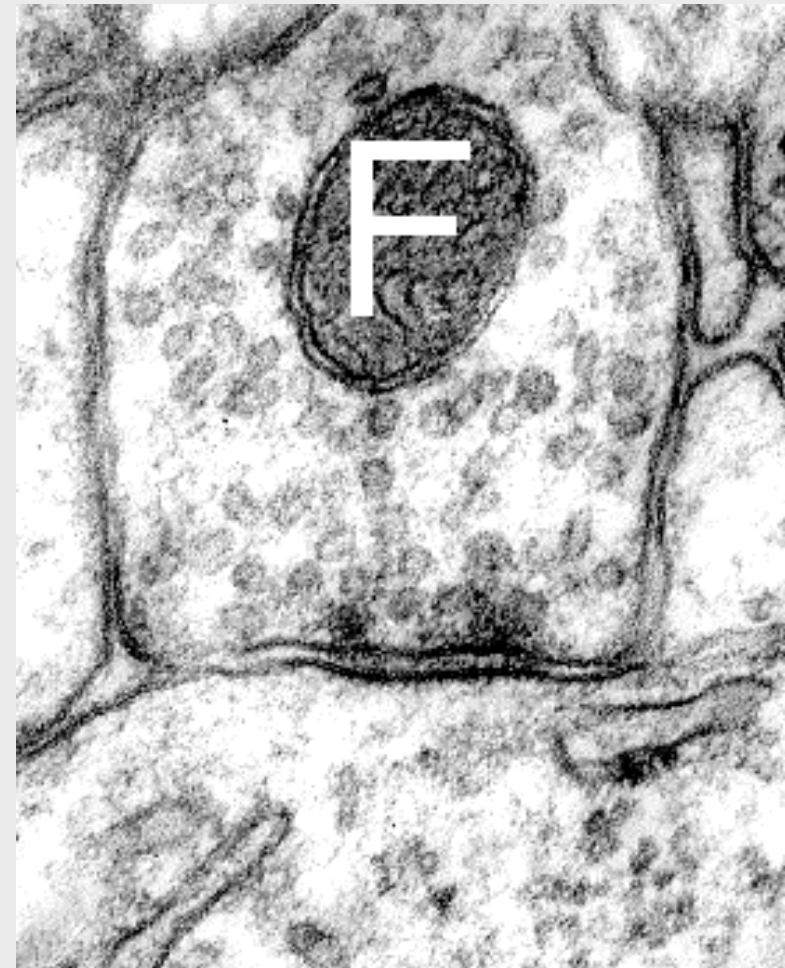
structure of the synapses

asymmetric synapse (Gray type I)
usually excitatory synapse



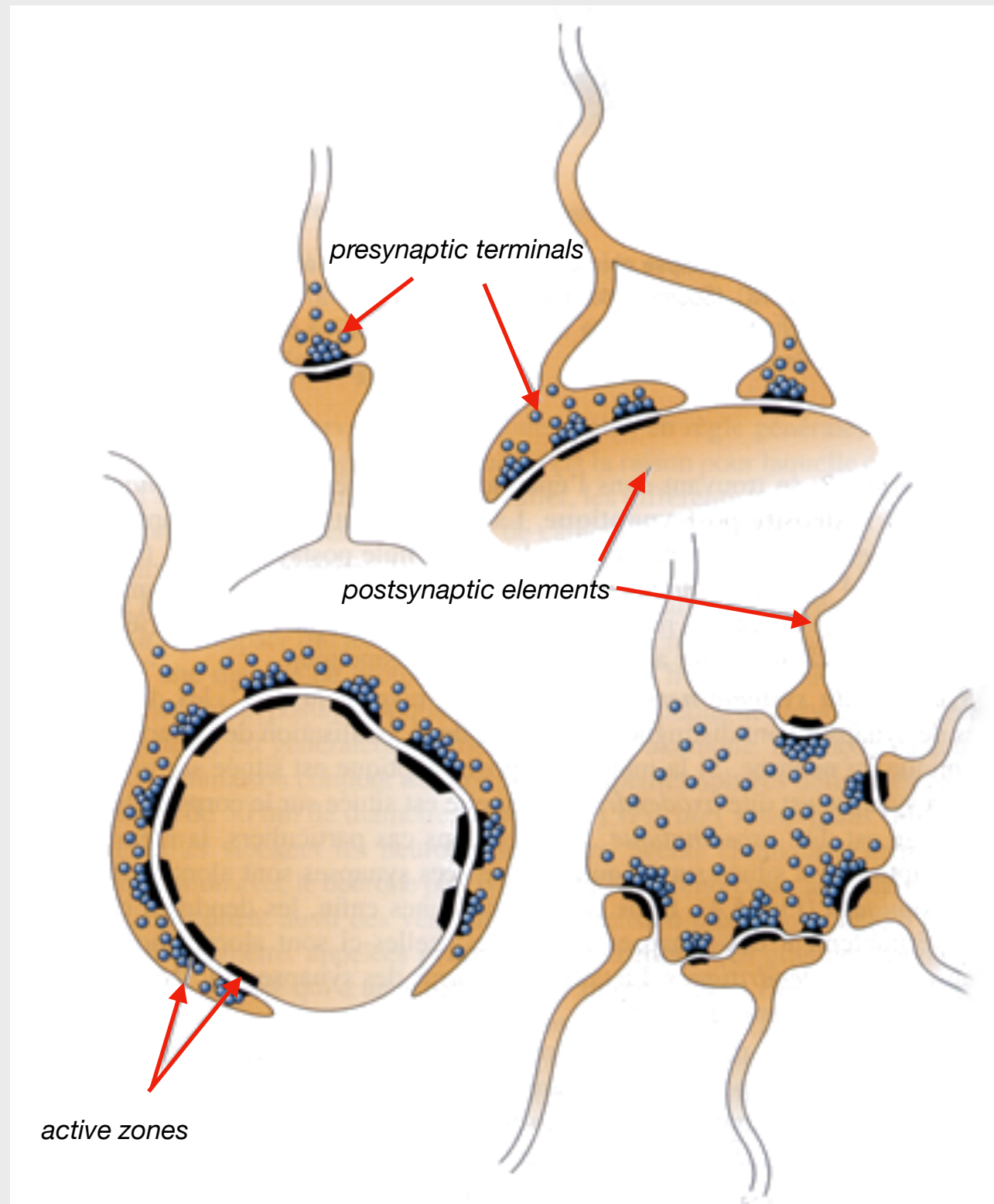
thalamic excitatory synapse

symmetric synapse (Gray type 2)
usually inhibitory synapse

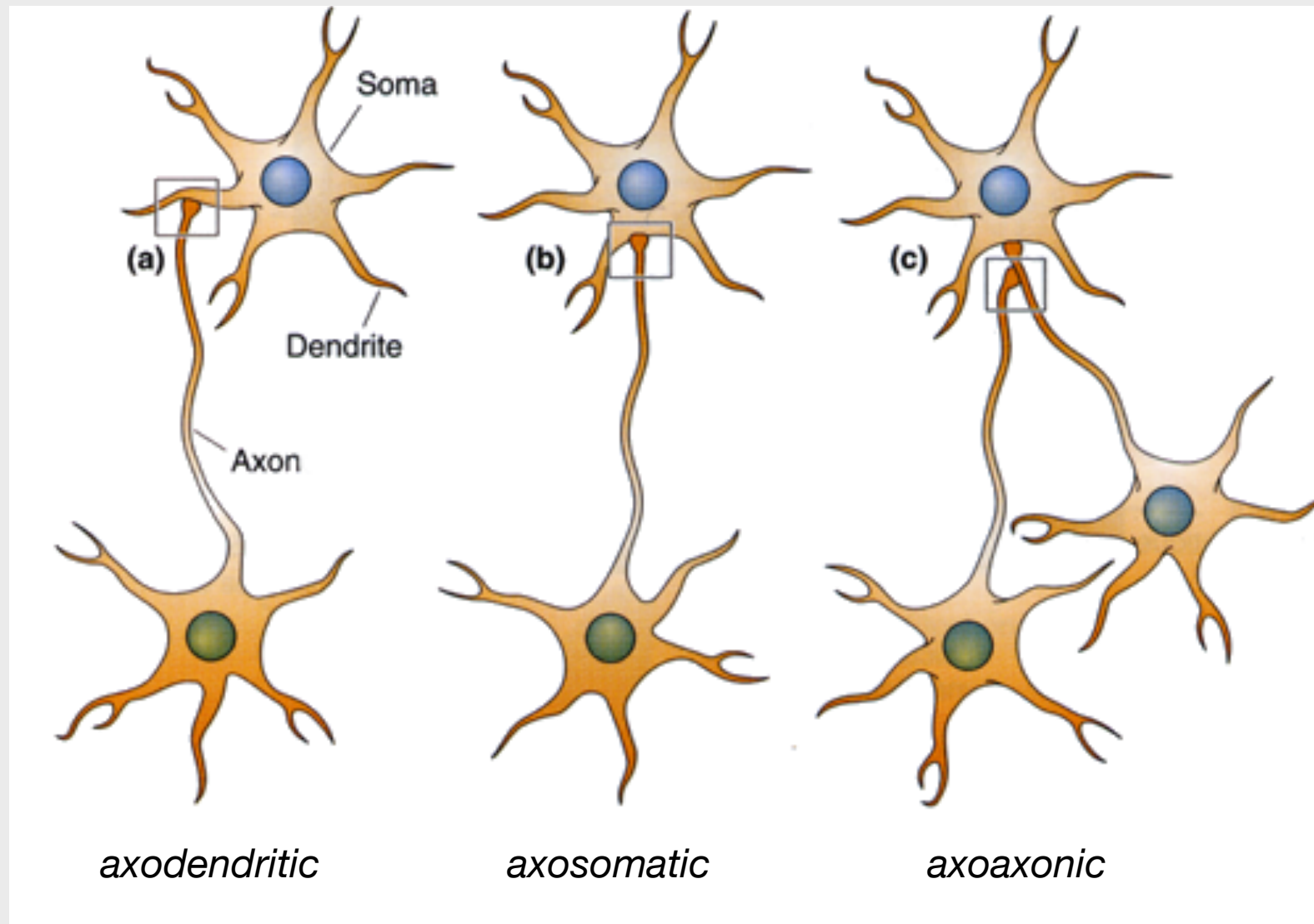


hippocampal inhibitory synapse

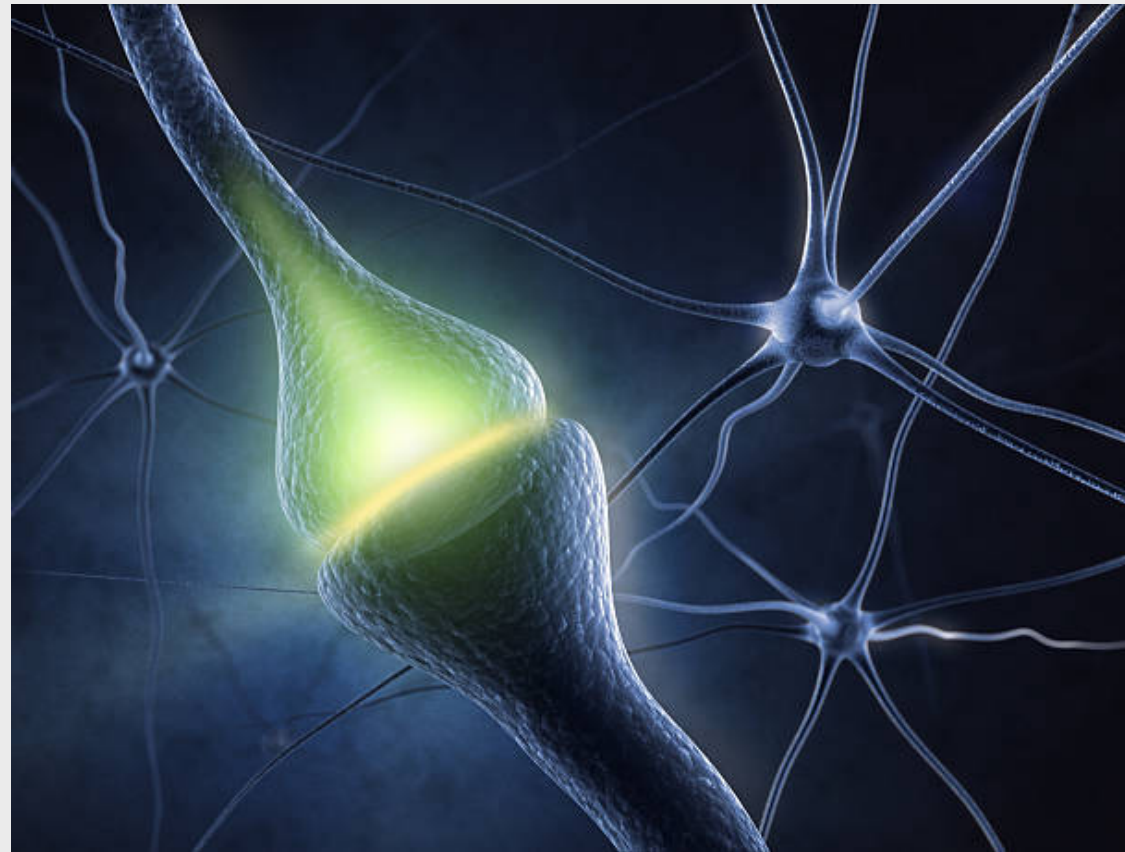
structure of the synapses



structure of the synapses



basic knowledge of synaptic function



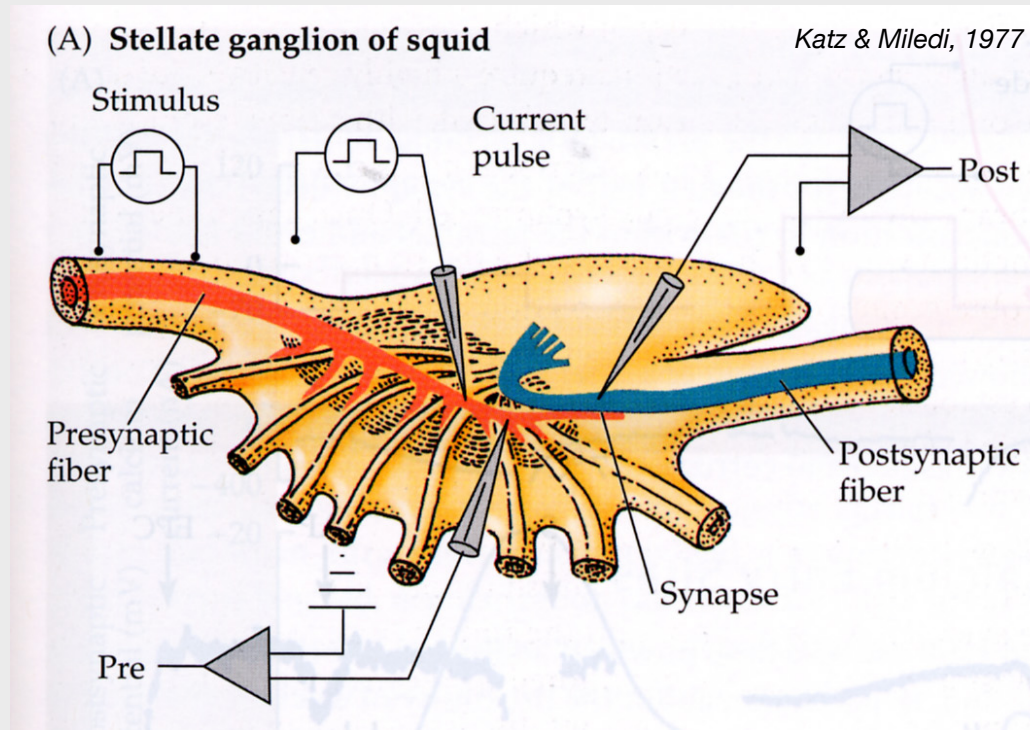
neurotransmitters are accumulated in vesicles (quanta)

entry of Ca^{2+} triggers vesicle fusion and release of the neurotransmitter into the synaptic cleft

Where's the evidence?

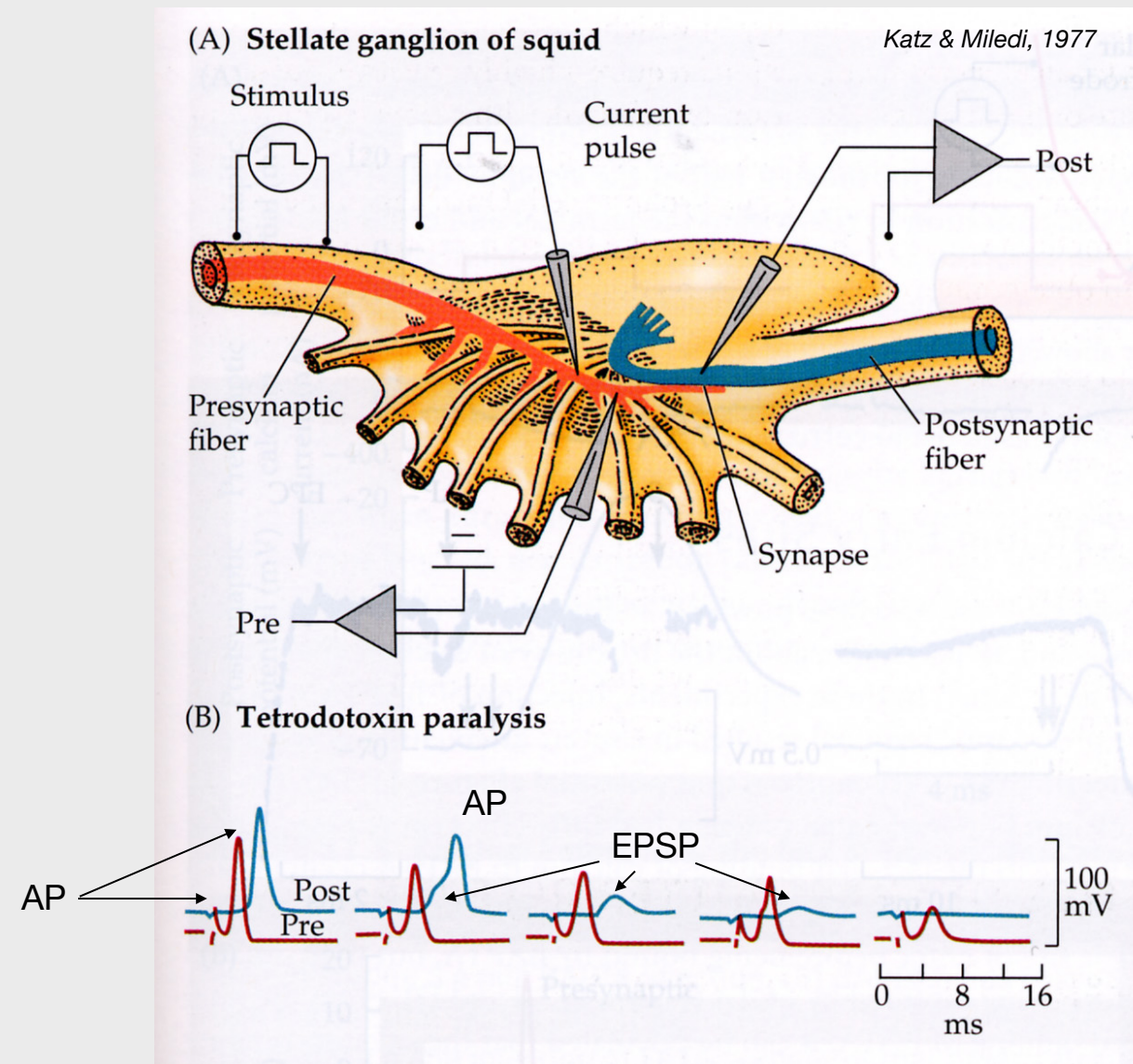
What is the role of the action potential ?

a model where both presynaptic and postsynaptic Vm can be recorded simultaneously



What is the role of the action potential ?

a model where both presynaptic and postsynaptic Vm can be recorded simultaneously



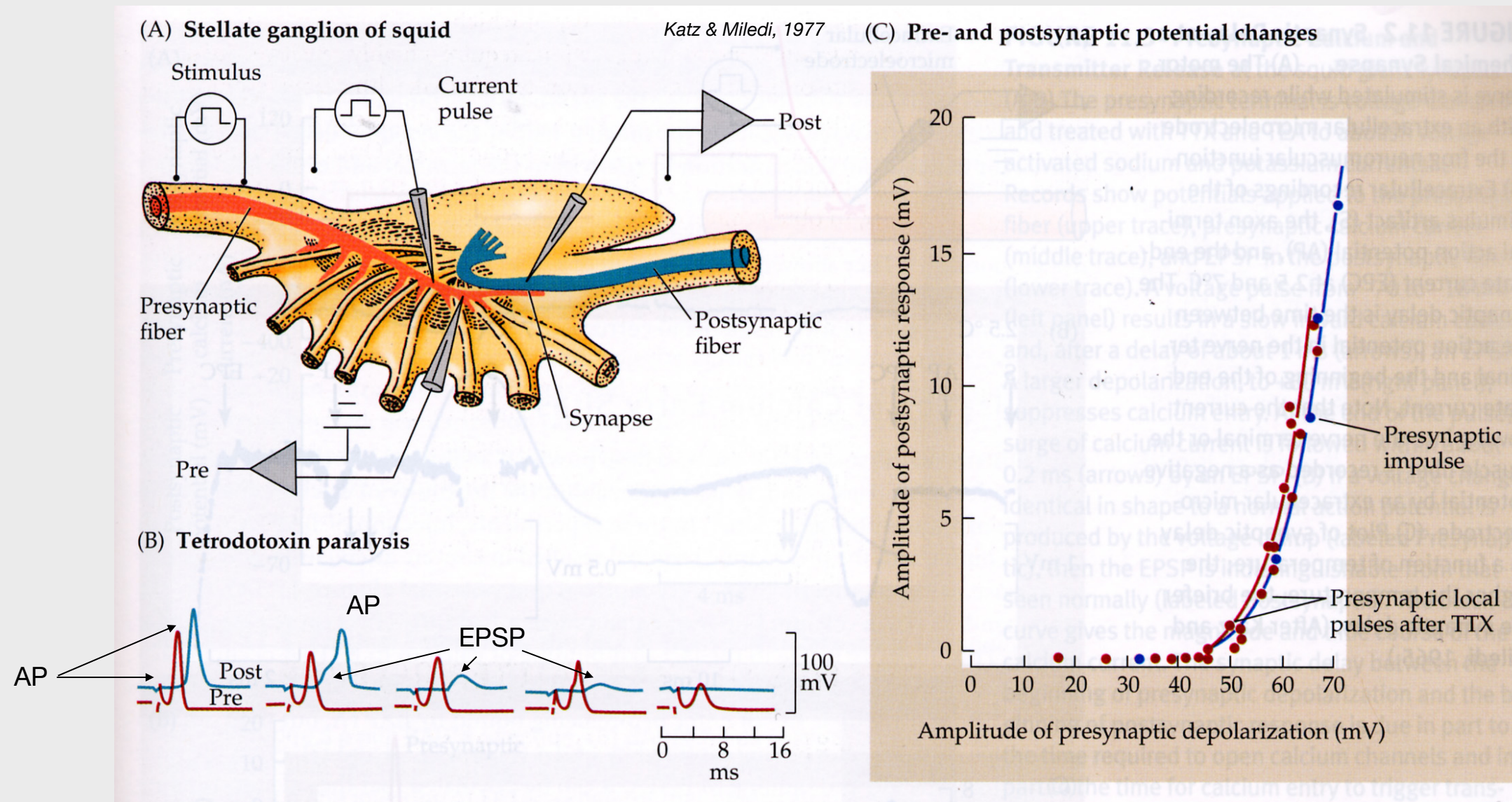
- bath application of TTX progressively suppress pre and postsynaptic AP
- when presynaptic depolarization is small, an EPSP is not observed anymore



presynaptic depolarization / Na⁺ entry ?

What is the role of the action potential ?

a model where both presynaptic and postsynaptic Vm can be recorded simultaneously

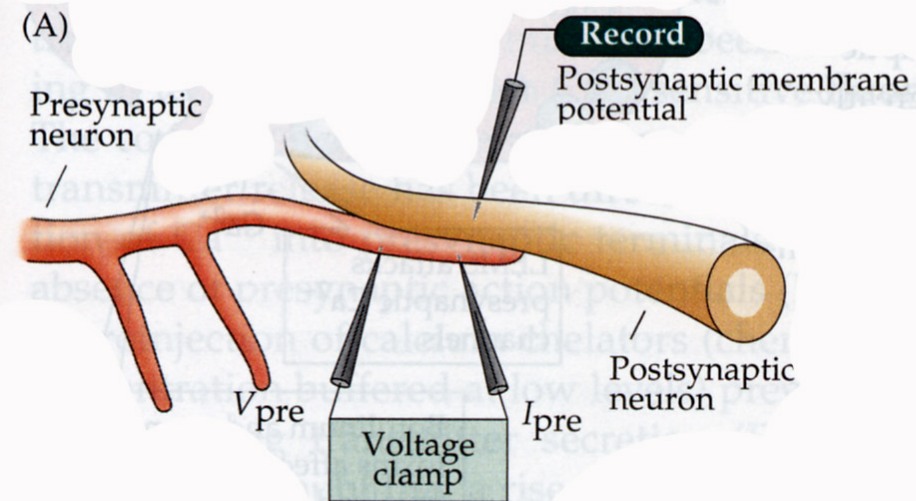


- bath application of TTX progressively suppress pre and postsynaptic AP
- when presynaptic depolarization is small, an EPSP is not observed anymore
- when the presynaptic element is depolarized by current injection (no Na^+ involved), similar EPSP amplitudes are observed

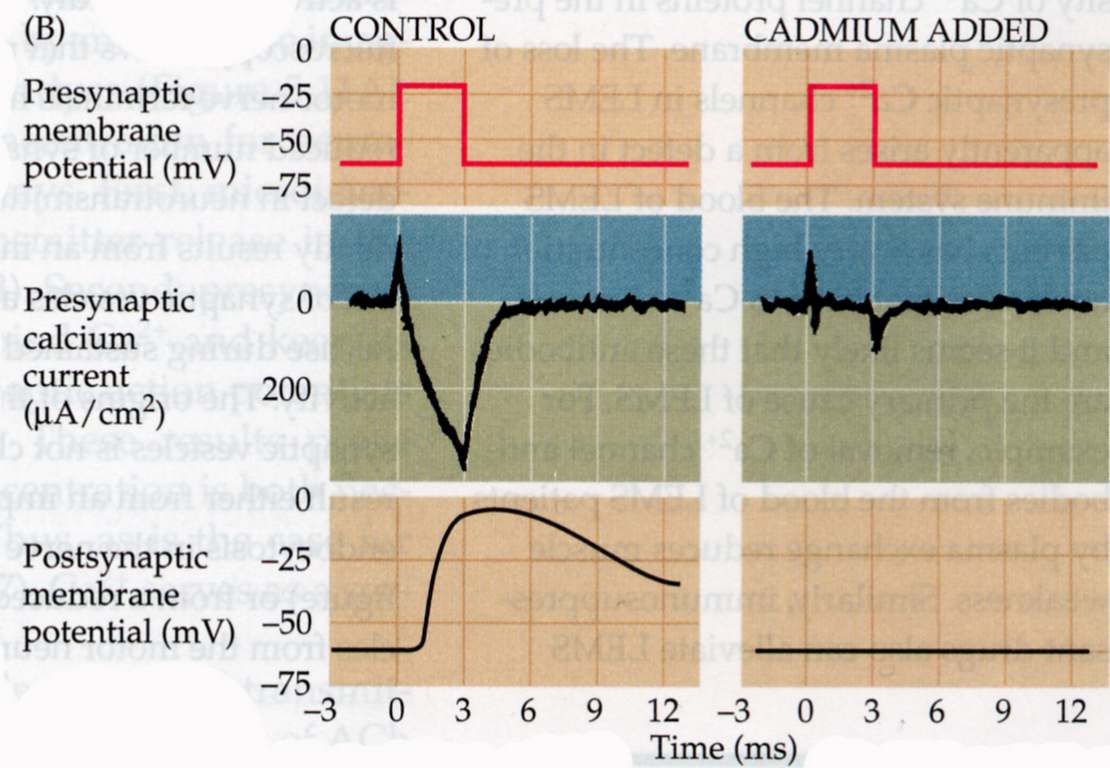


presynaptic depolarization / Na^+ entry?

Ca²⁺ ions

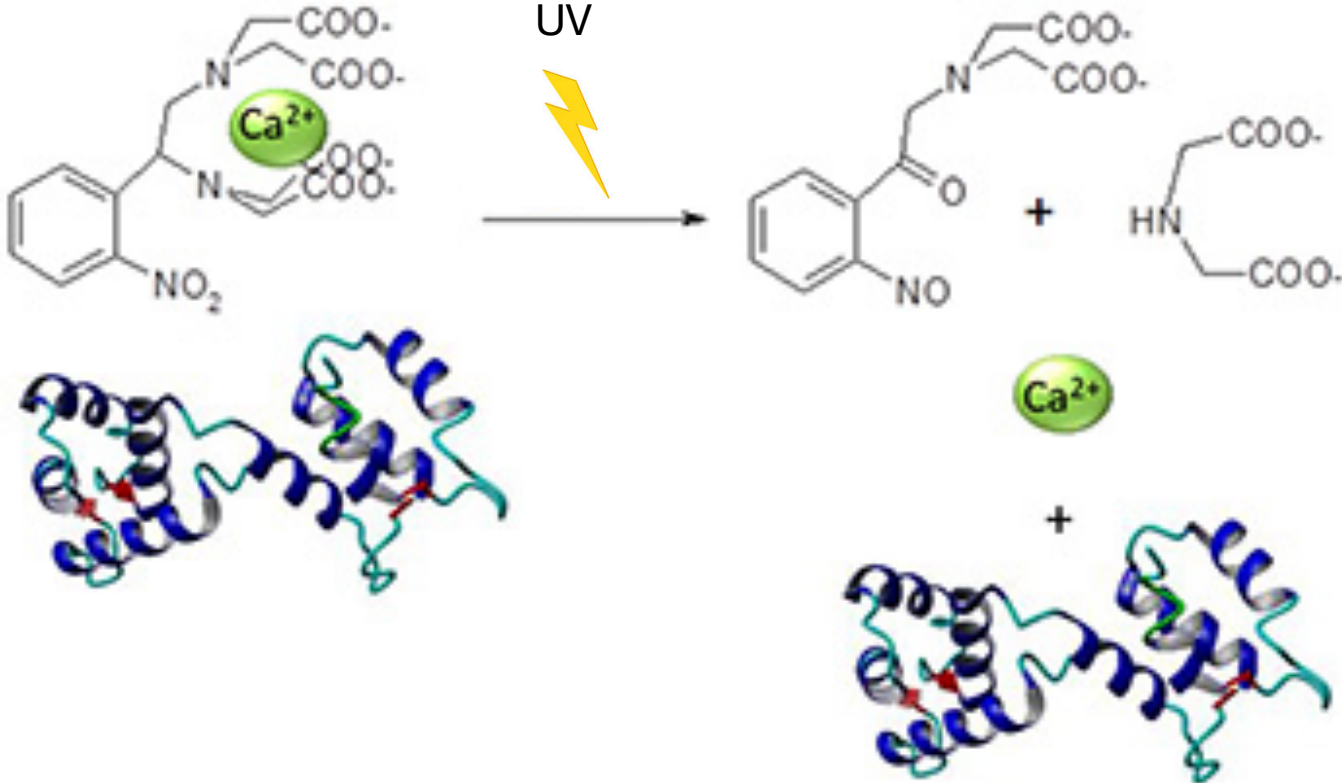
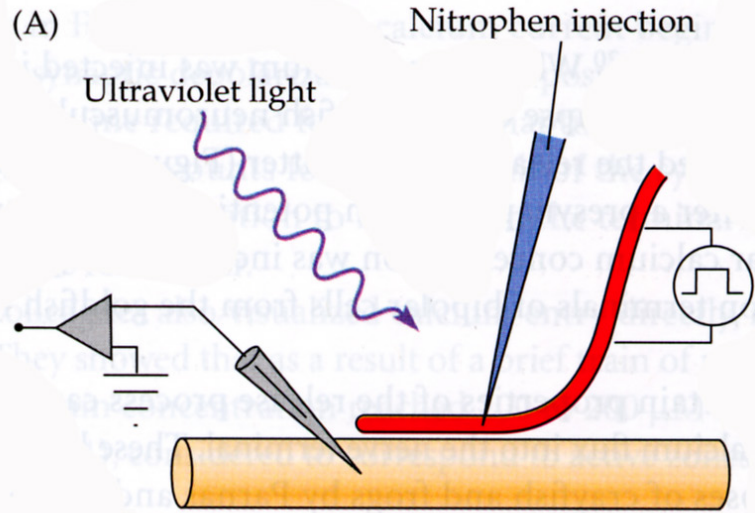


Augustine & Eckert, 1984

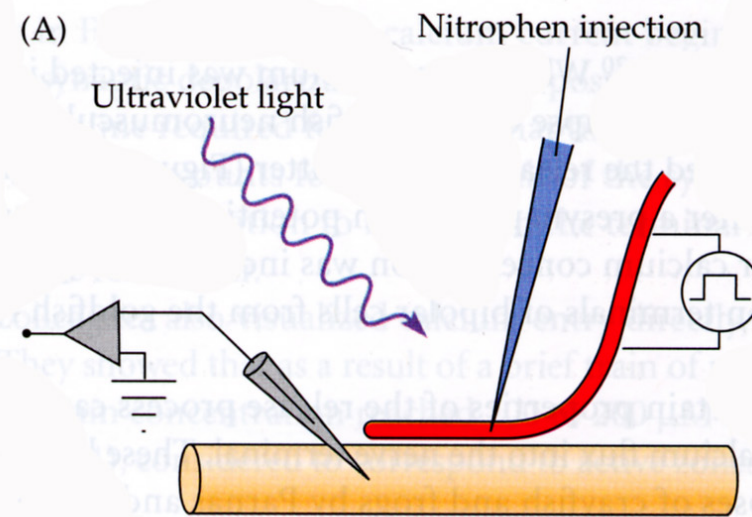


- *Ca²⁺ entry is required*
- *is it sufficient ?*

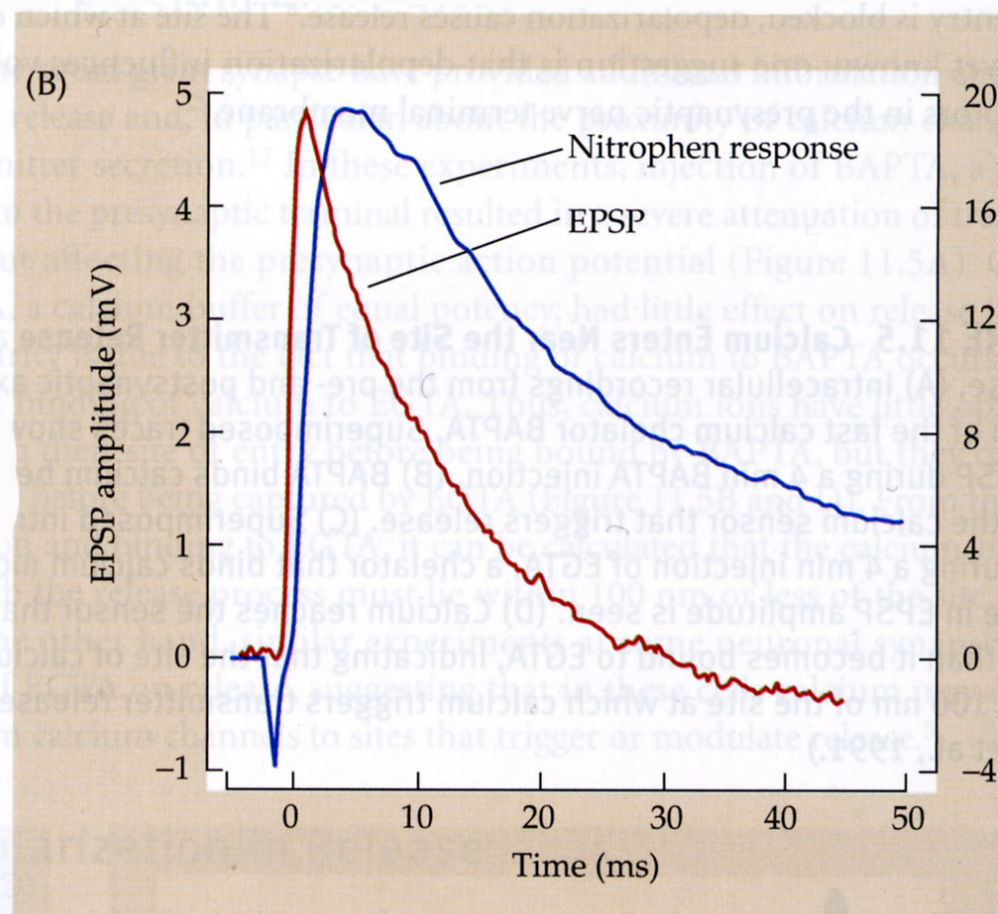
Ca²⁺ ions



Ca²⁺ ions



Zucker, 1993

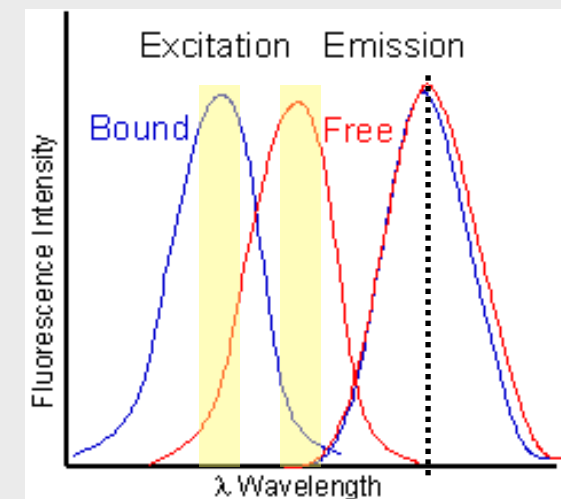
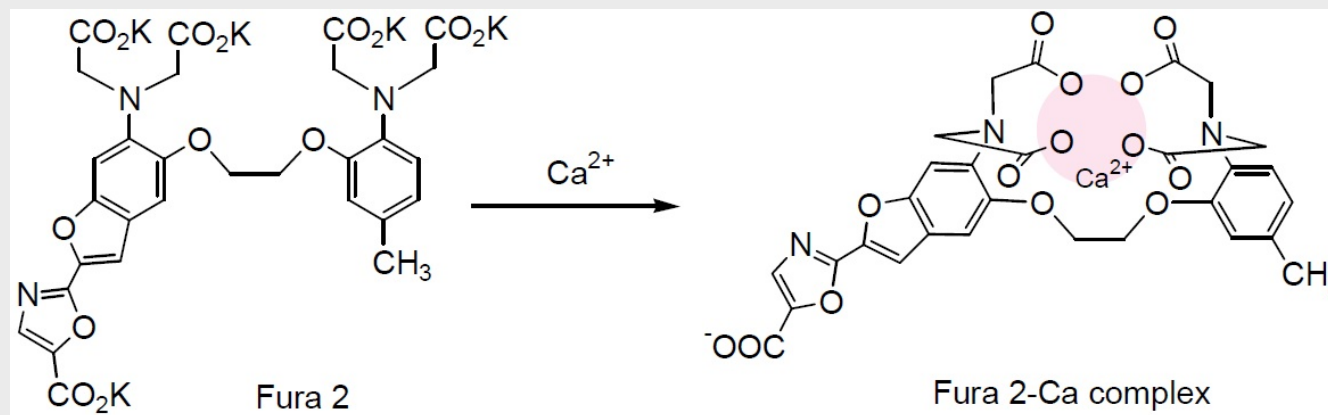


intracellular Ca²⁺ rise is required and sufficient

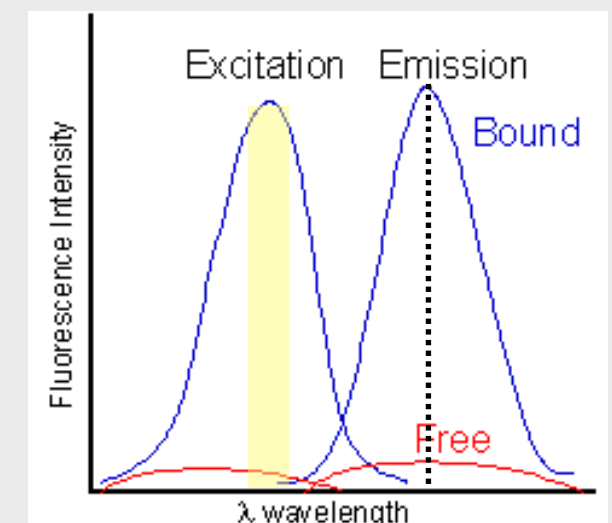
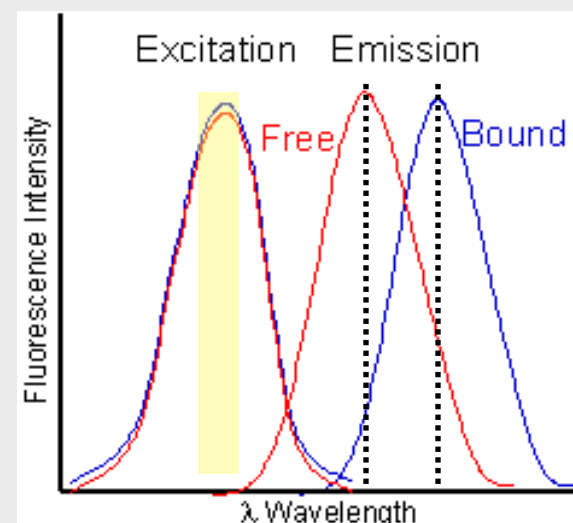
Ca²⁺ ions



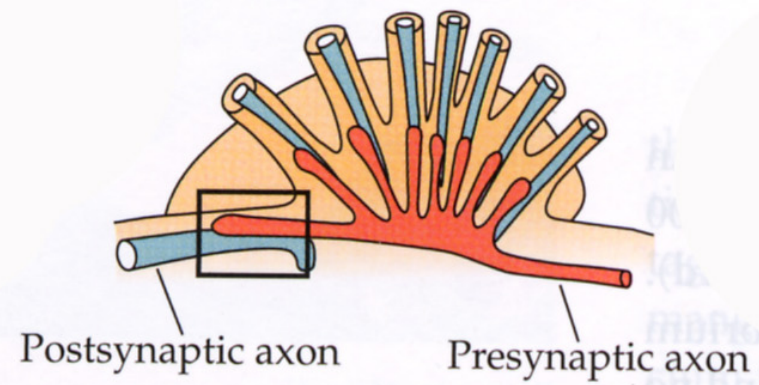
fluorescent Ca²⁺ dyes



Fluo4, 5, ..., calcium green, ...

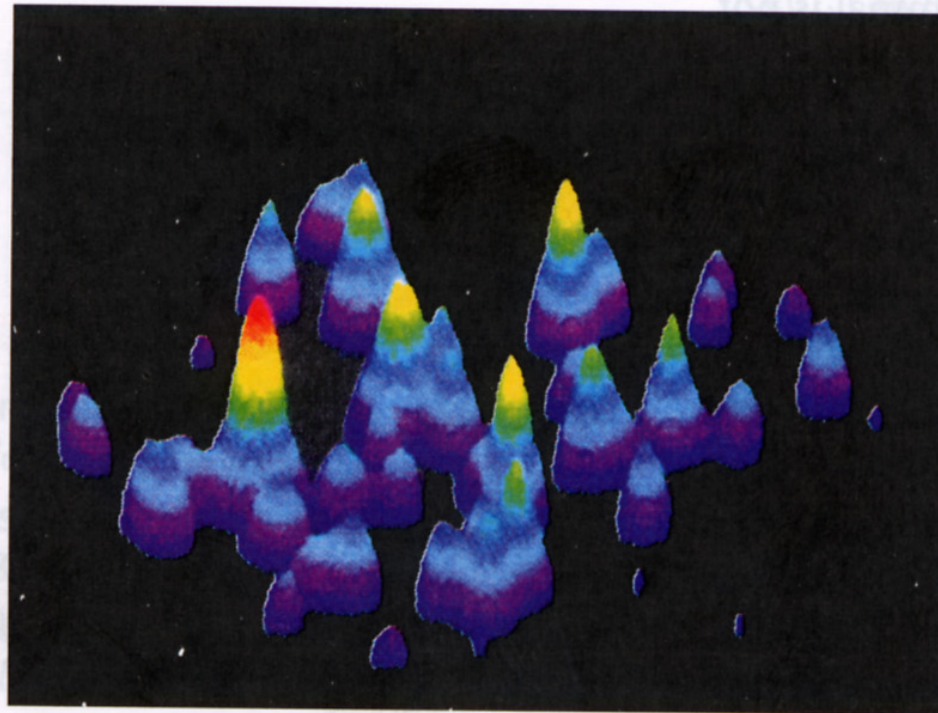


Ca²⁺ ions



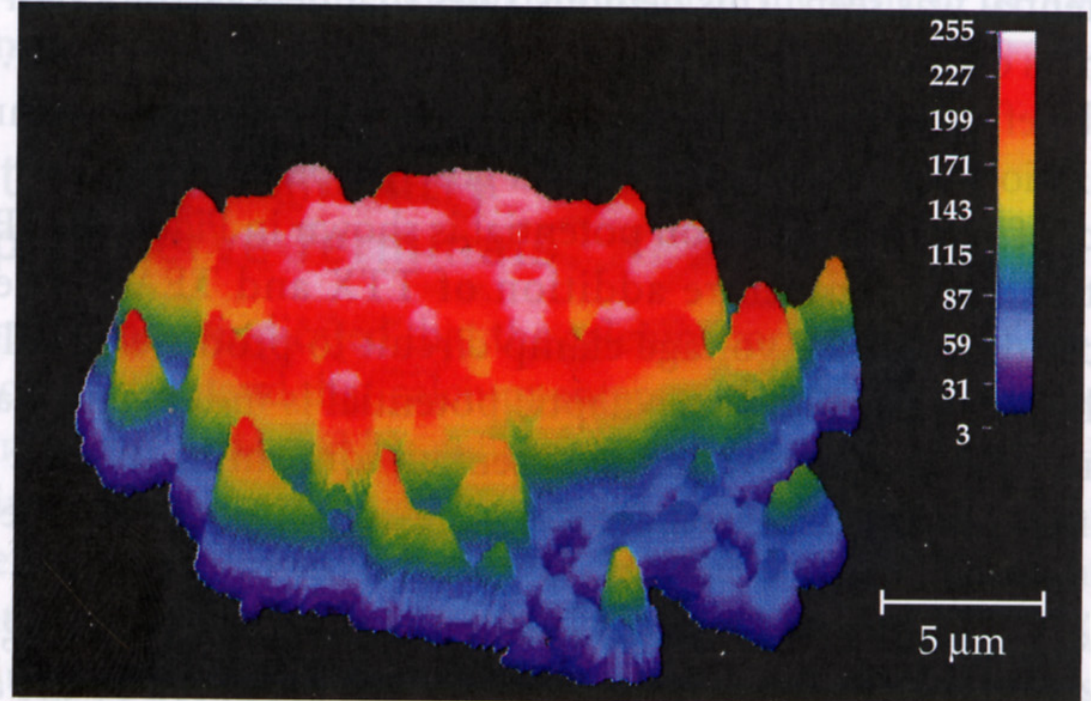
Llinas et al., 1992

(A)



no AP

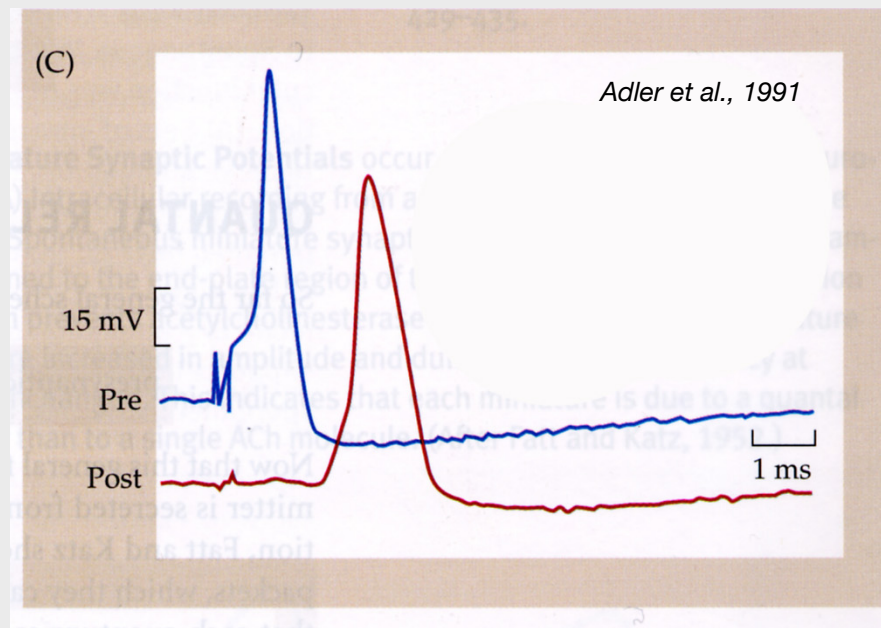
(B)



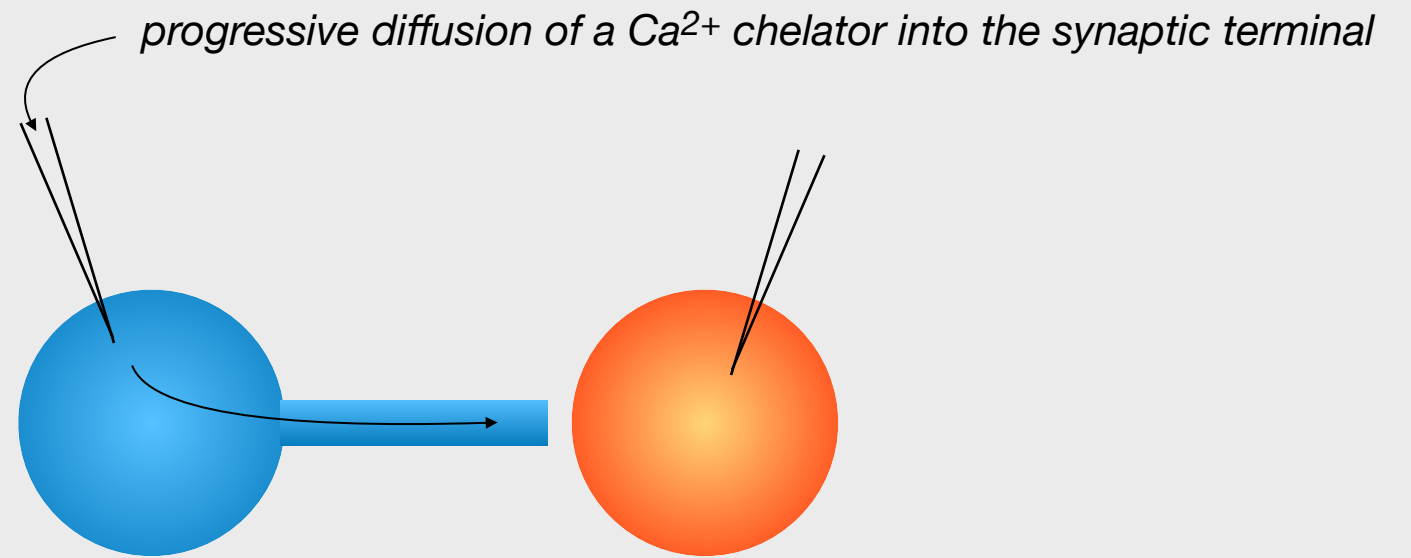
AP

global increase of Ca²⁺ concentration during action potentials (APs)

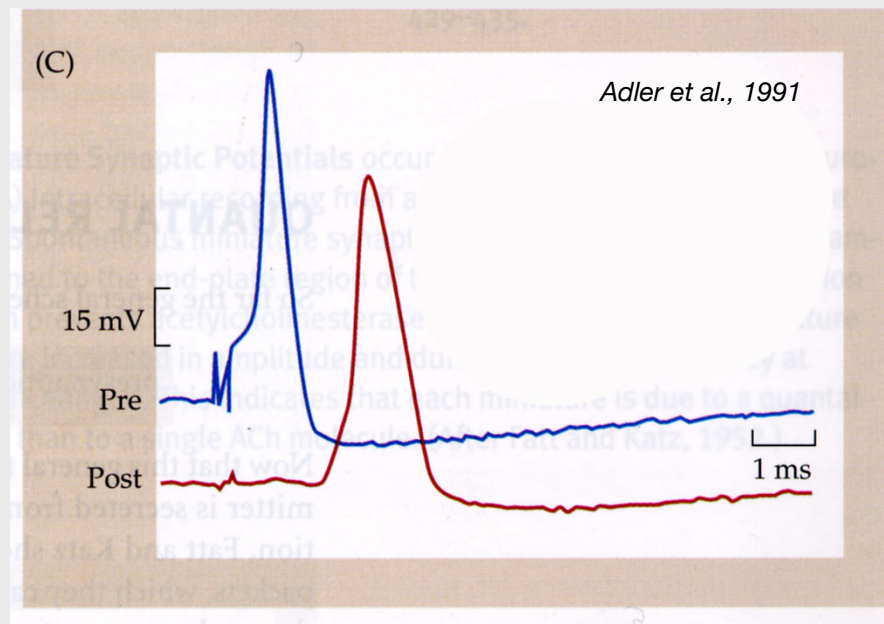
Ca²⁺ ions



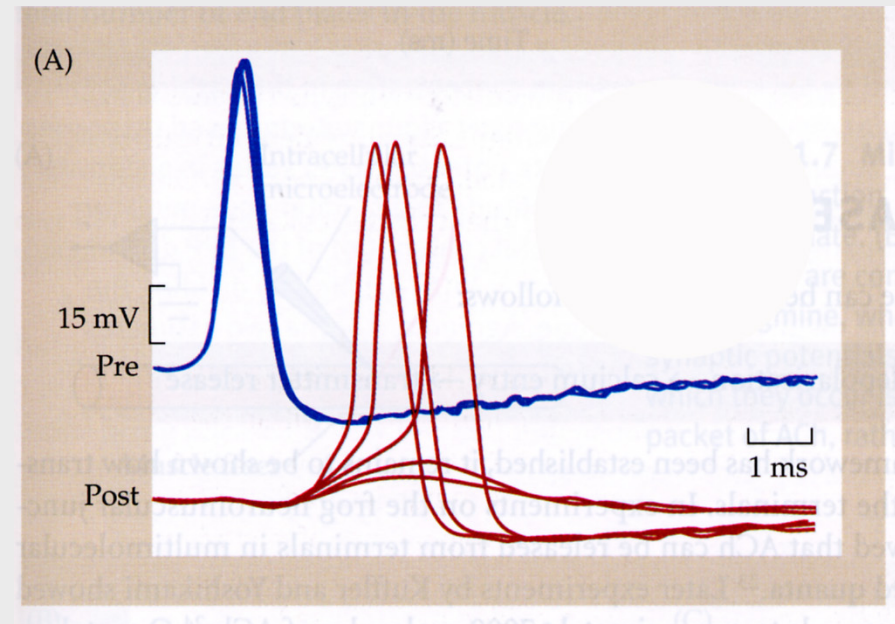
EGTA : « slow » Ca²⁺ chelator



Ca²⁺ ions



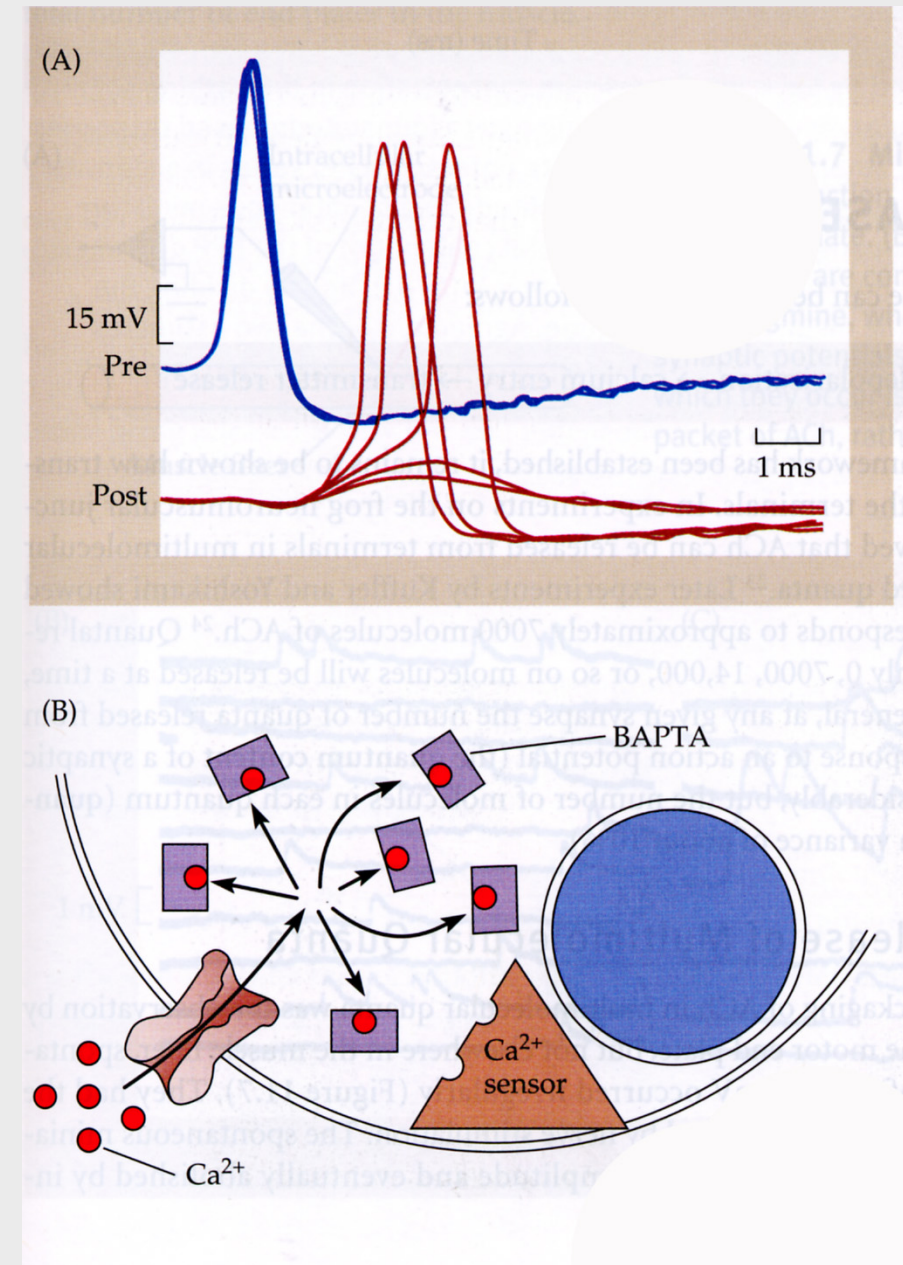
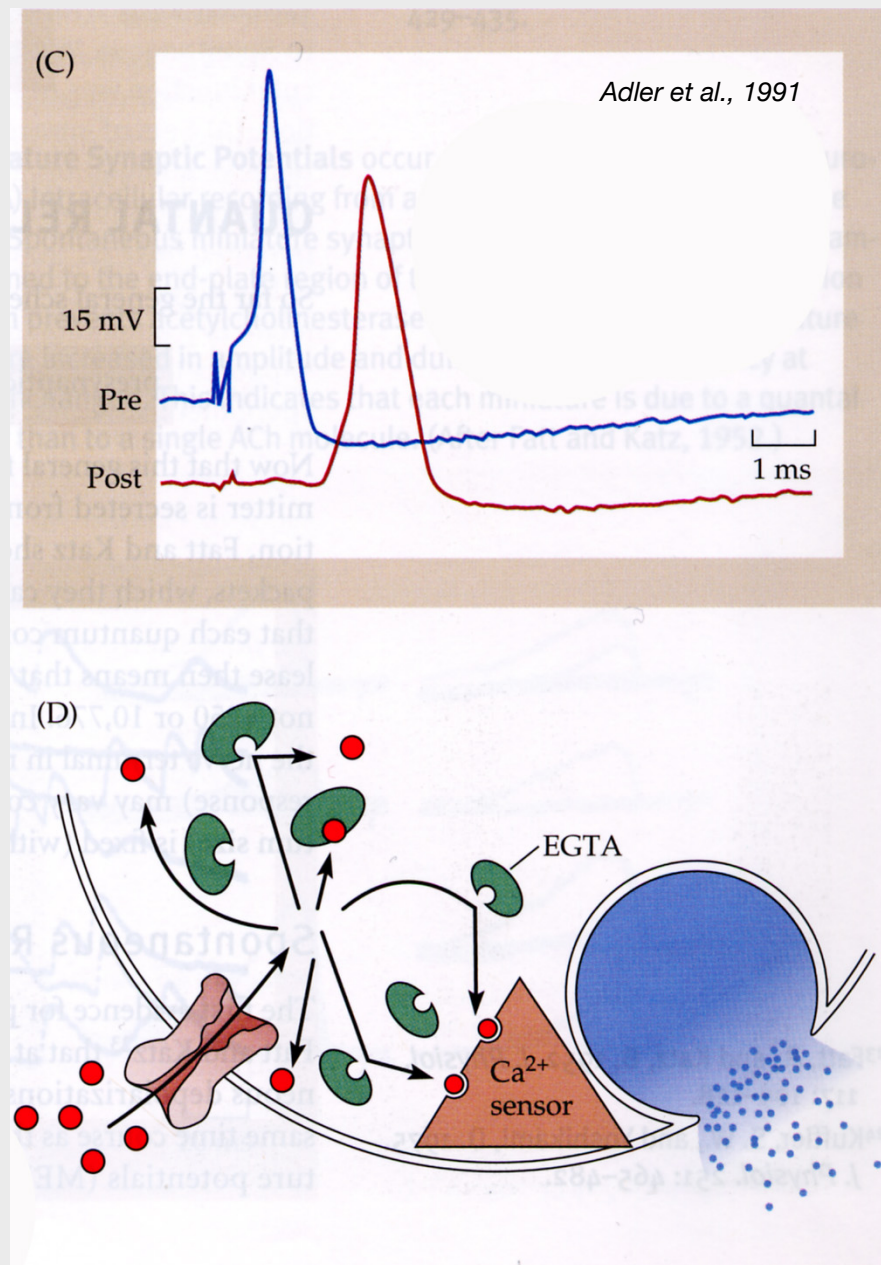
EGTA : « slow » Ca²⁺ chelator



BAPTA : « fast » Ca²⁺ chelator

?

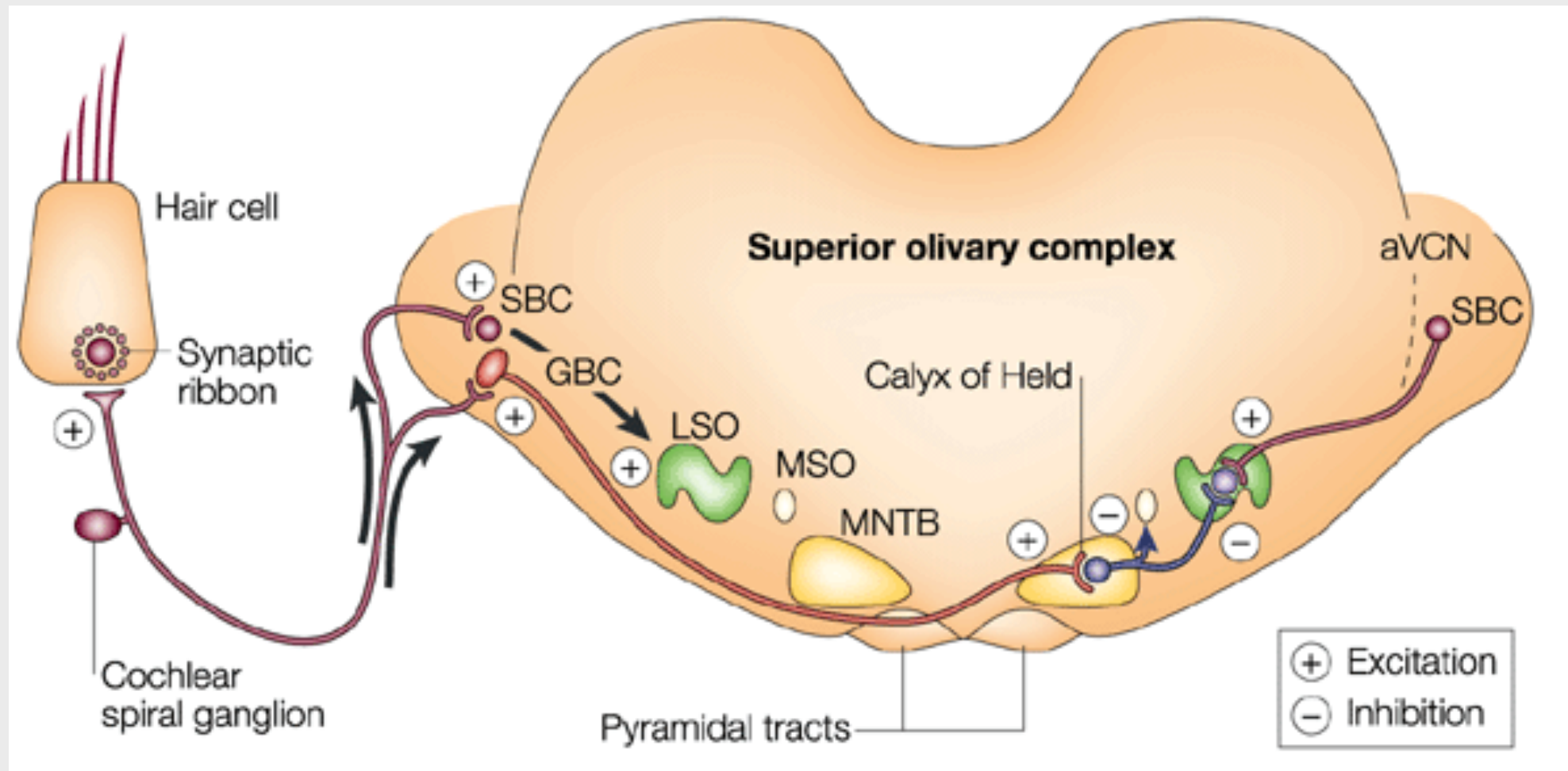
Ca²⁺ ions



Ca²⁺ channels are close to the Ca²⁺ sensor

Calcium concentration triggering the neurotransmitter release

a model where both presynaptic and postsynaptic Vm or currents can be recorded simultaneously in the vertebrate CNS

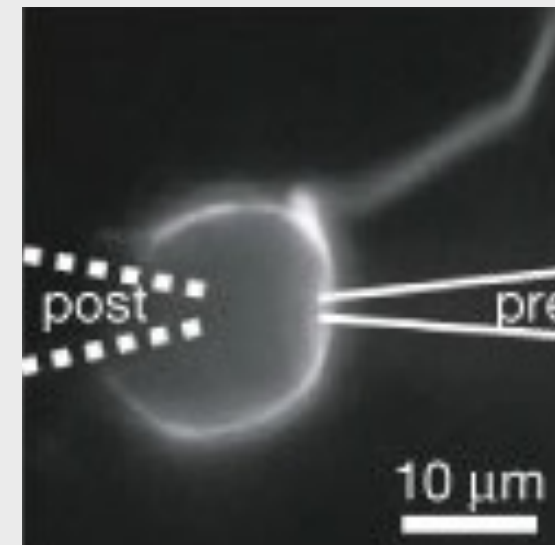
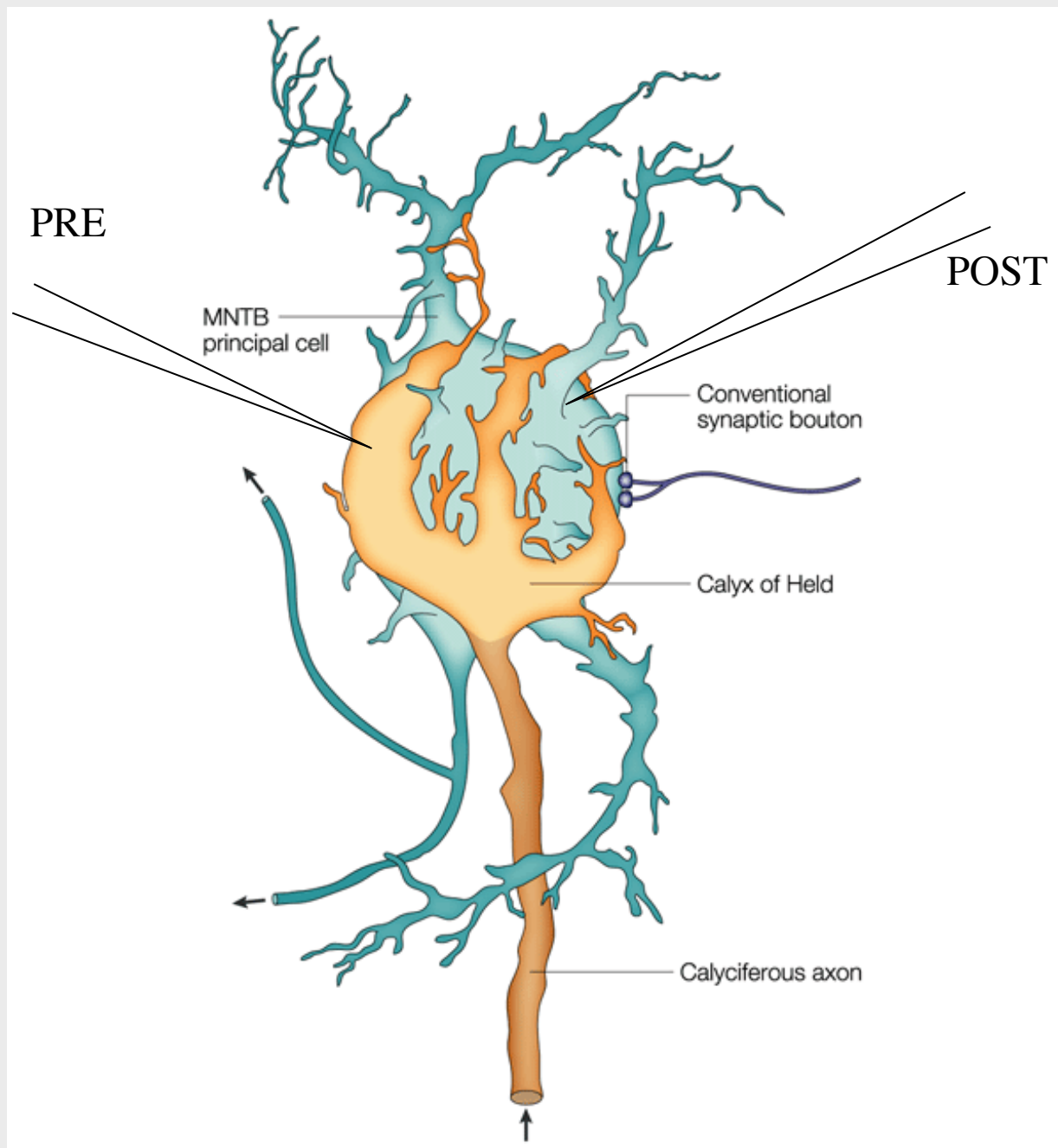


- **SBC** : Spherical Bushi Cell
- **GBC** : Globular Bushi Cell
- **LSO** : Lateral Superior Olive
- **MSO** : Medial Superior Olive
- **MNTB** : Medial Nucleus of the Trapezoid Body
- **VCN** : Ventral Cochlear Nucleus

- Sound coming from a lateral direction reaches one cochlea first, and then, with a short delay, the other cochlea, at a slightly attenuated intensity because our head acts as a sound barrier.
- The microsecond interaural time differences in the arrival of sound are used to determine the spatial location of low-frequency (<1–2 kHz) sounds, whereas differences in the intensity of sound arriving at each cochlea are used to locate high-frequency sounds.
- The superior olivary complex (SOC) of the mammalian brainstem is involved in computing sound localization from these two binaural cues, and the precise timing of action potential (AP) firing is crucial for this task.
- To preserve the timing of APs across several synaptic connections, some neurons of the SOC show remarkable morphological and biophysical adaptations, such as thick-calibre axons, large nerve terminals and fast-activating ion channels.

Calcium concentration triggering the neurotransmitter release

a model where both presynaptic and postsynaptic Vm or currents can be recorded simultaneously in the vertebrate CNS



fluorescent Ca^{2+} sensitive dye



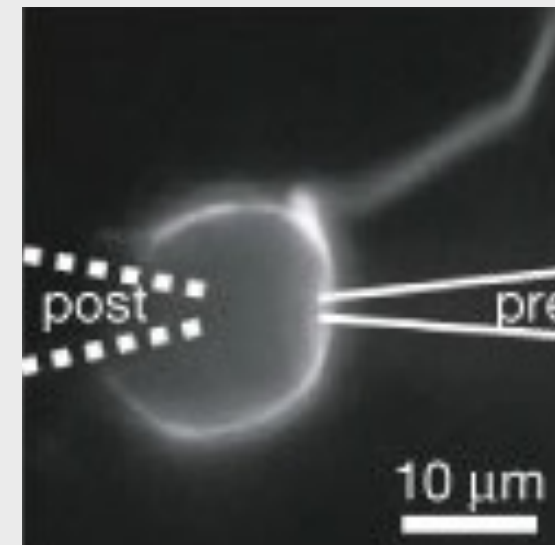
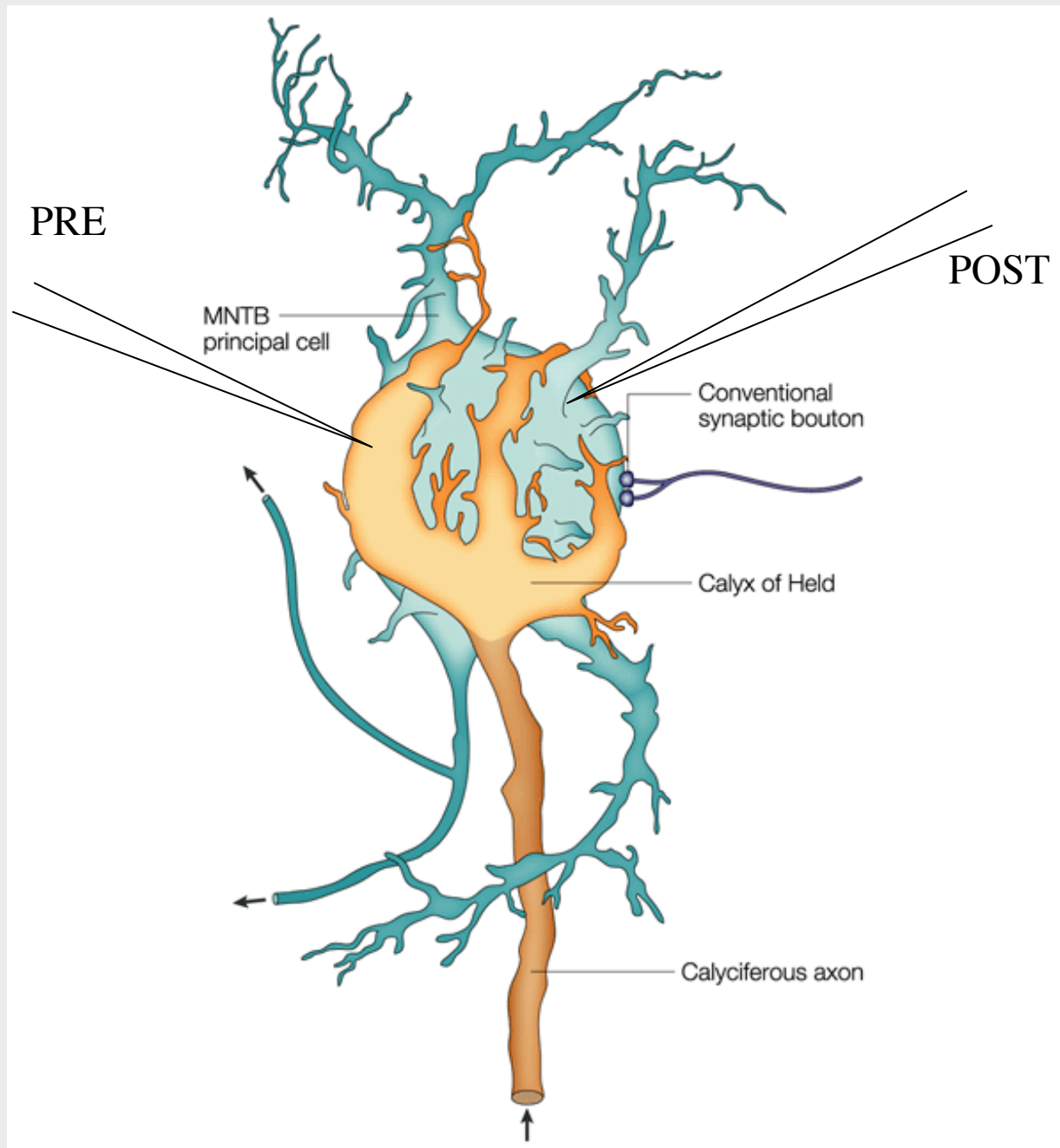
mean Ca^{2+} concentration



mean Ca^{2+} concentration =
local Ca^{2+} concentration at the
level of the Ca^{2+} sensor

Calcium concentration triggering the neurotransmitter release

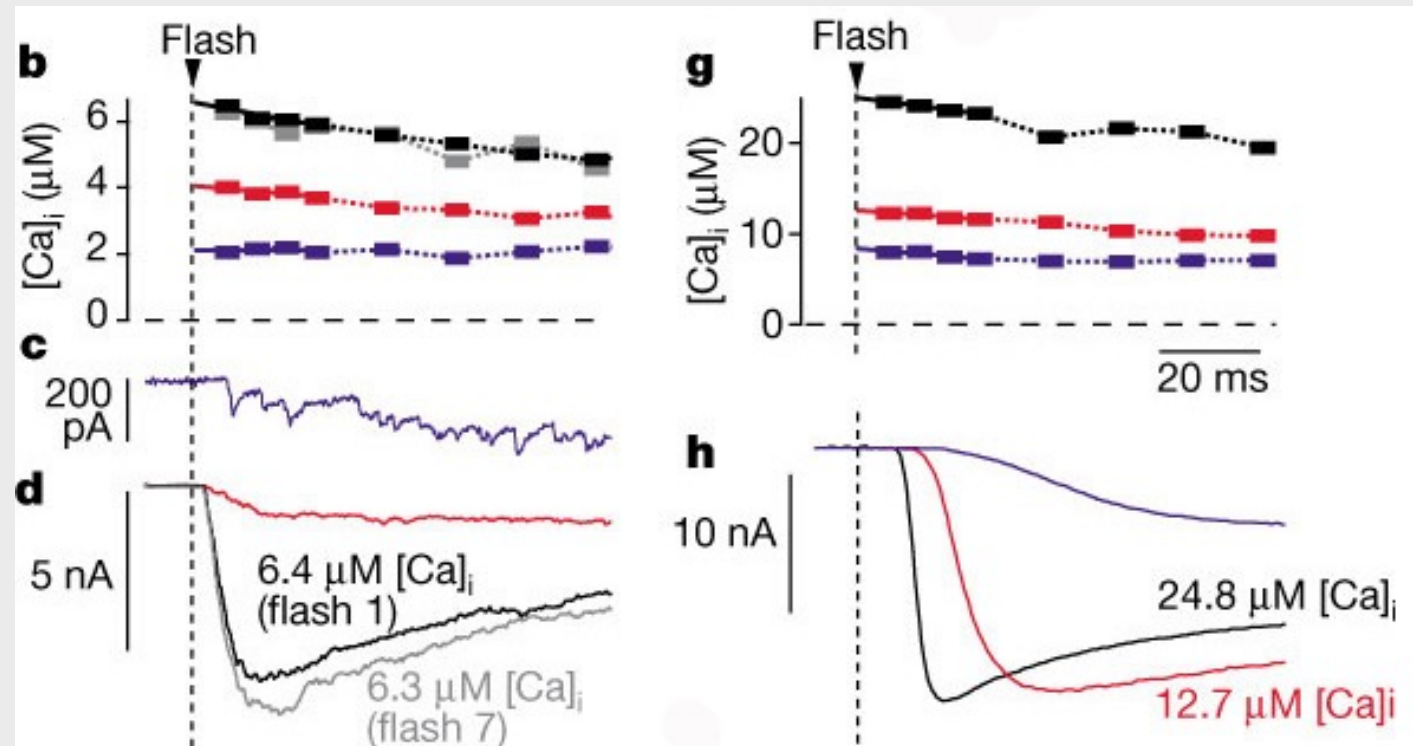
a model where both presynaptic and postsynaptic Vm or currents can be recorded simultaneously in the vertebrate CNS



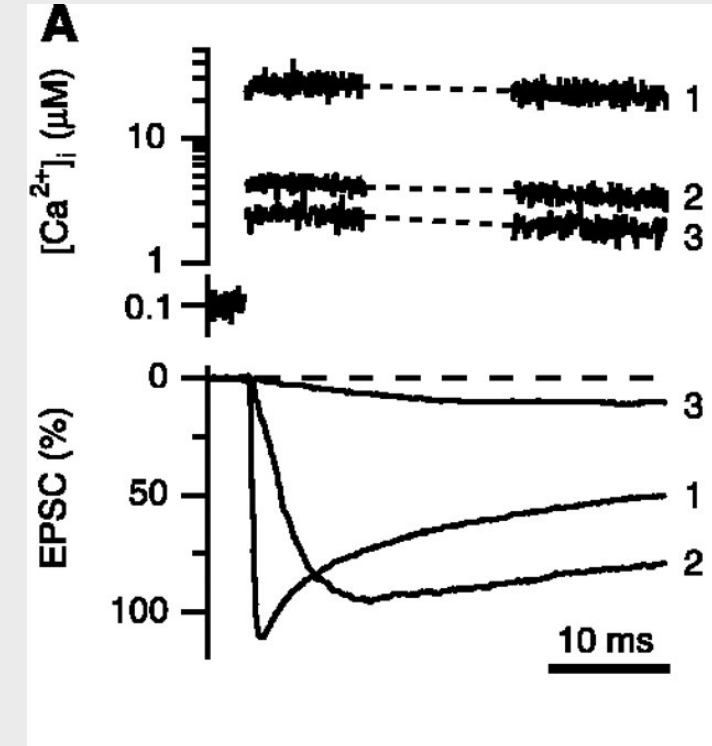
fluorescent Ca^{2+} sensitive dye
Nitrophen (caged Ca^{2+})

Calcium concentration triggering the neurotransmitter release

a model where both presynaptic and postsynaptic Vm or currents can be recorded simultaneously in the vertebrate CNS



Schneggenburger & Neher, 2000



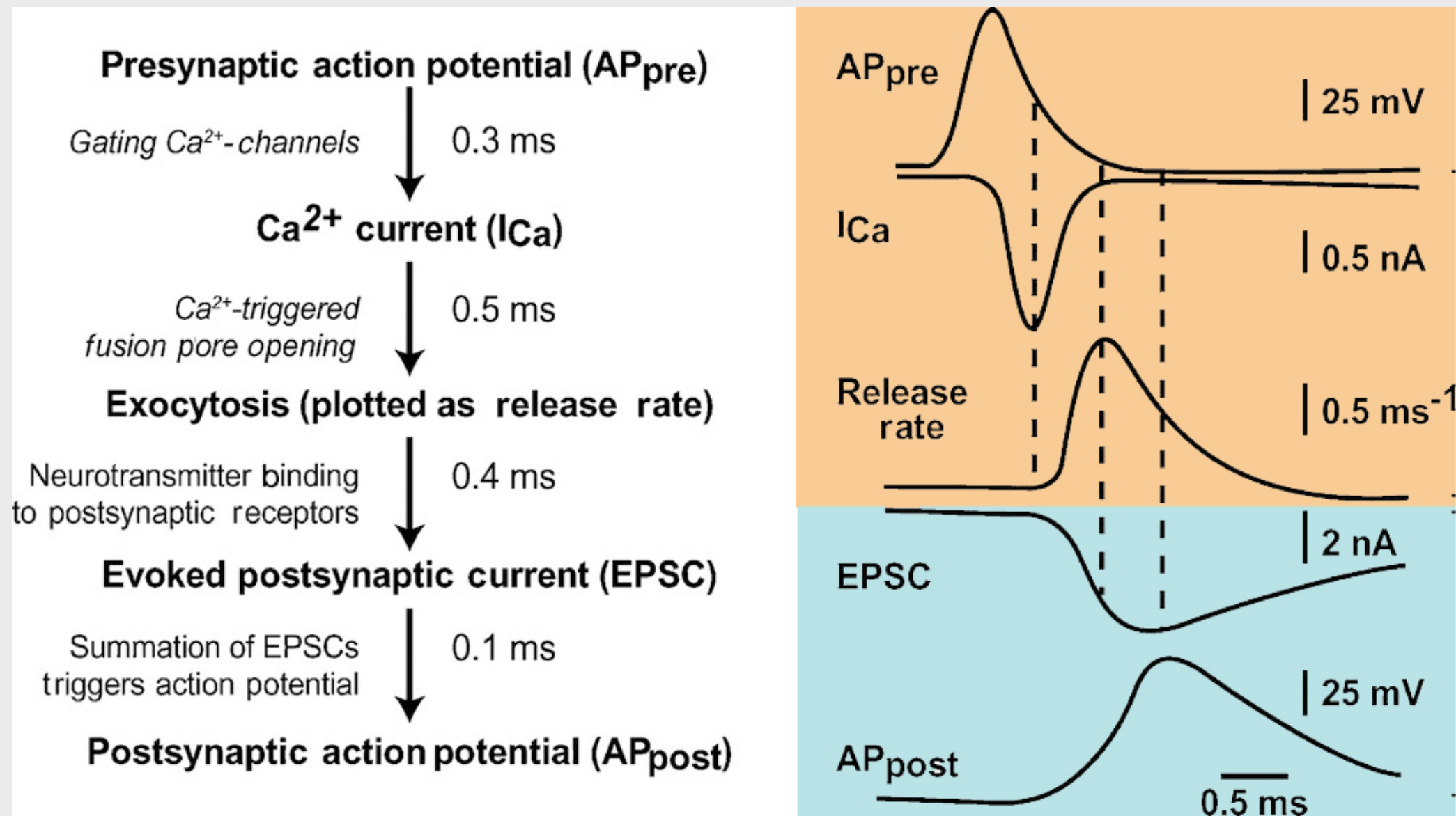
Bollmann et al., 2000

- $< 1 \mu M$: undetectable
- $1-2 \mu M$: observable
- $> 5 \mu M$: normal
- $> 20 \mu M$: maximal

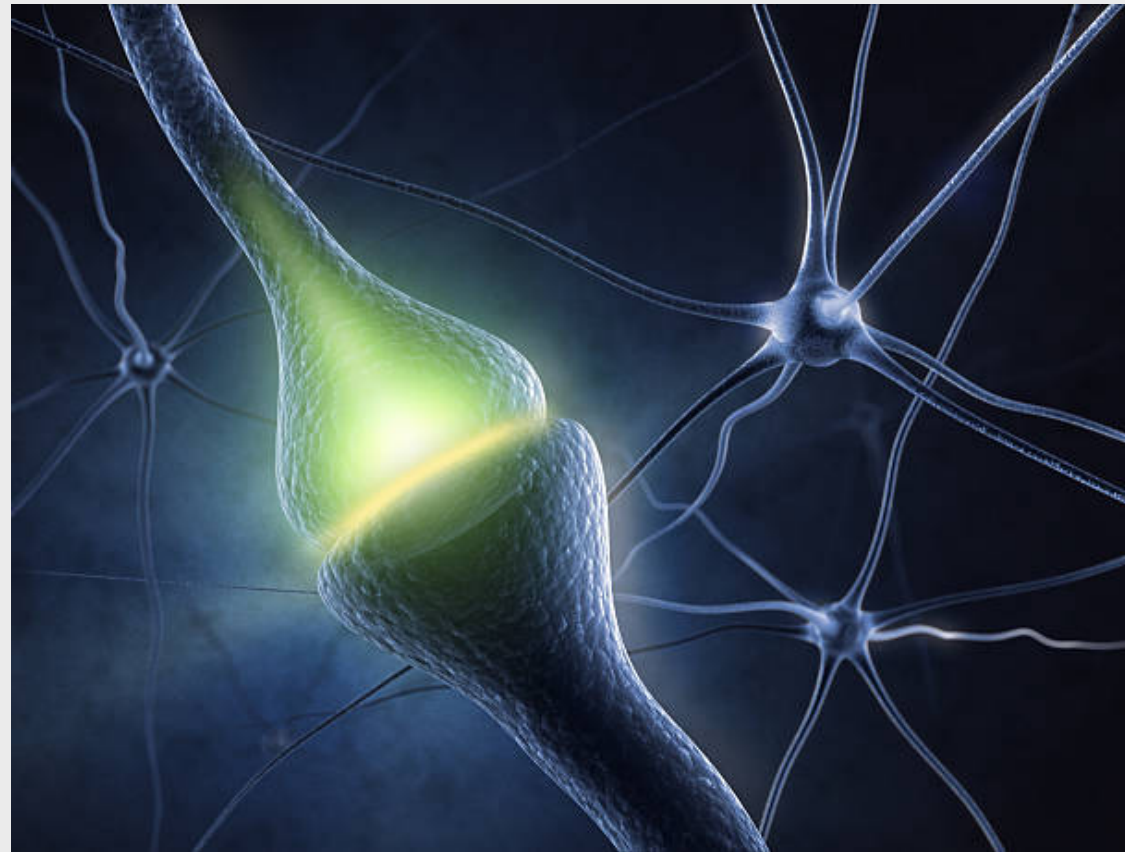


Kd: 5–25 μM of the Ca^{2+} sensor

synaptic transmission takes time



basic knowledge of synaptic function

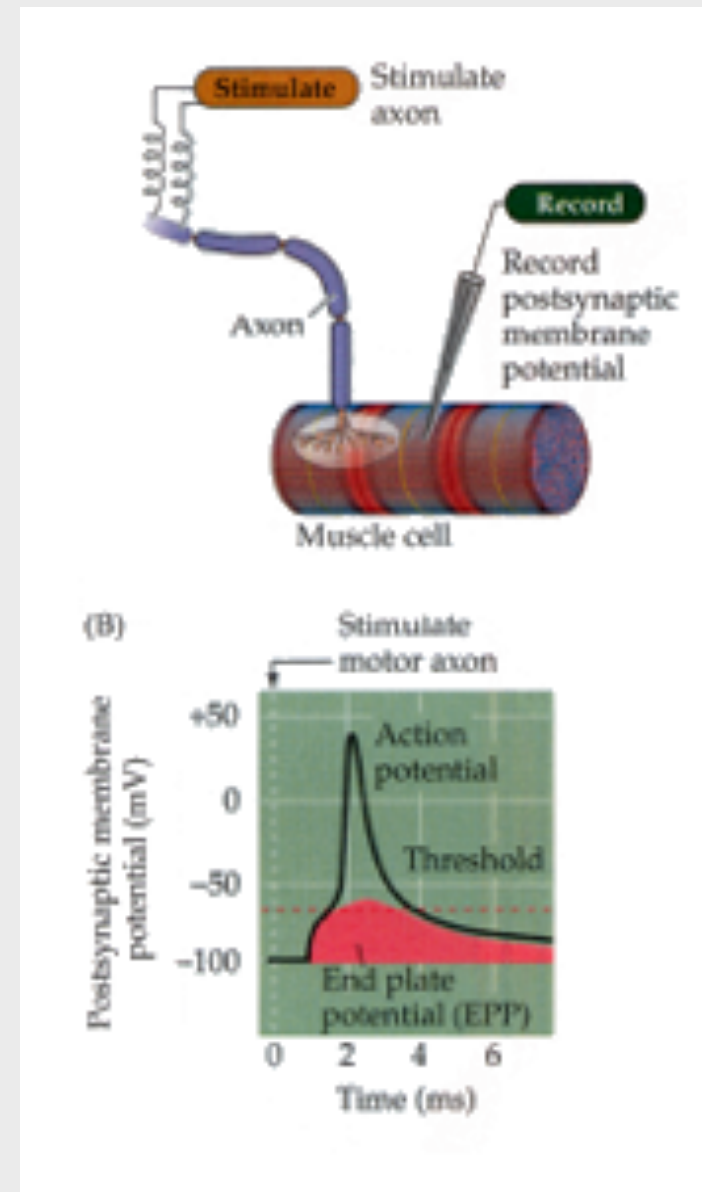
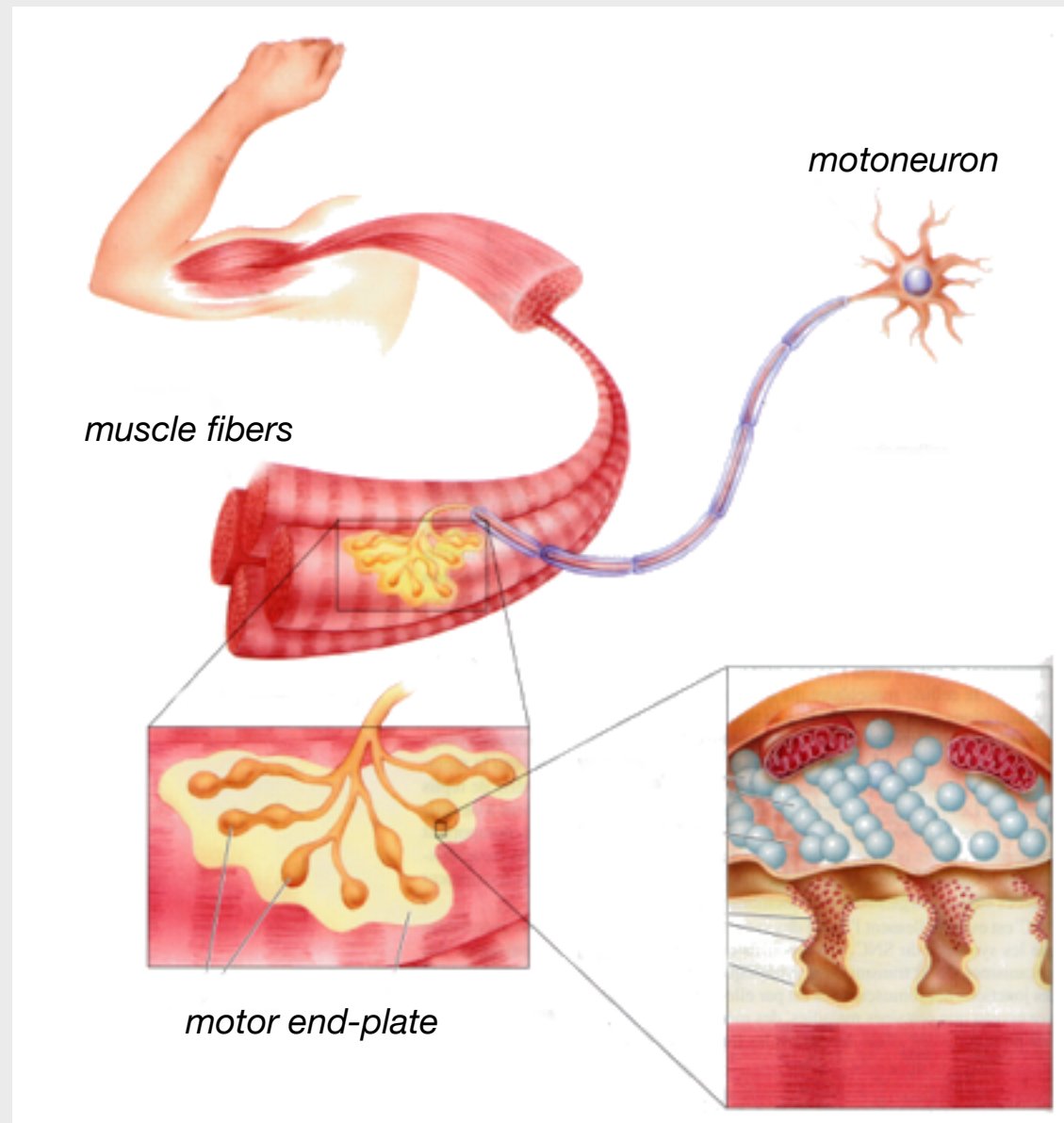


neurotransmitters are accumulated in vesicles (quanta)

entry of Ca^{2+} triggers vesicle fusion and release of the neurotransmitter into the synaptic cleft ✓

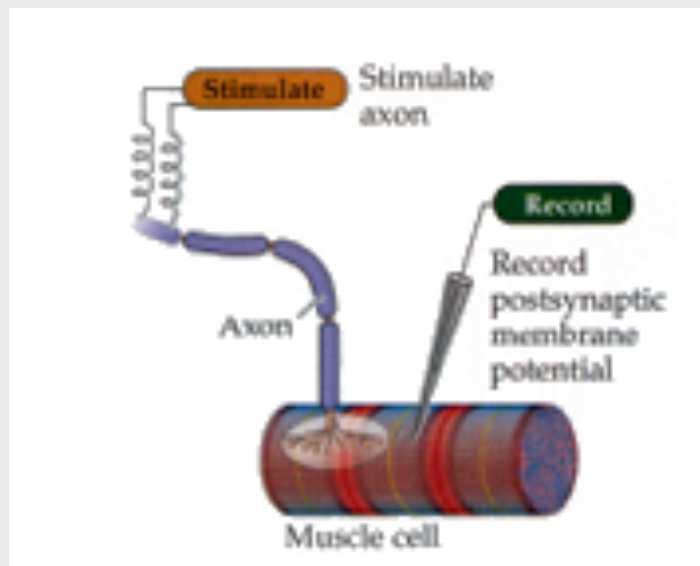
Where's the evidence?

quantal release at the neuromuscular junction

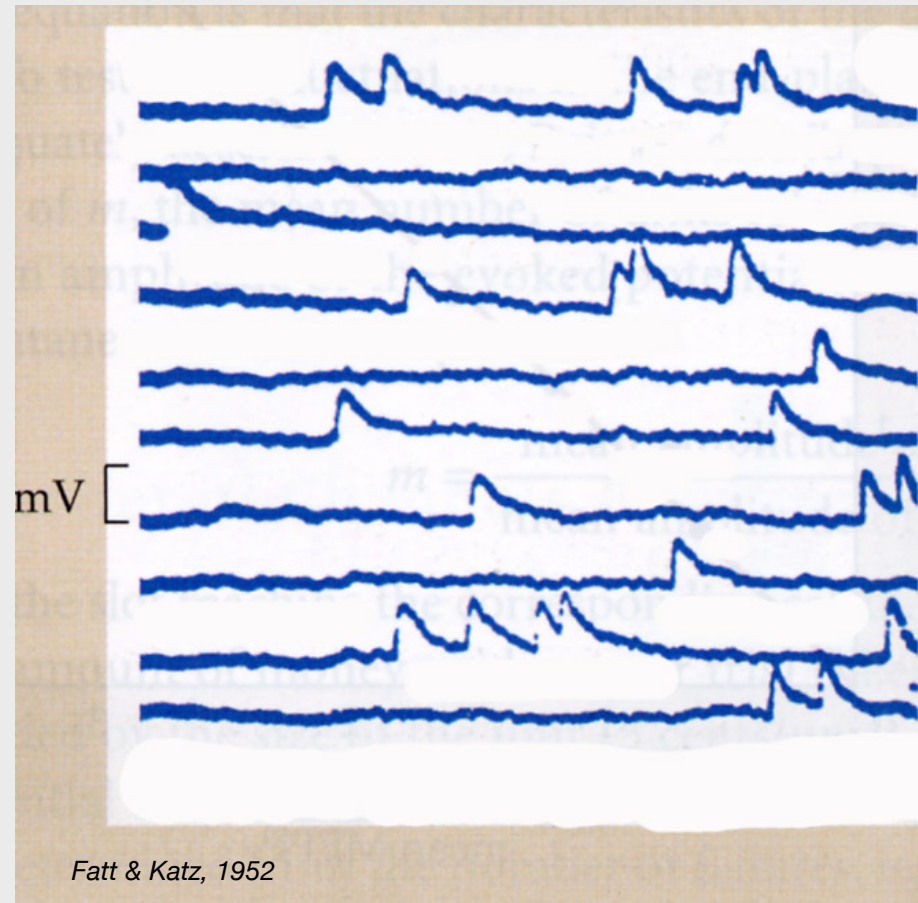


- *neurotransmitter* : Acetylcholine
- *receptor* : nicotinic

quantal release at the neuromuscular junction



+ **TTX** (no action potential)



spontaneous activity : *miniatures*

Do they represent the leakage of one or more acetylcholine molecules?

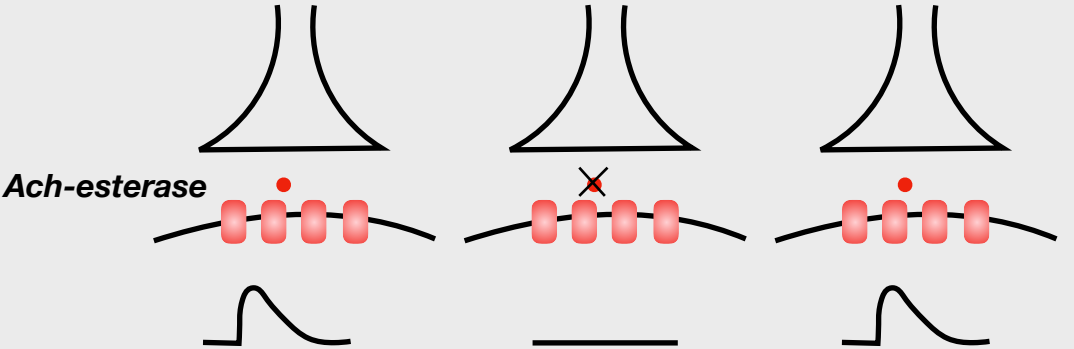
application of prostigmine (acetylcholinesterase blocker)

quantal release at the neuromuscular junction

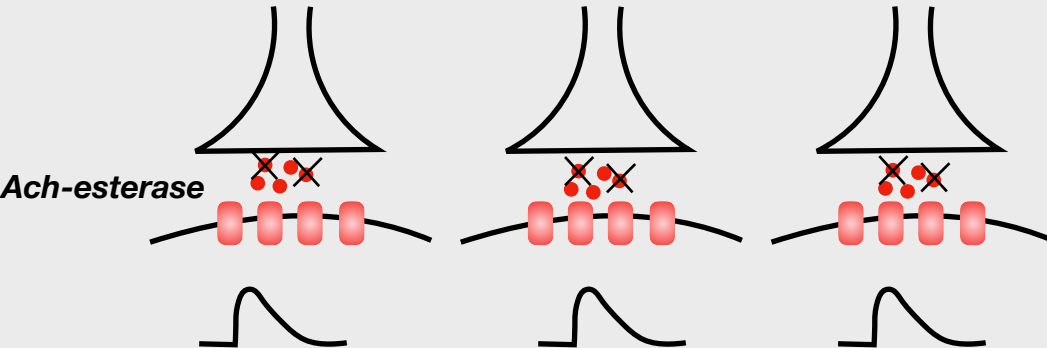
one molecule of Ach

one quantum of Ach

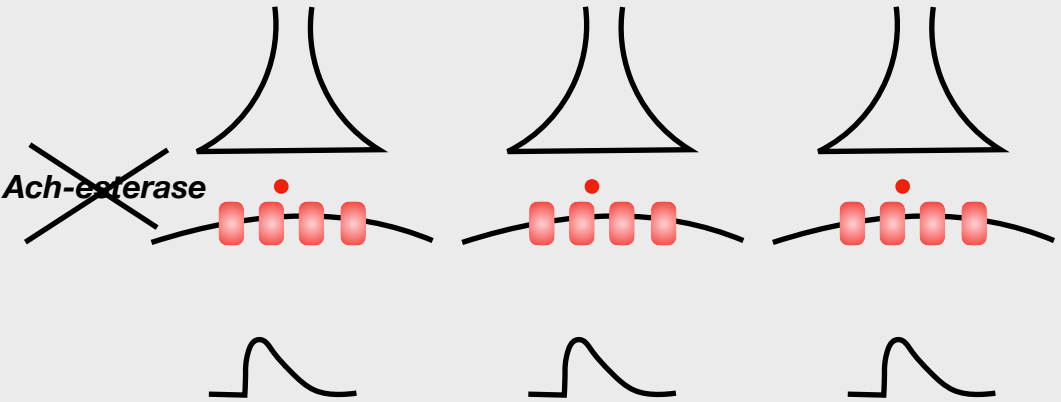
+ TTX



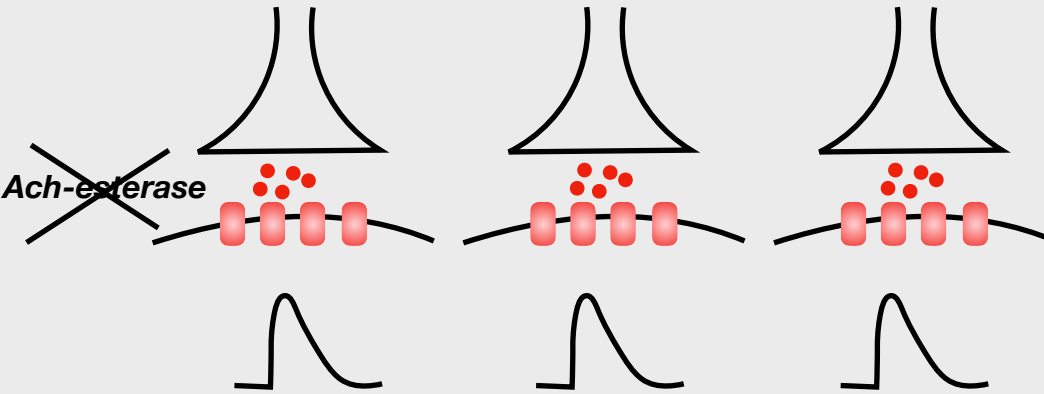
+ TTX



+ TTX + prostigmine



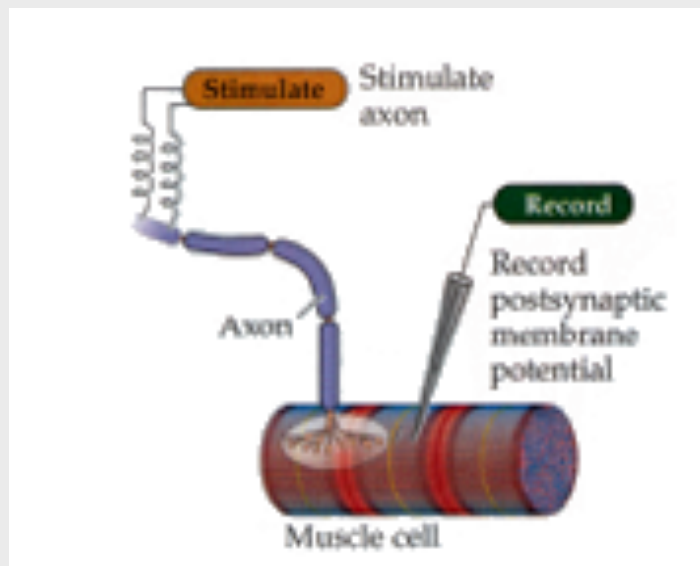
+ TTX + prostigmine



increase in frequency

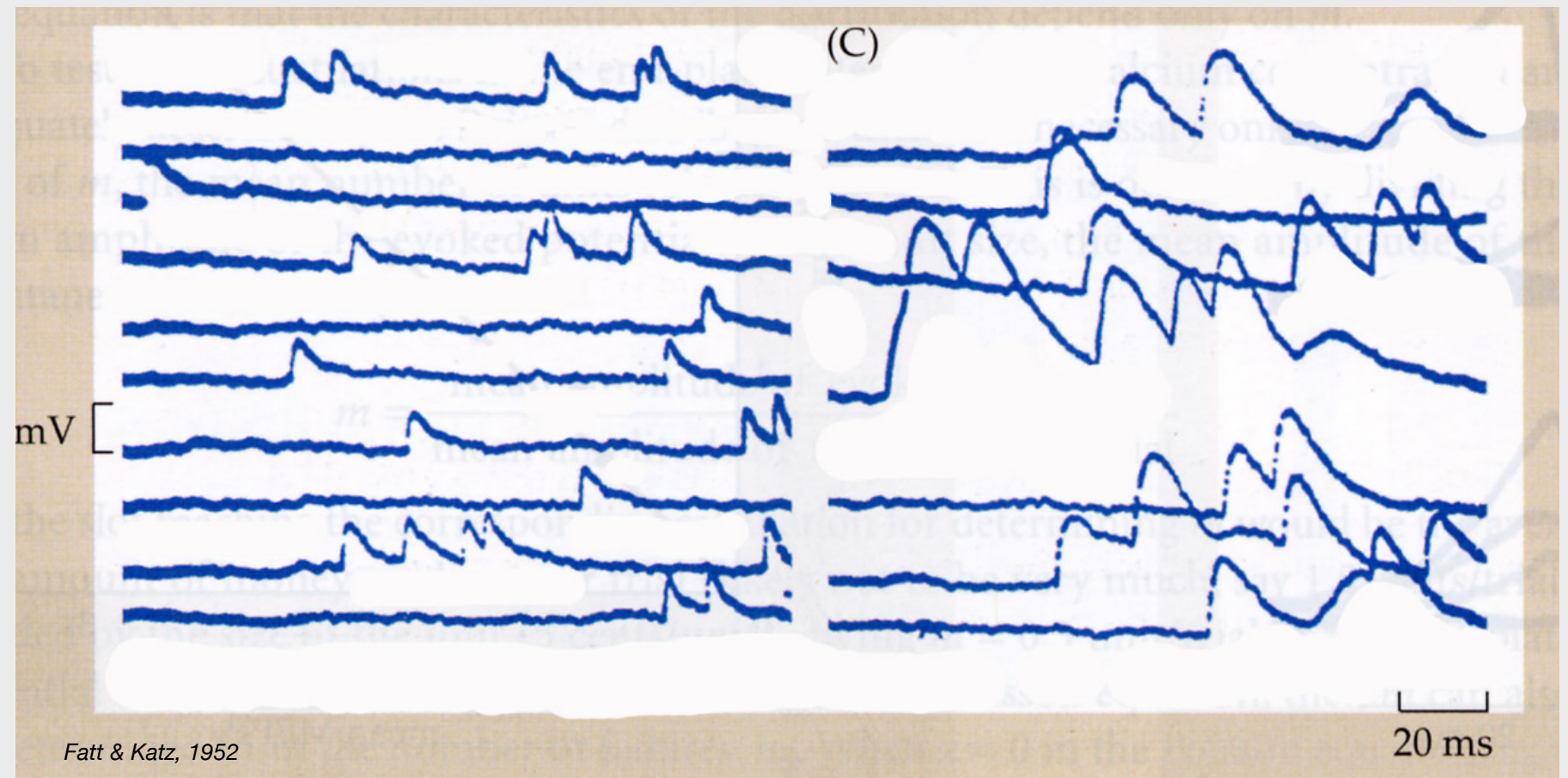
increase in amplitude

quantal release at the neuromuscular junction



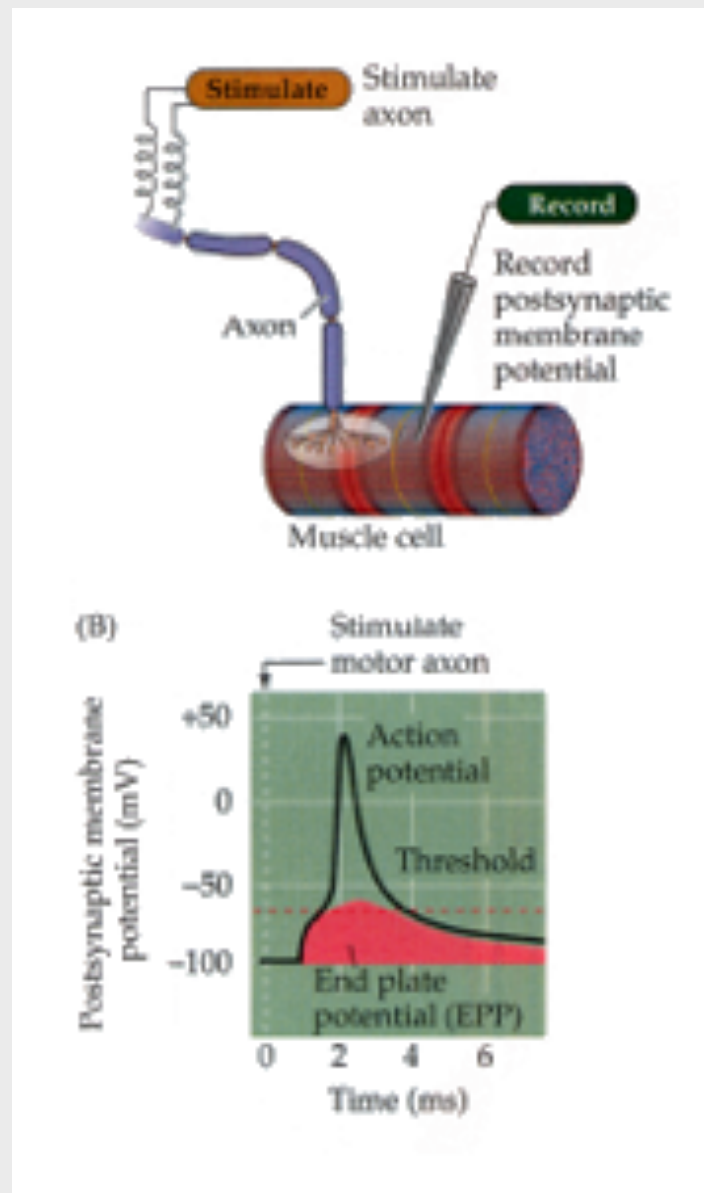
+ **TTX** (no action potential)

+ **TTX** + **prostigmine**



spontaneous activity : *miniatures*

quantal release at the neuromuscular junction



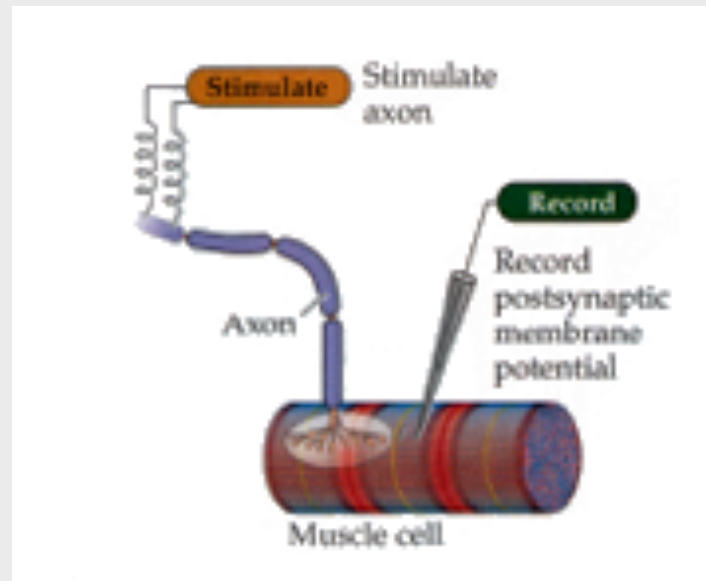
the systematic occurrence of a muscle fiber action potential preclude the measurement of the EPSP amplitude



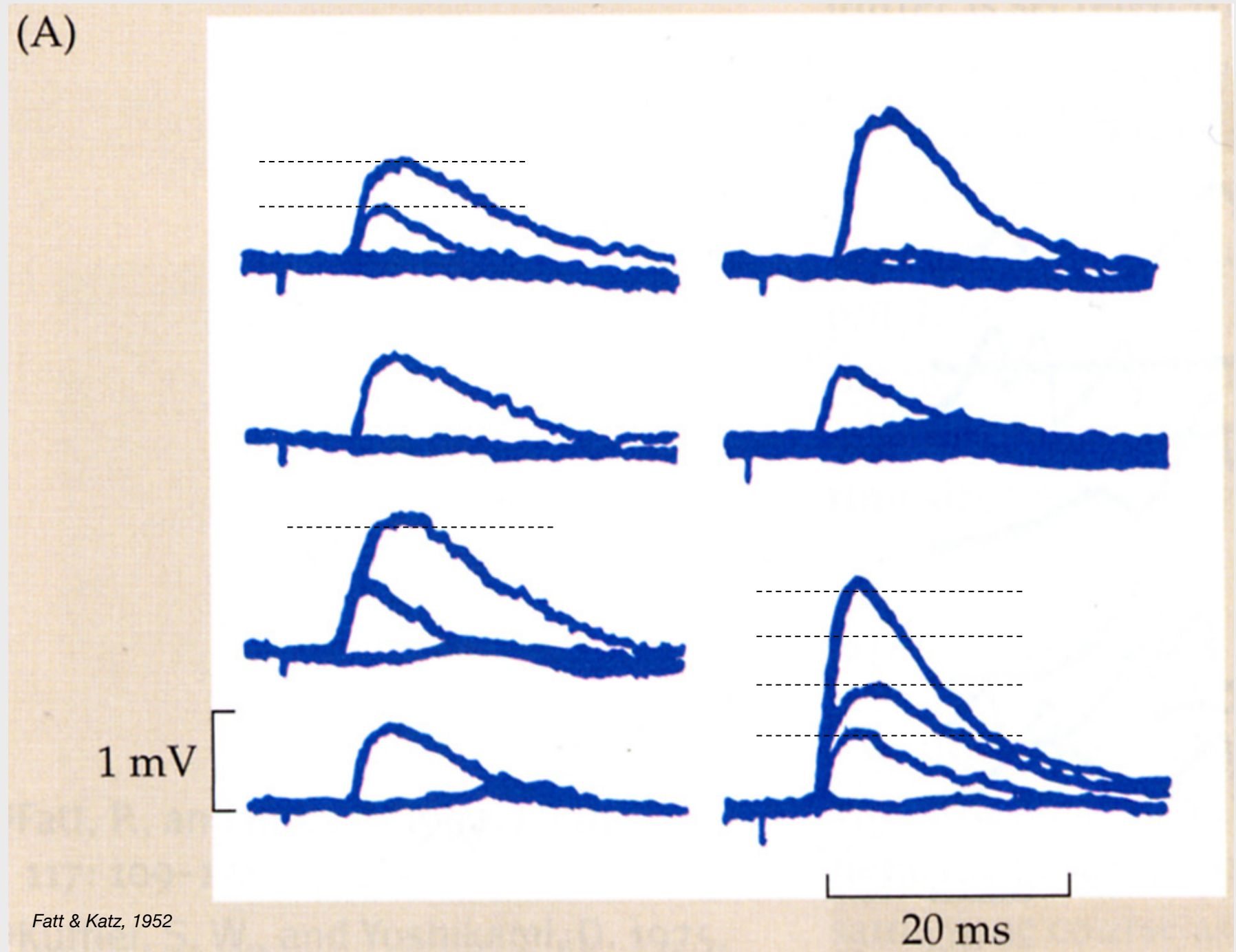
decreasing the extracellular Ca^{2+} concentration decreases synaptic efficiency and suppress action potential generation

quantal release at the neuromuscular junction

variability in EPSP amplitudes

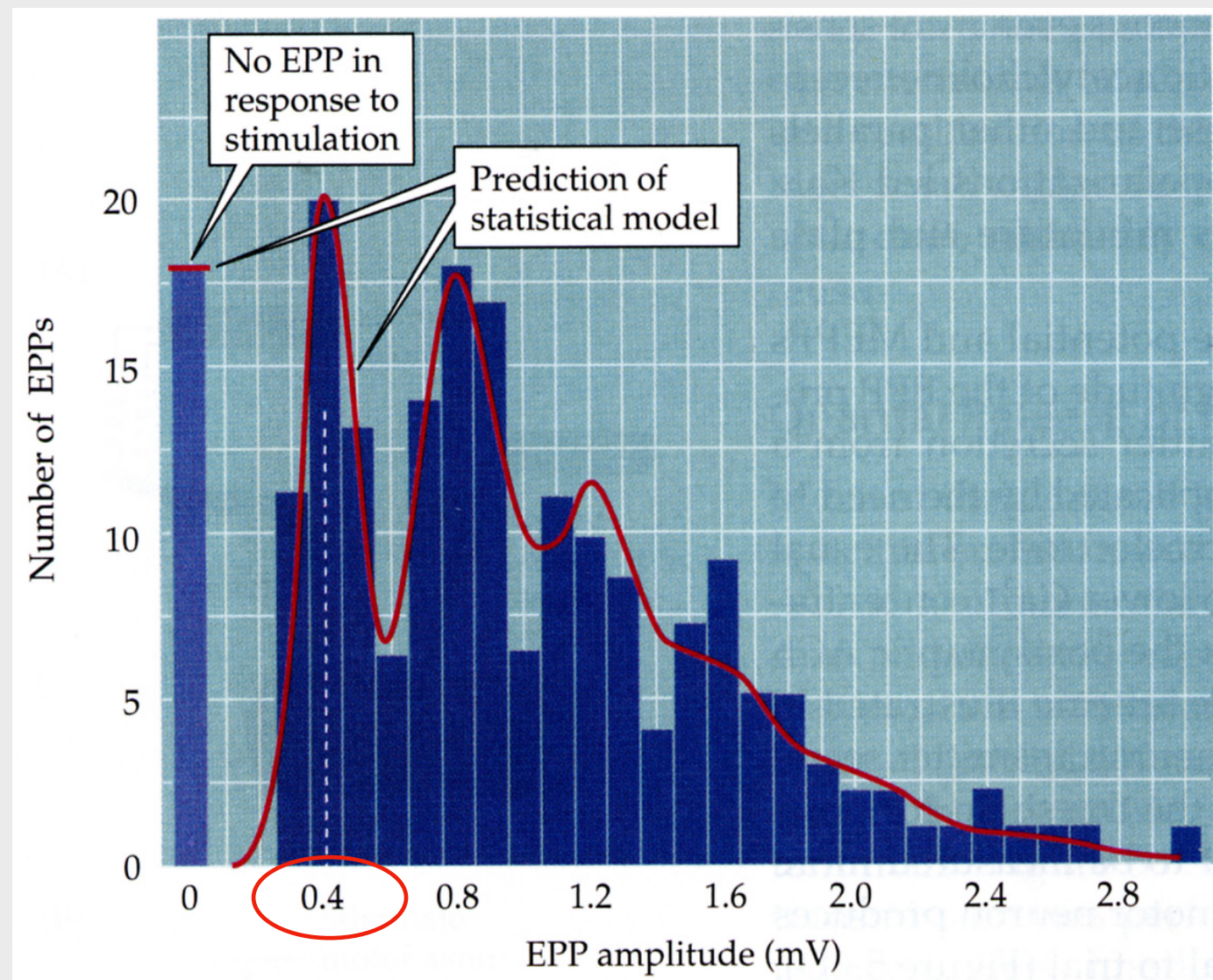


low $[Ca^{2+}]_{ex}$

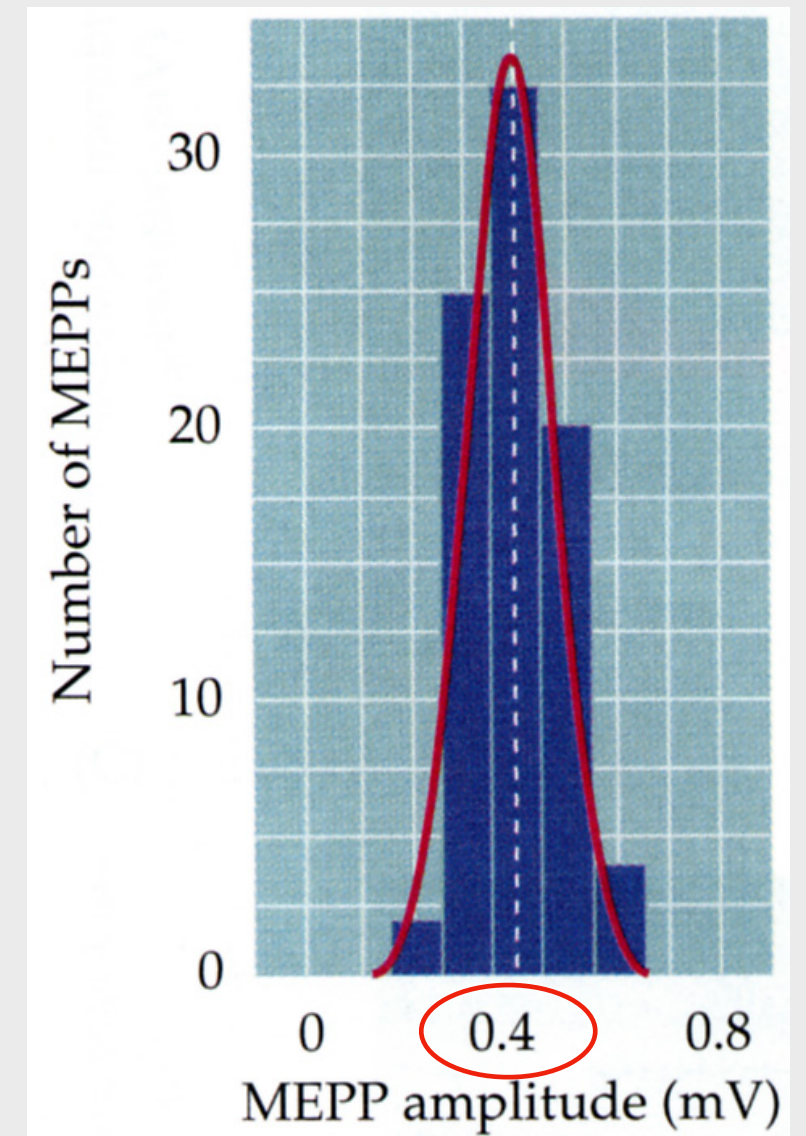


quantal release at the neuromuscular junction

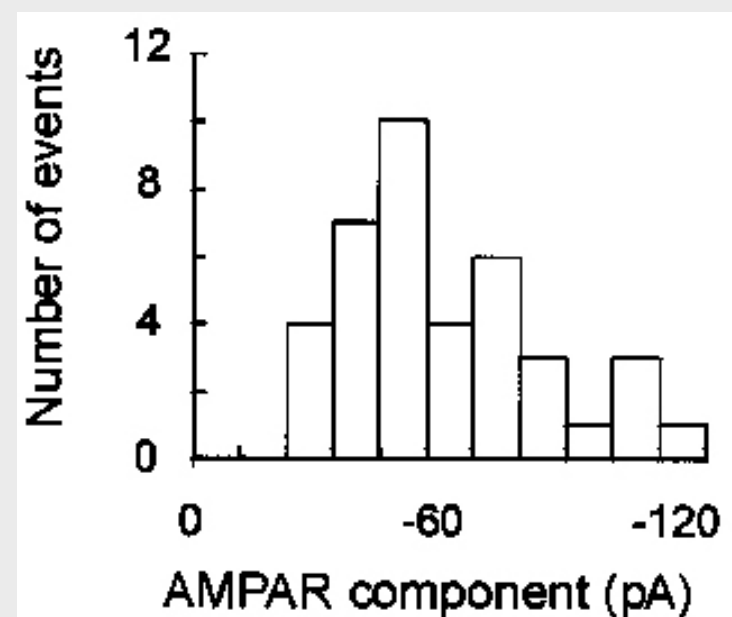
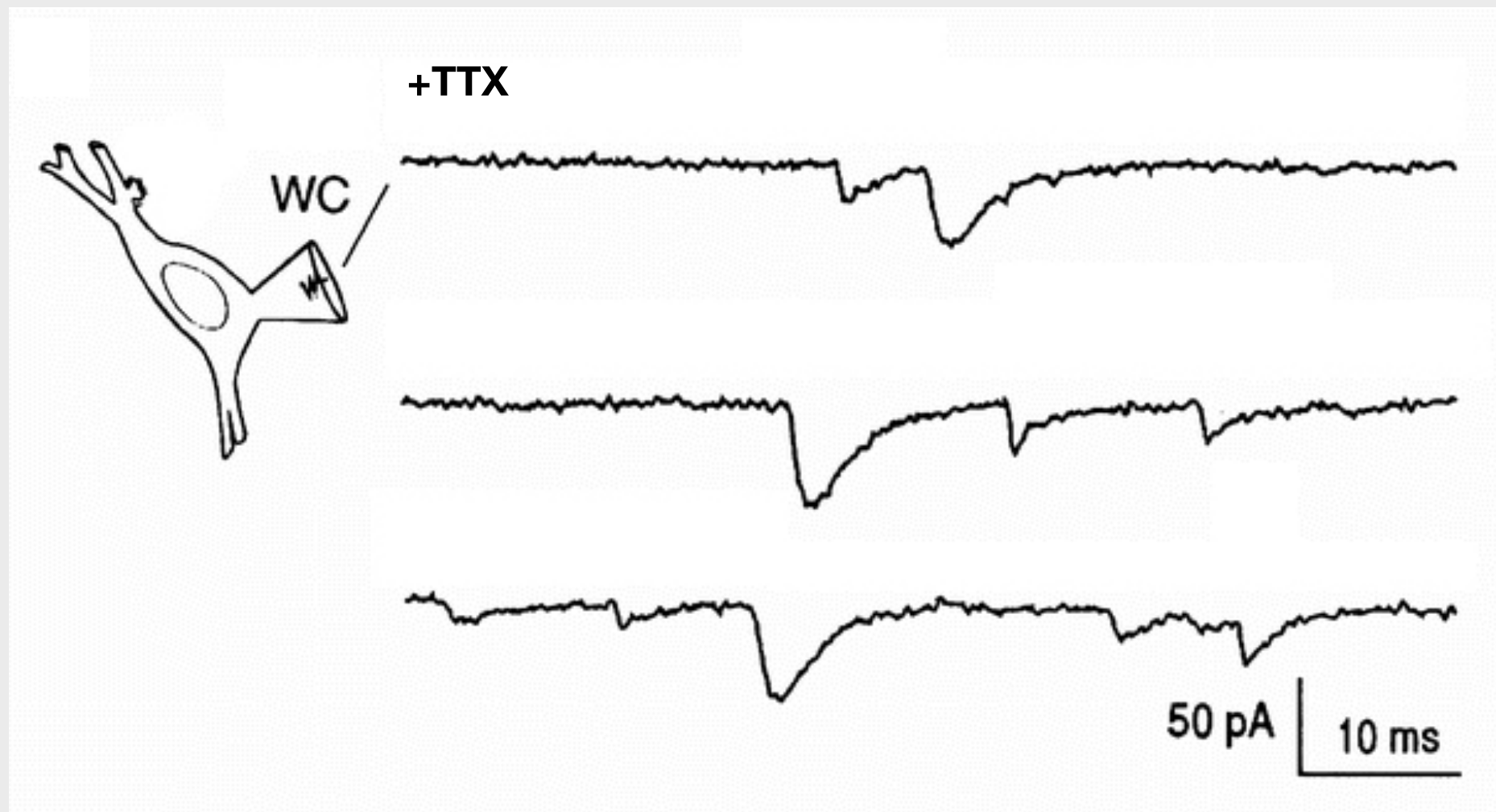
variability in EPSP amplitudes



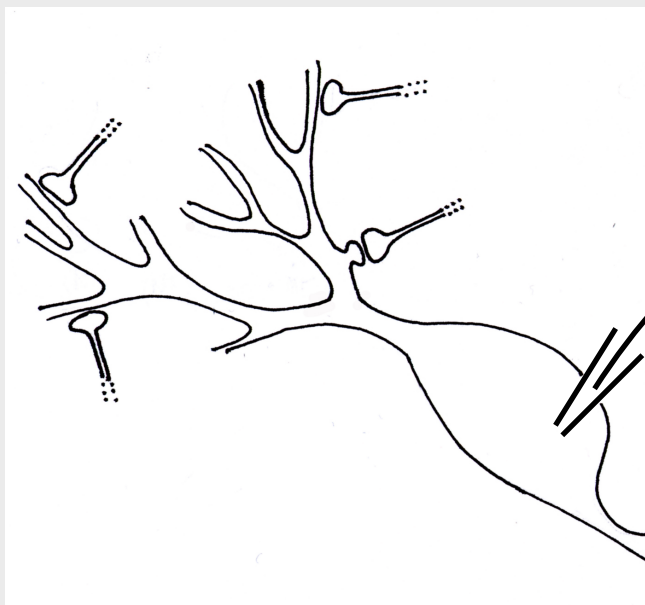
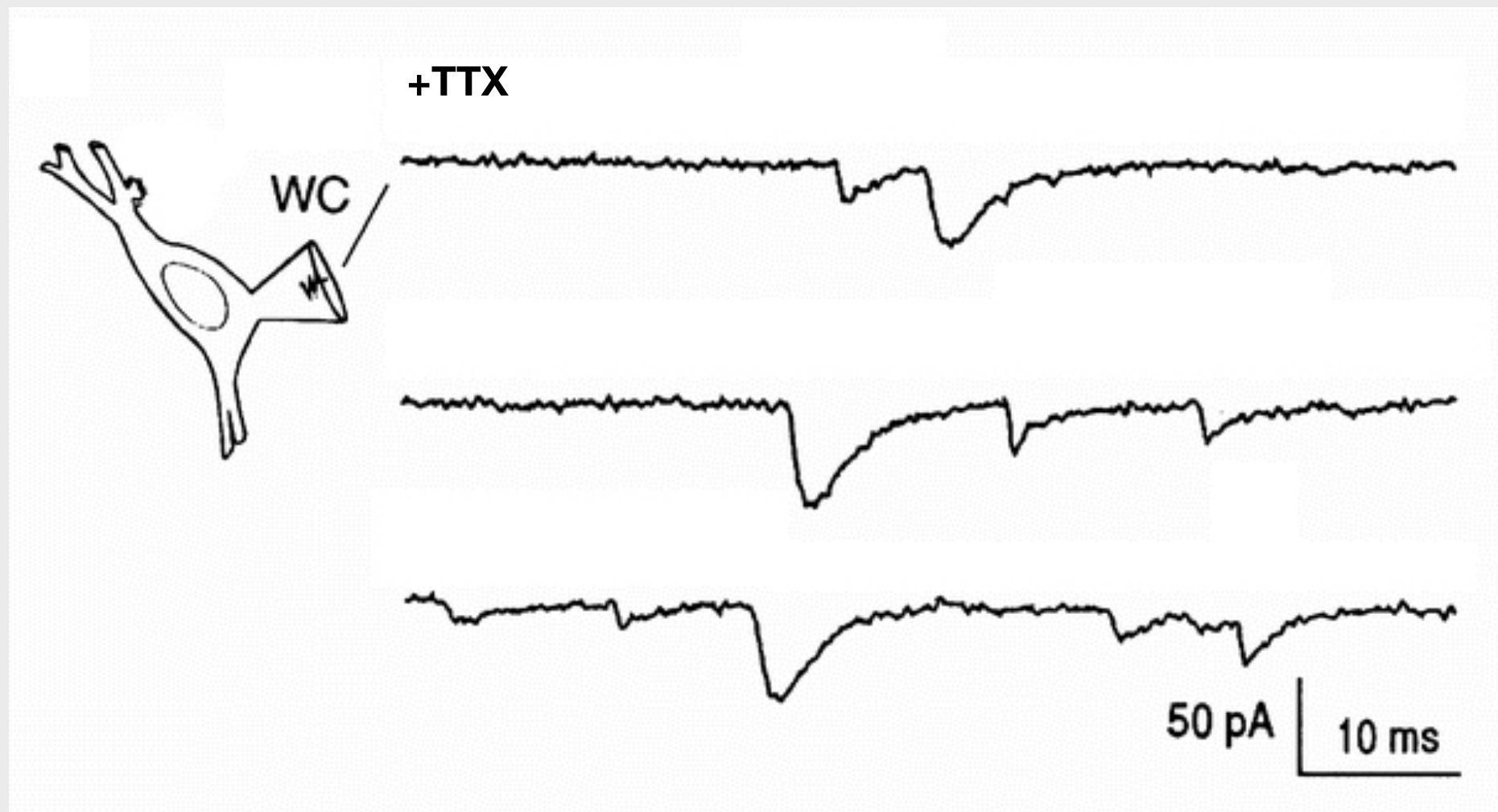
miniature EPSP amplitude



quantal release in the CNS



quantal release in the CNS



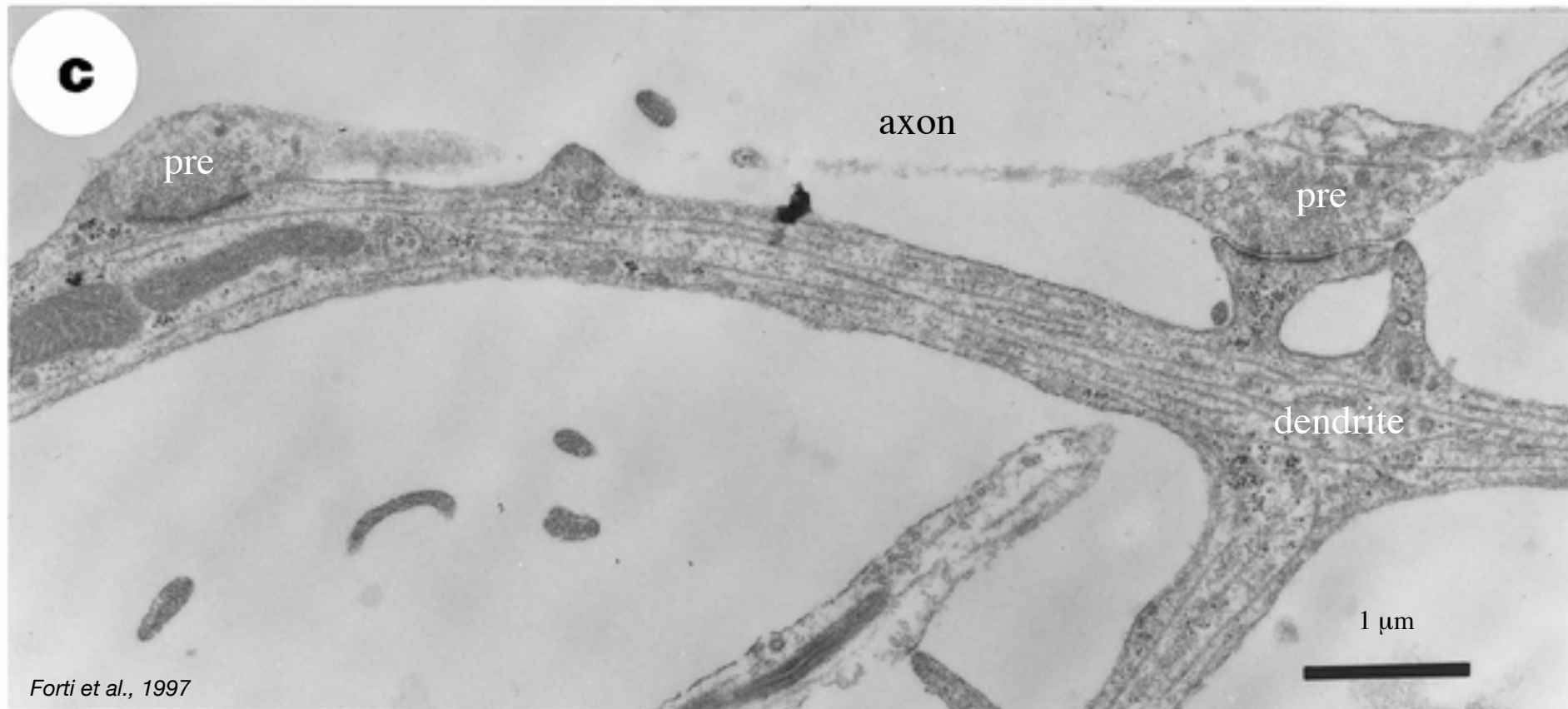
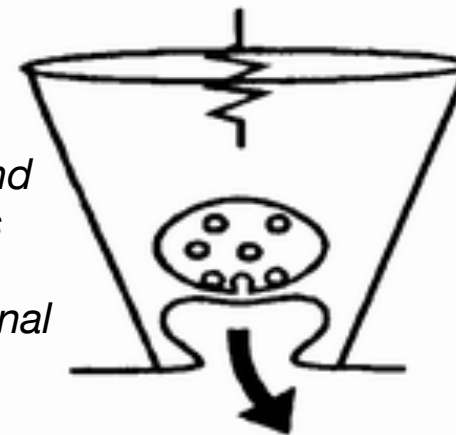
somatic recording : the variability of miniature amplitudes may result from the different location of synapses or the different content of quanta in a given synapse

quantal release in the CNS

culture of hippocampal neurons



*opening of ligand
gated receptors
creates an
extracellular signal*

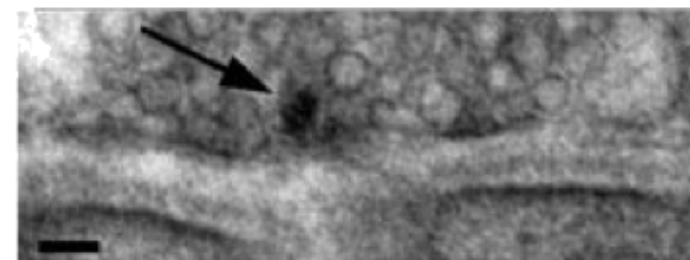
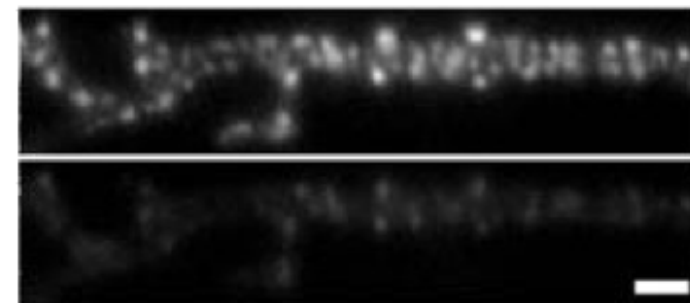
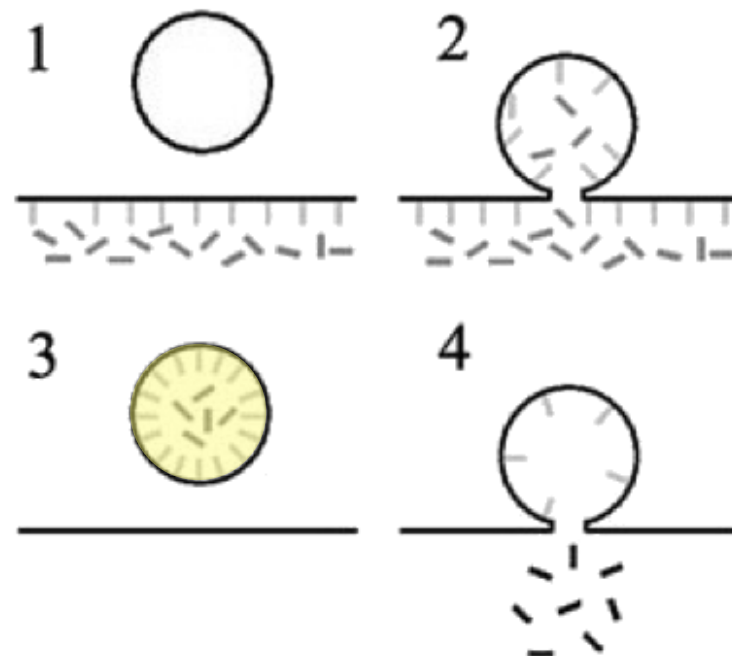
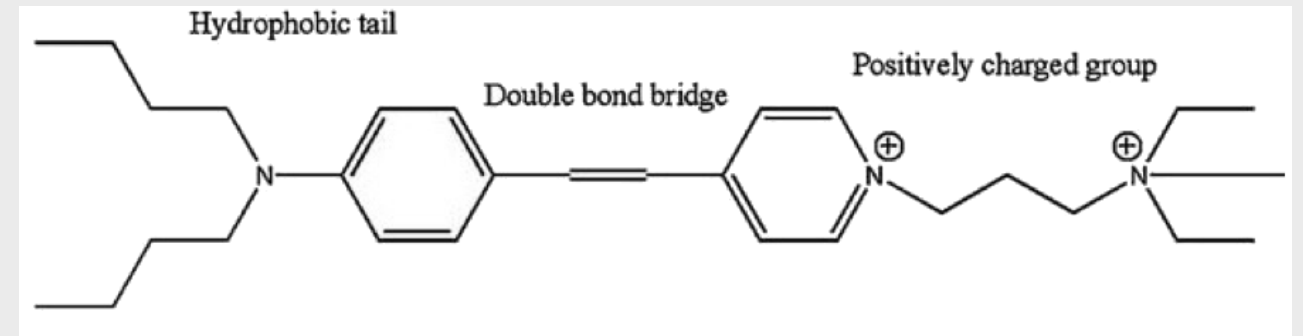


quantal release in the CNS

culture of hippocampal neurons

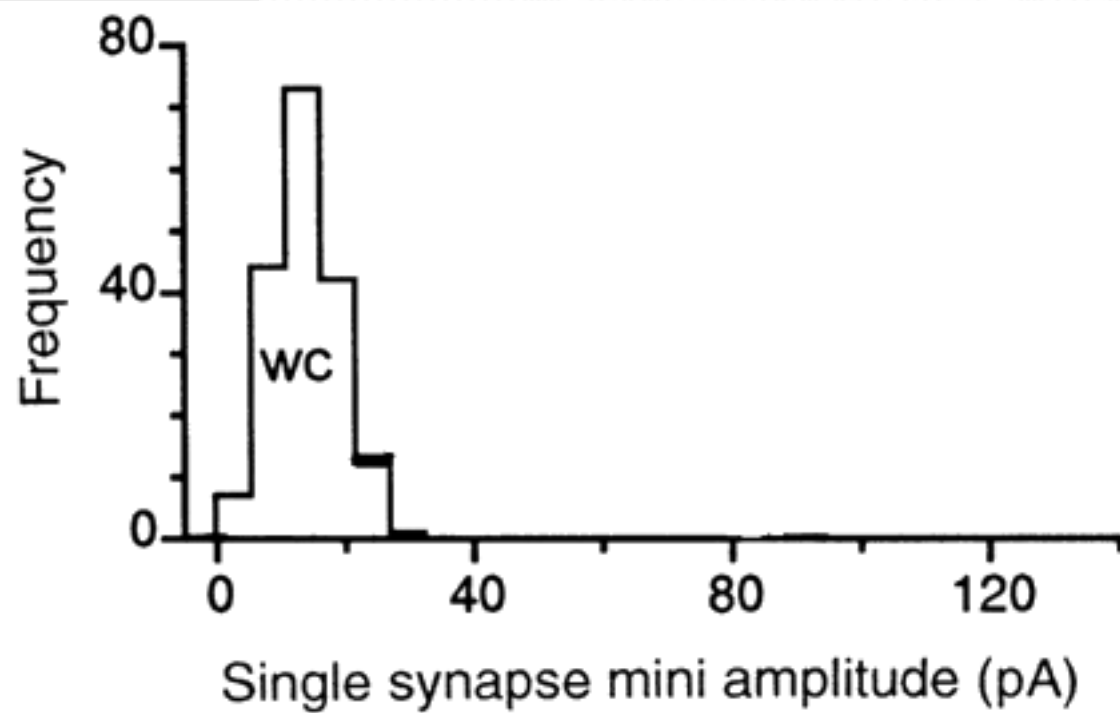
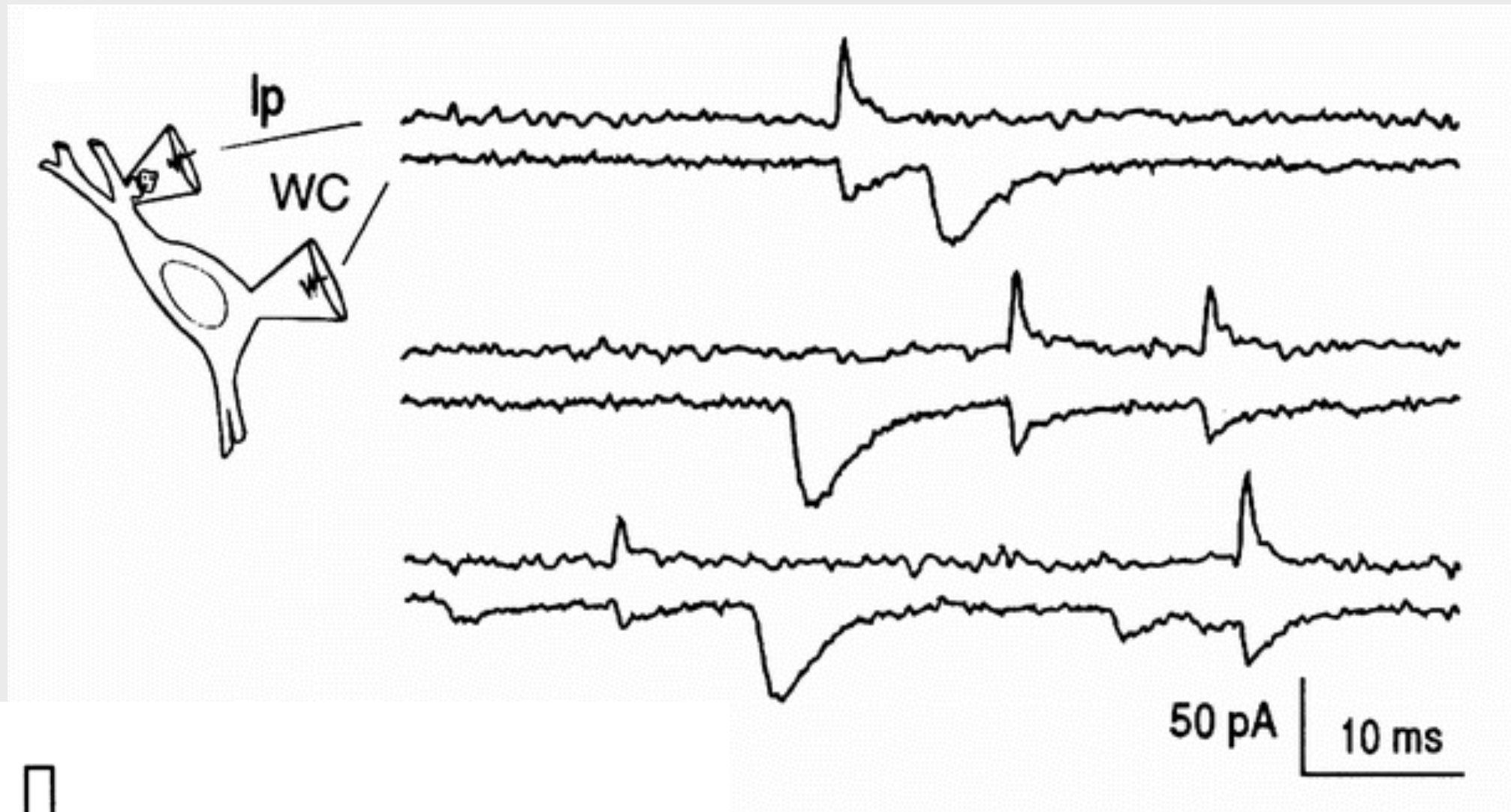


FM1-43 : fluorescent in hydrophobe medium



quantal release in the CNS

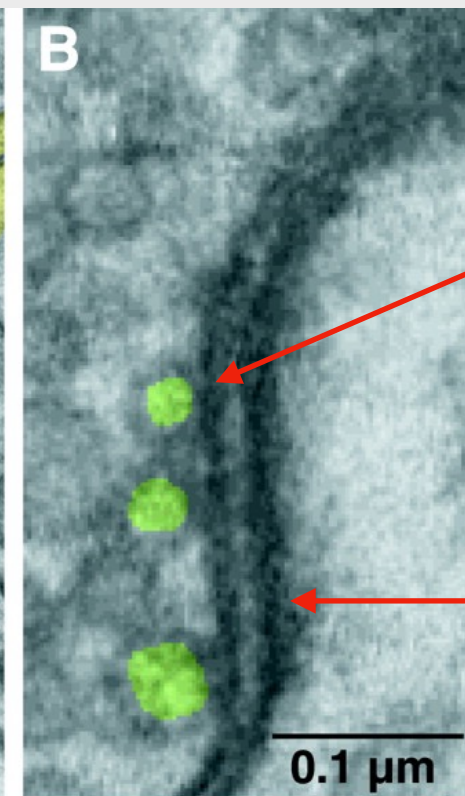
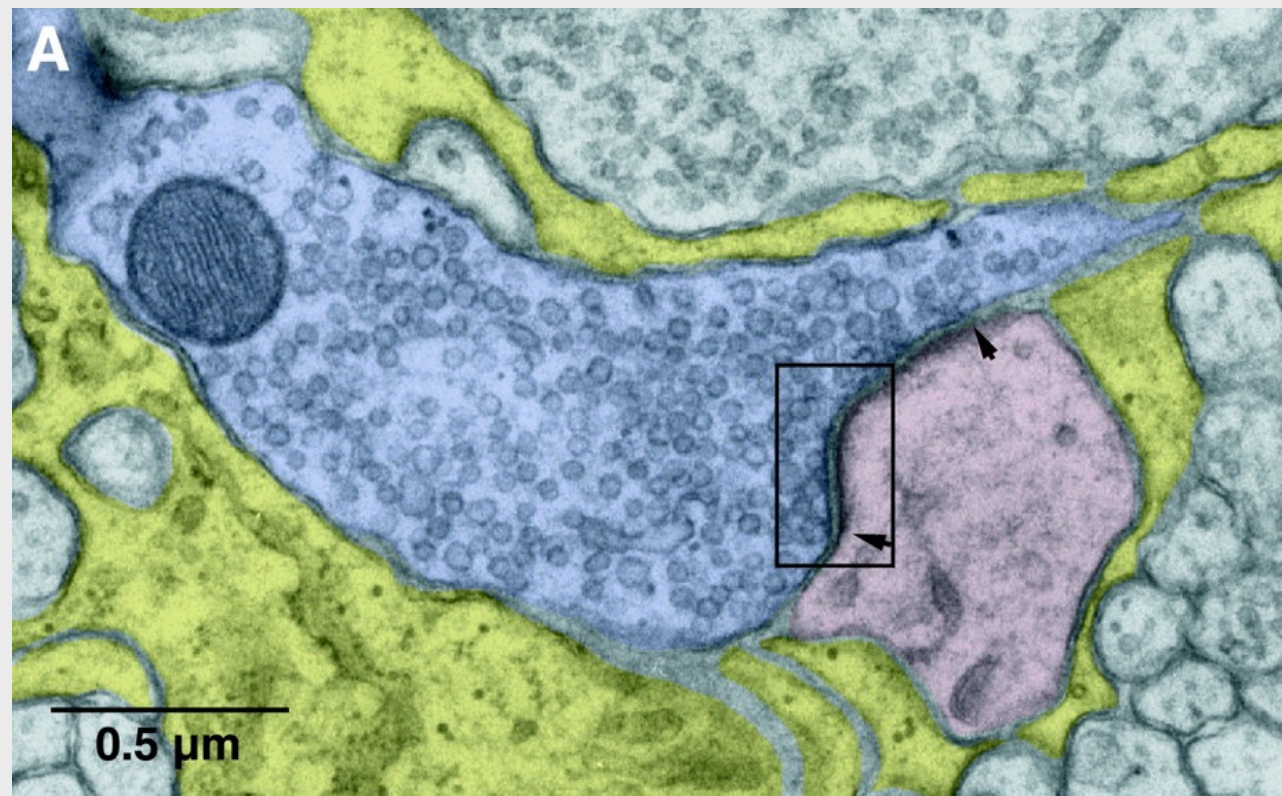
culture of hippocampal neurons



the synaptic vesicle cycle

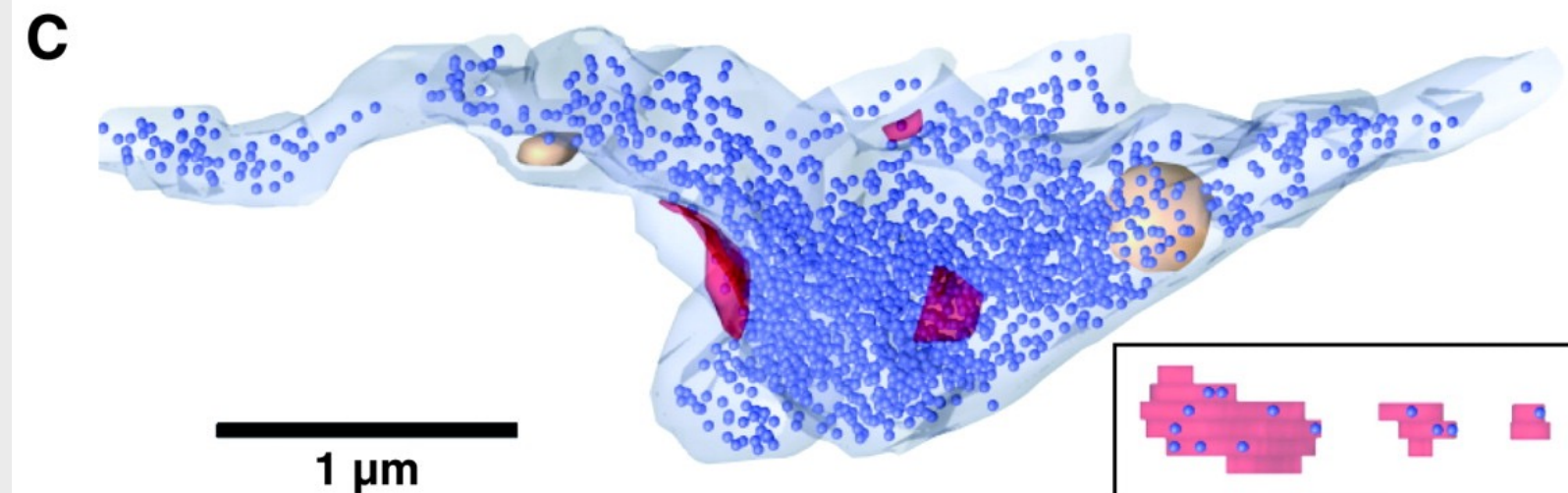
reconstruction of the climbing fiber synapse from EM images

- blue : climbing fiber
- pink : Purkinje cell
- yellow : glial cell



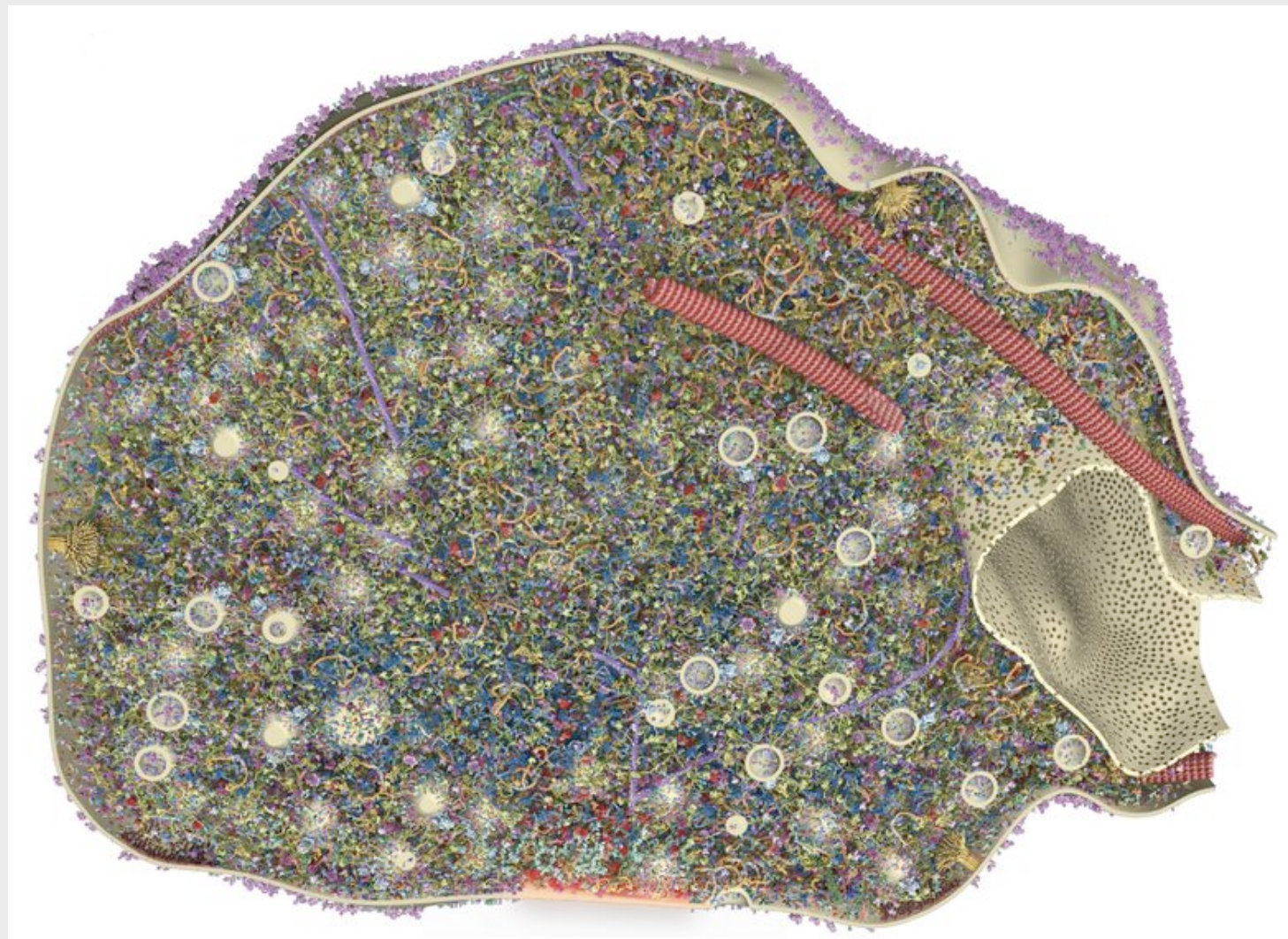
vesicle docked
at the active zone

post-synaptic density

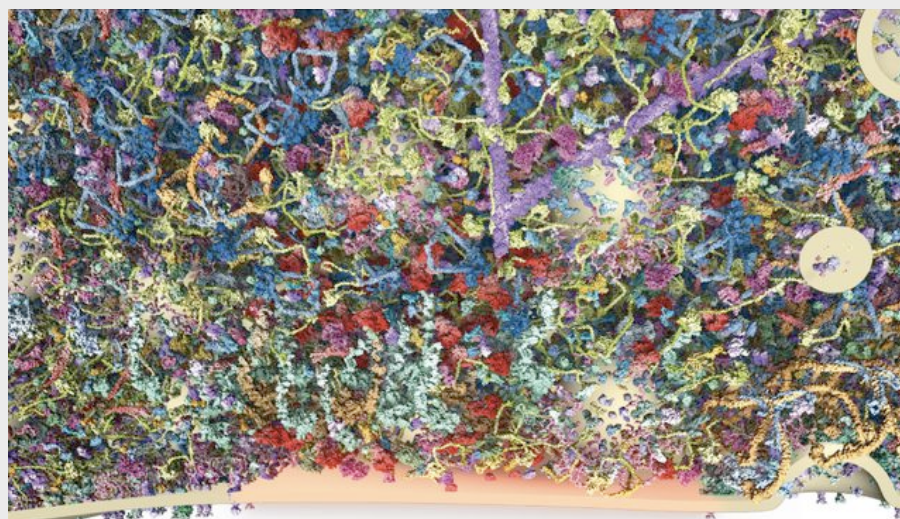
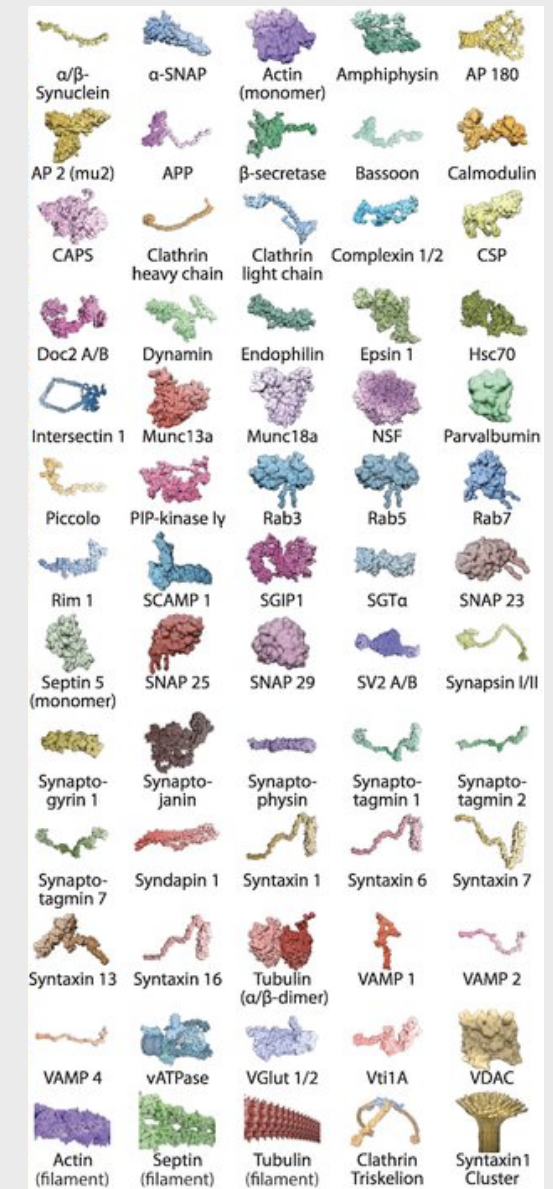


- blue : vesicles
- brown : mitochondria
- pink : active zones

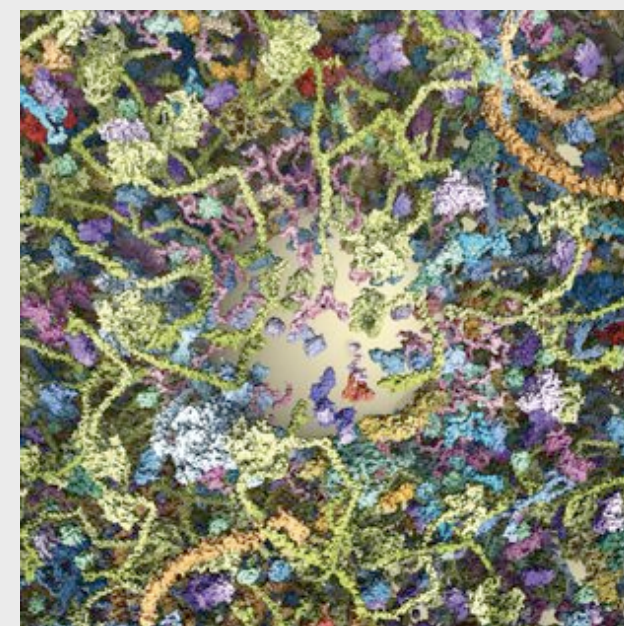
A 3D model of synaptic architecture.



the different proteins



active zone area

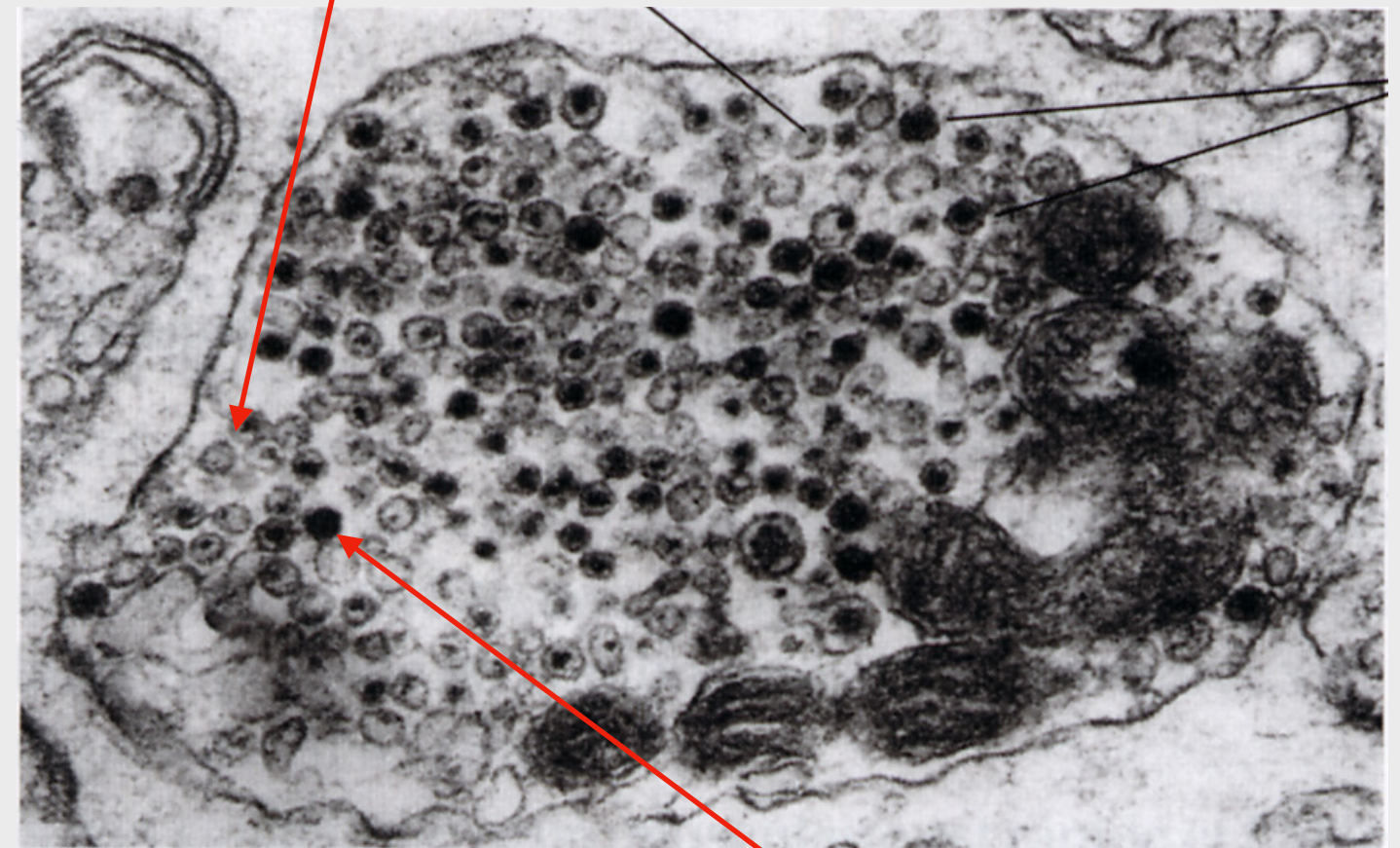
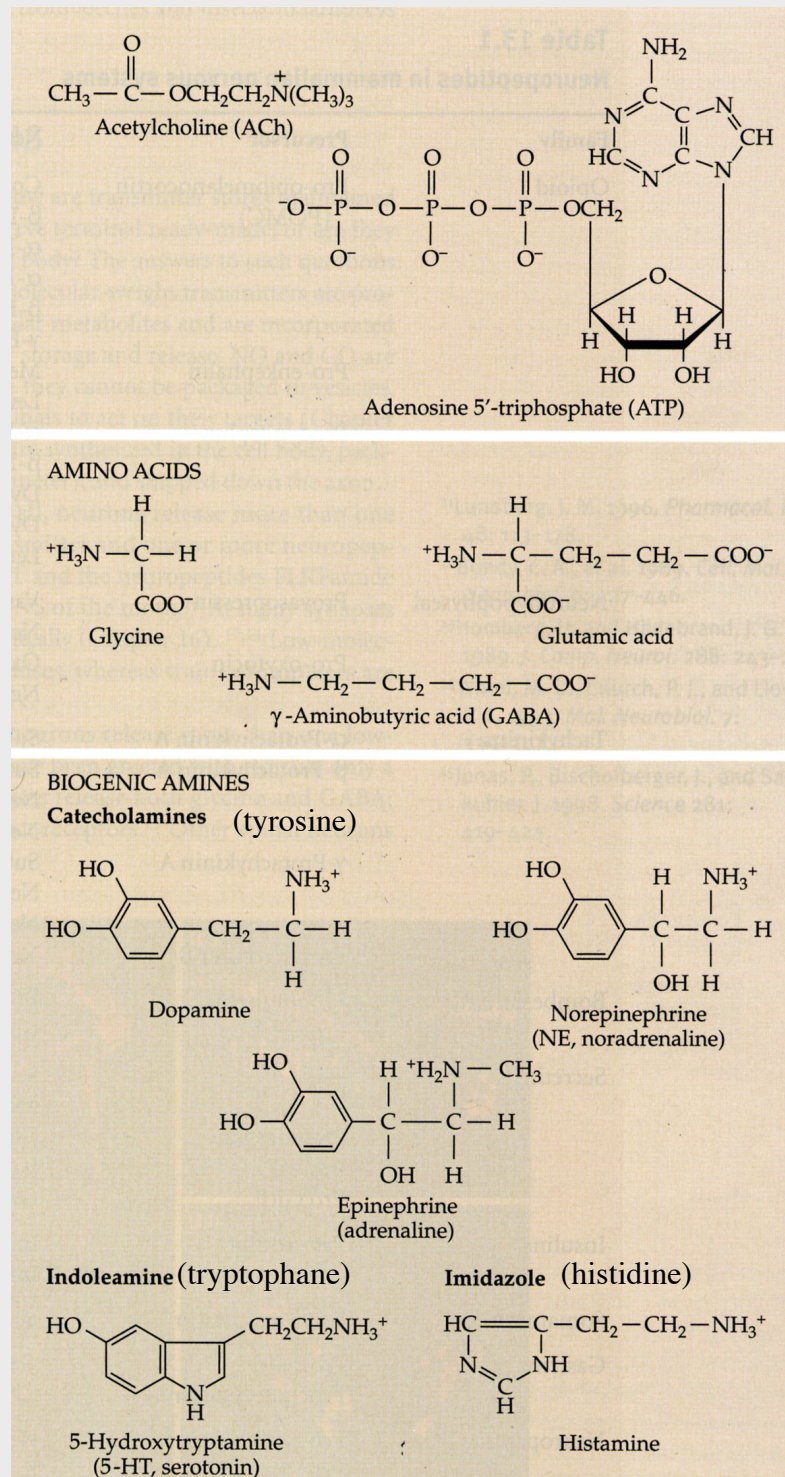


one vesicle within the vesicle cluster

vesicle diameter = 42nm

the synaptic vesicle cycle

small molecular weight

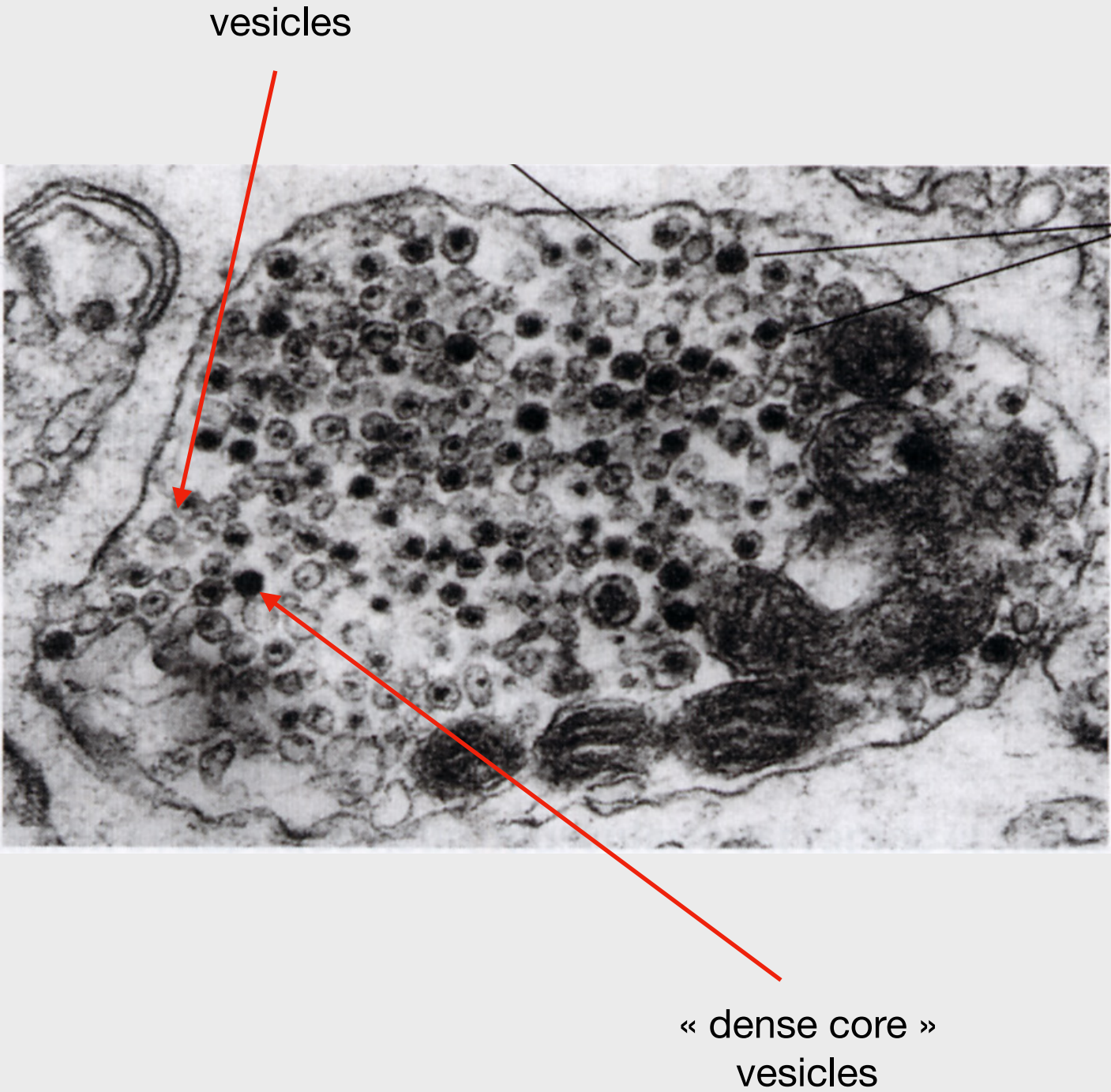


vesicles

« dense core »
vesicles

the synaptic vesicle cycle

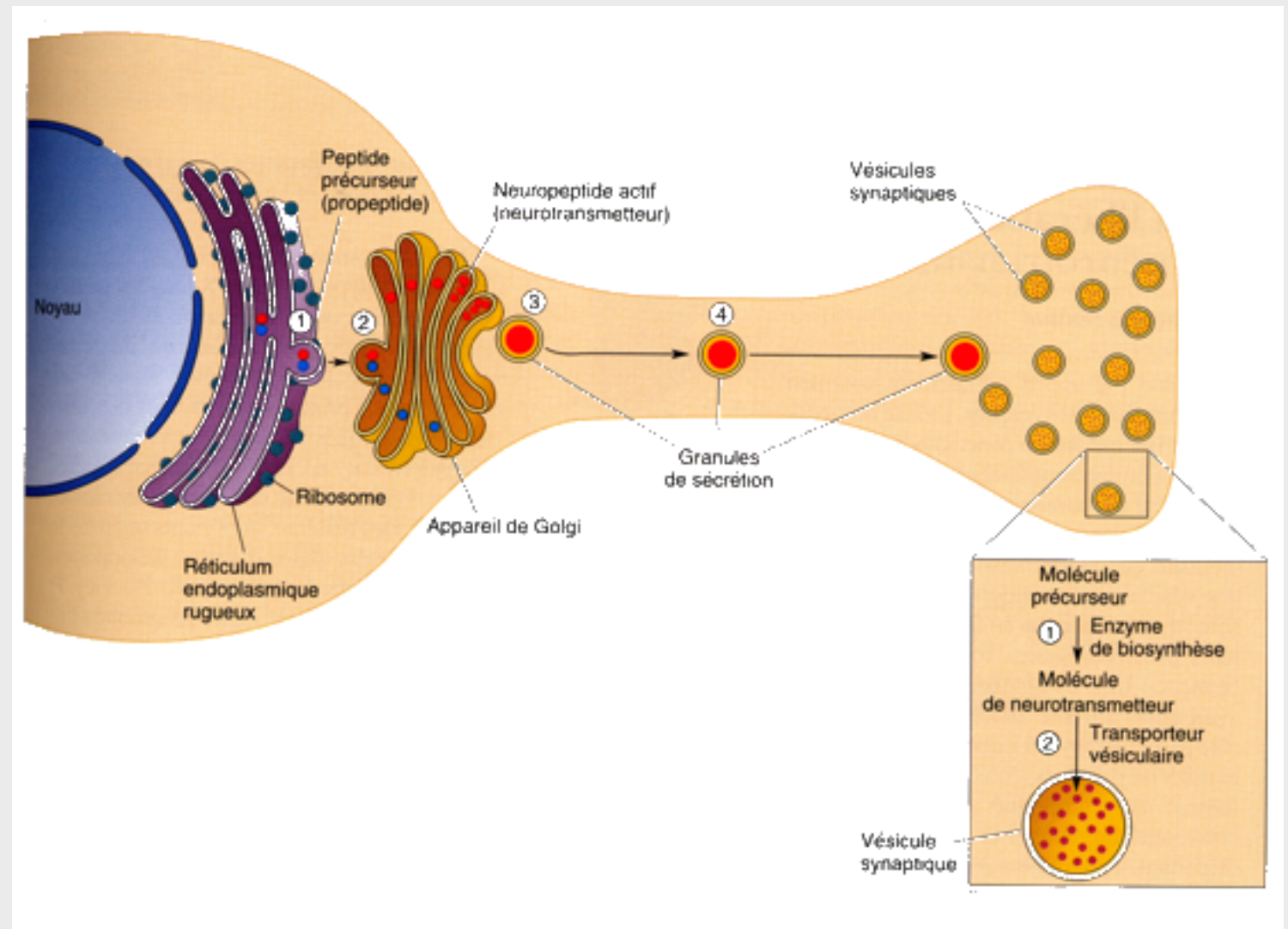
Neuropeptides in mammalian nervous systems		
Family	Precursor	Neuropeptide
Opioid	Pro-opiomelanocortin (POMC)	Corticotropin (ACTH)
		β-Lipotropin
		α-MSH
		α-Endorphin
		β-Endorphin
		γ-Endorphin
	Pro-enkephalin	Met-enkephalin
	Prodynorphin	Leu-enkephalin
		α-Neoendorphin
		β-Neoendorphin
		Dynorphin A
		Dynorphin B (rimorphin)
		Leumorphin
Neurohypophyseal	Provasopressin	Vasopressin
	Pro-oxytocin	Neurophysin II
		Oxytocin
Tachykinins	α-Protachykinin A	Substance P
	β-Protachykinin A	Substance P
		Neurokinin A
		Neuropeptide K
	γ-Protachykinin A	Substance P
		Neurokinin A
		Neuropeptide γ
		Neurokinin B
	Protachykinin B	
Bombesin/GRP	Probombesin	Bombesin
	Pro GRP	Gastrin releasing peptide (GRP)
Secretins	—	Secretin
	—	Motilin
	Proglucagon	Glucagon
	Pro VIP	Vasoactive intestinal peptide (VIP)
	Pro GRF	Growth hormone–releasing factor (GRF)
Insulins	Pro-insulin	Insulin
	—	Insulin-like growth factors
Somatostatins	Prosomatostatin	Somatostatin
Gastrins	Progastrin	Gastrin
	Procholecystokinin	Cholecystokinin (CCK)
Neuropeptide Y	Pro NPY	Neuropeptide Y (NPY)
	Pro PP	Pancreatic polypeptide (PP)
	Pro PYY	Peptide YY (PYY)
Other	Pro CRF	Corticotropin-releasing factor (CRF)
	Procalcitonin	Calcitonin
	Pro CGRP	Calcitonin gene-related peptide (CGRP)
	Proangiotensin	Angiotensin
	Probradykinin	Bradykinin
	Pro TRH	Thyrotropin-releasing hormone (TRH)
	—	Neurotensin
	—	Galanin
		Luteinizing hormone–releasing hormone (LHRH)



the synaptic vesicle cycle

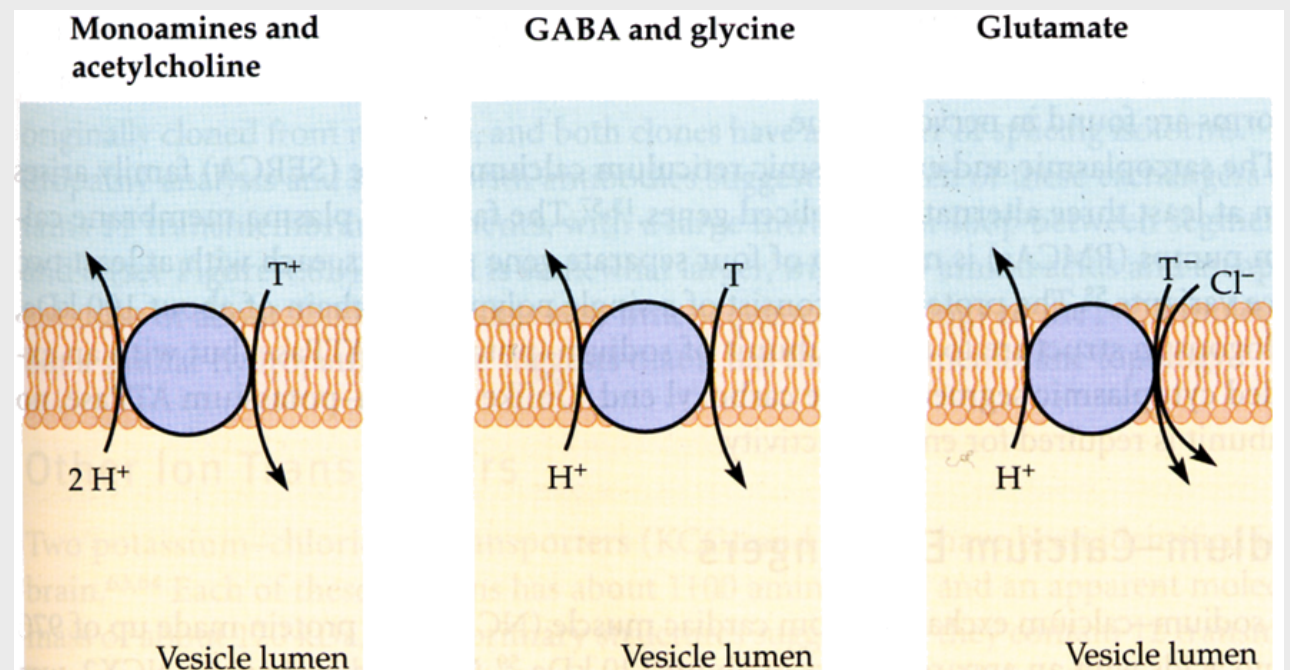
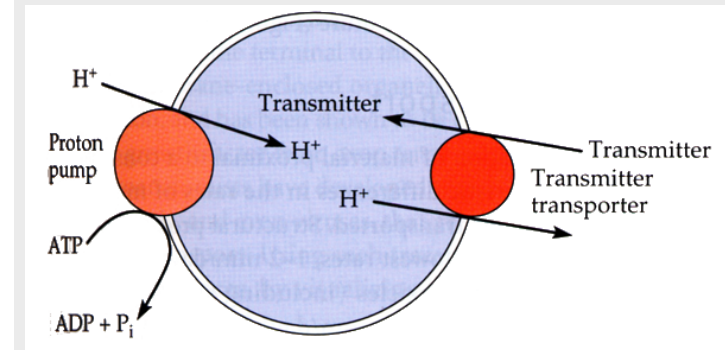
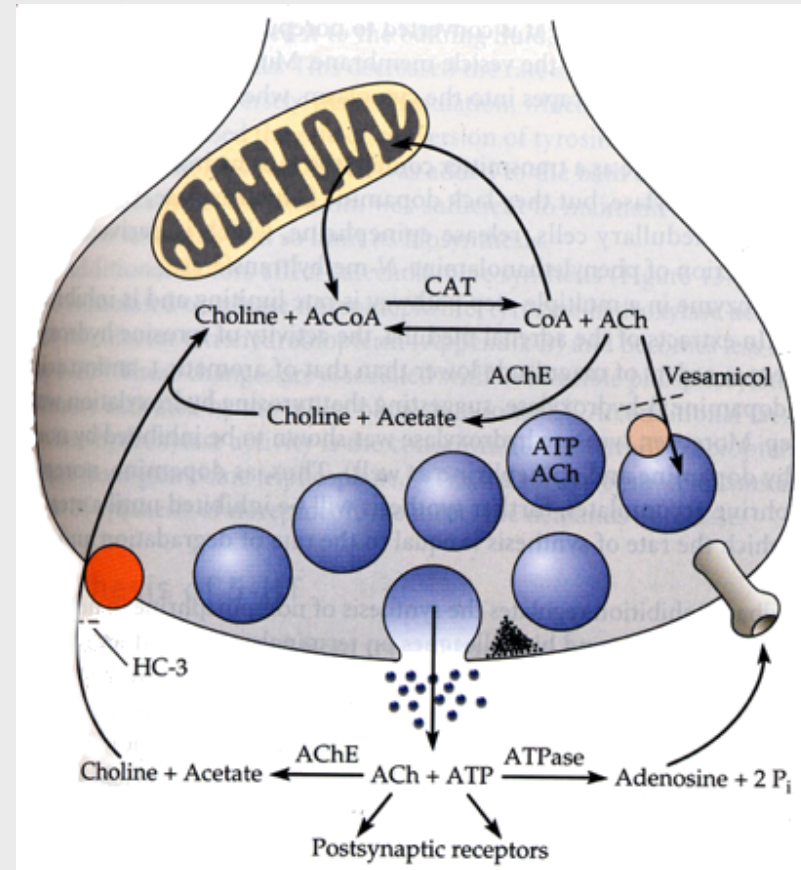
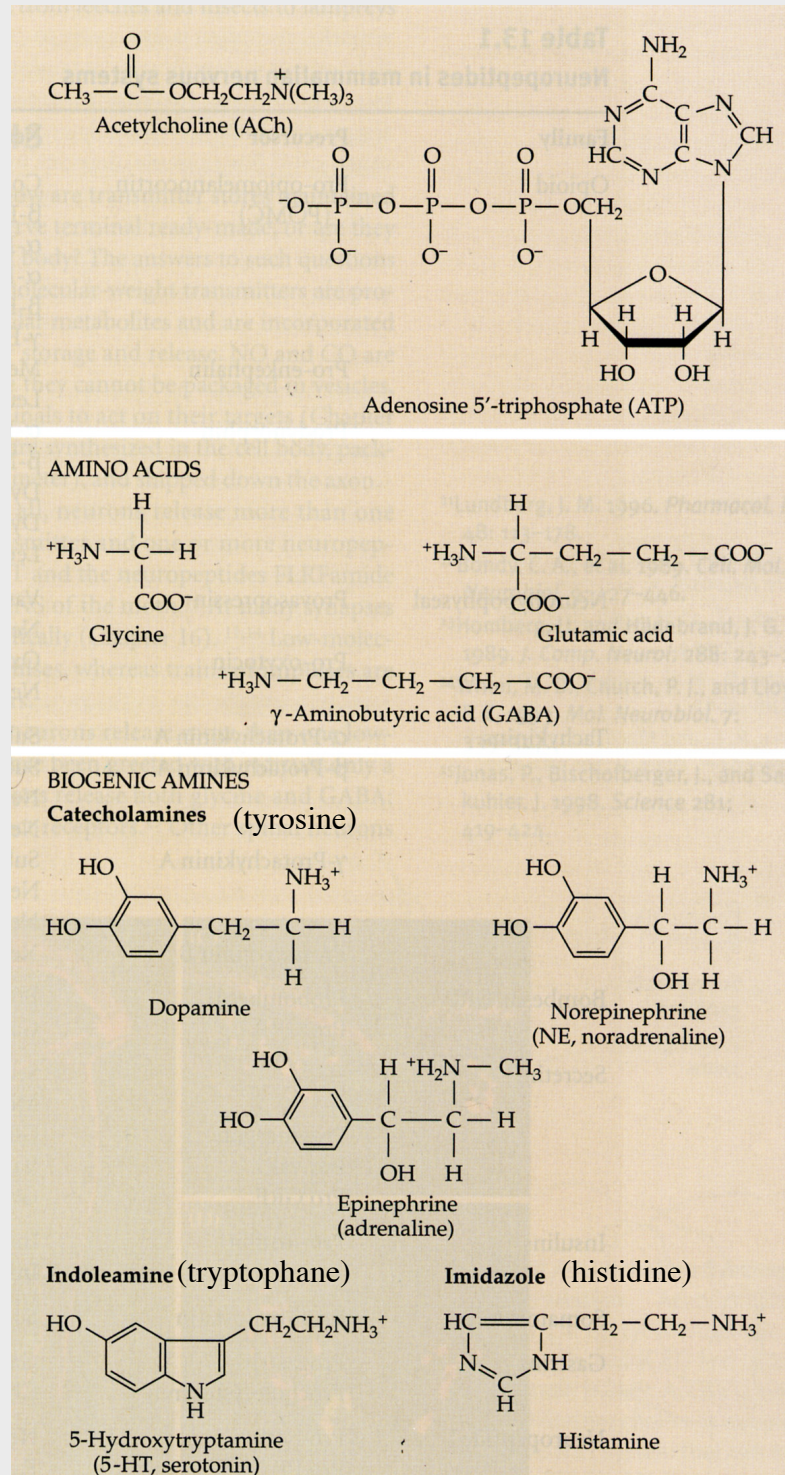
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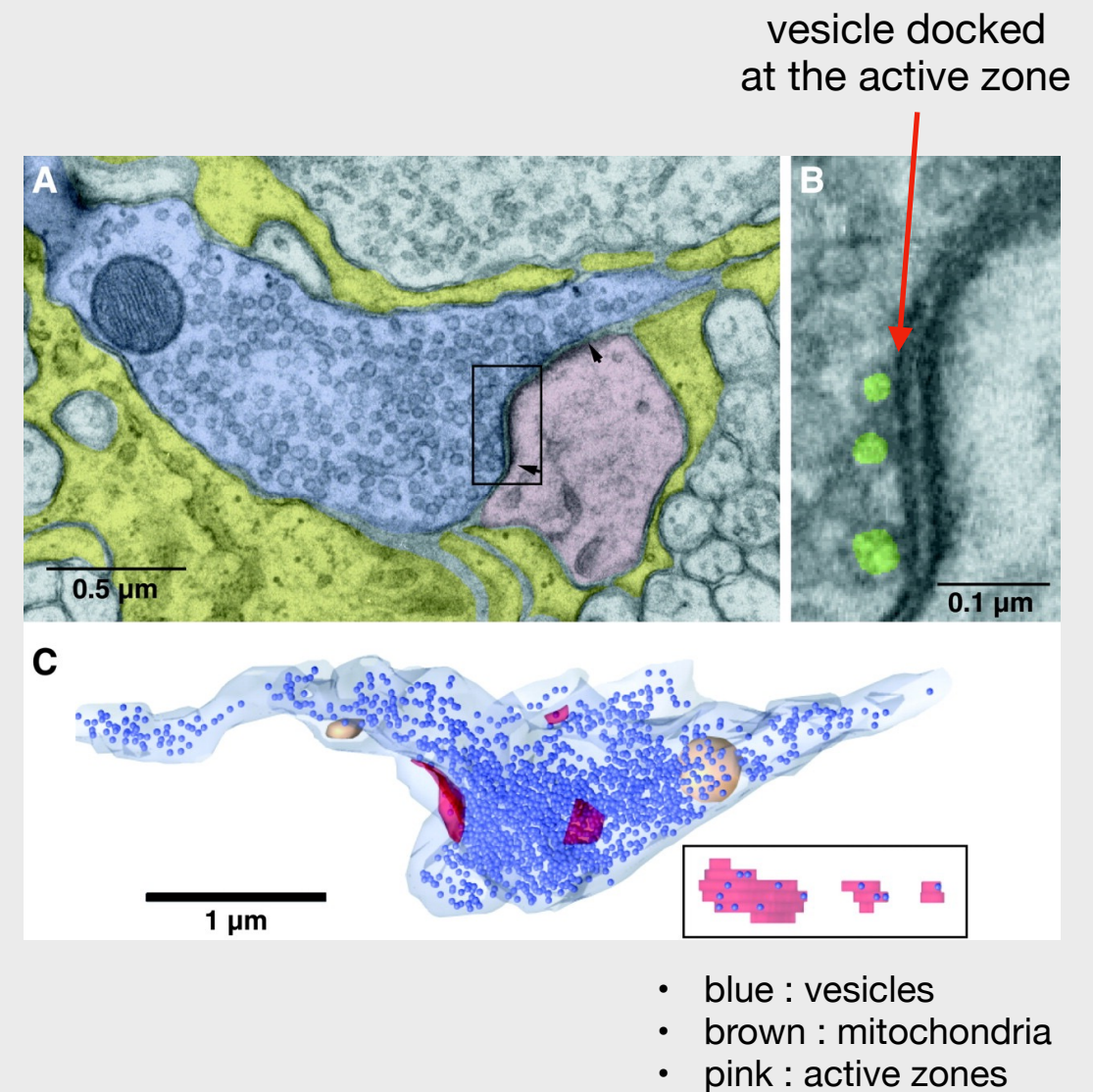
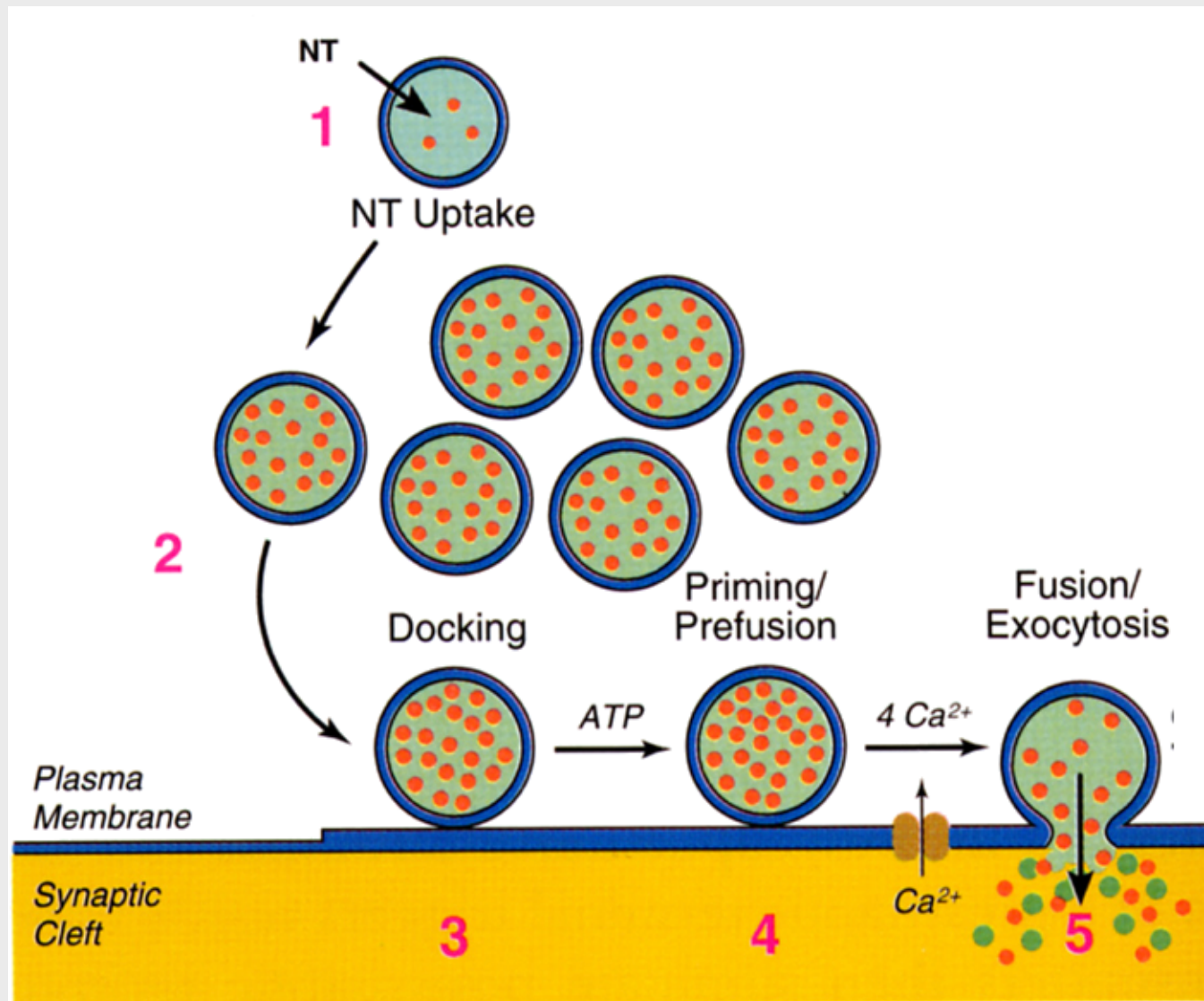


the synaptic vesicle cycle

small molecular weight

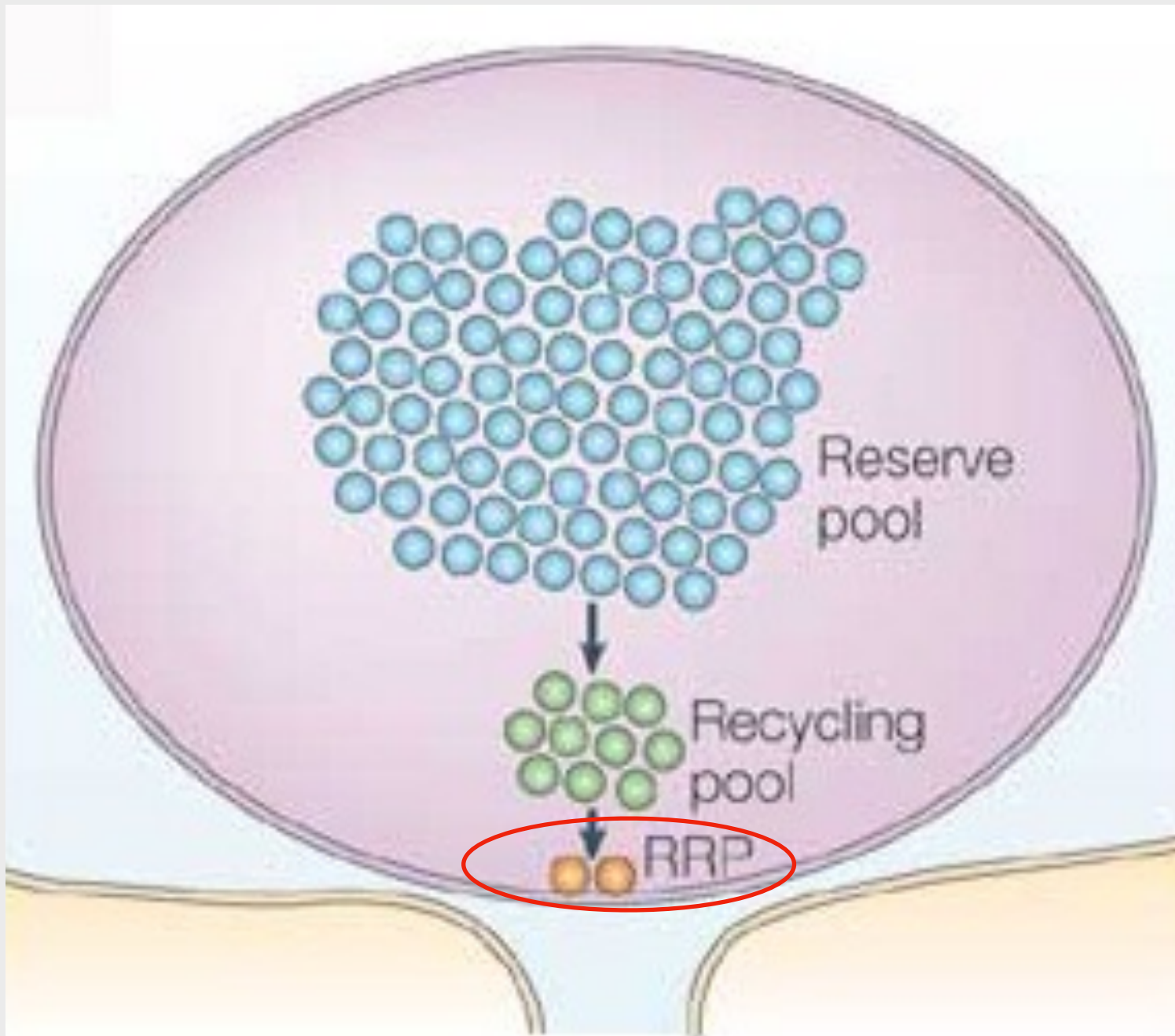


the synaptic vesicle cycle



the synaptic vesicle cycle

3 synaptic vesicle pools

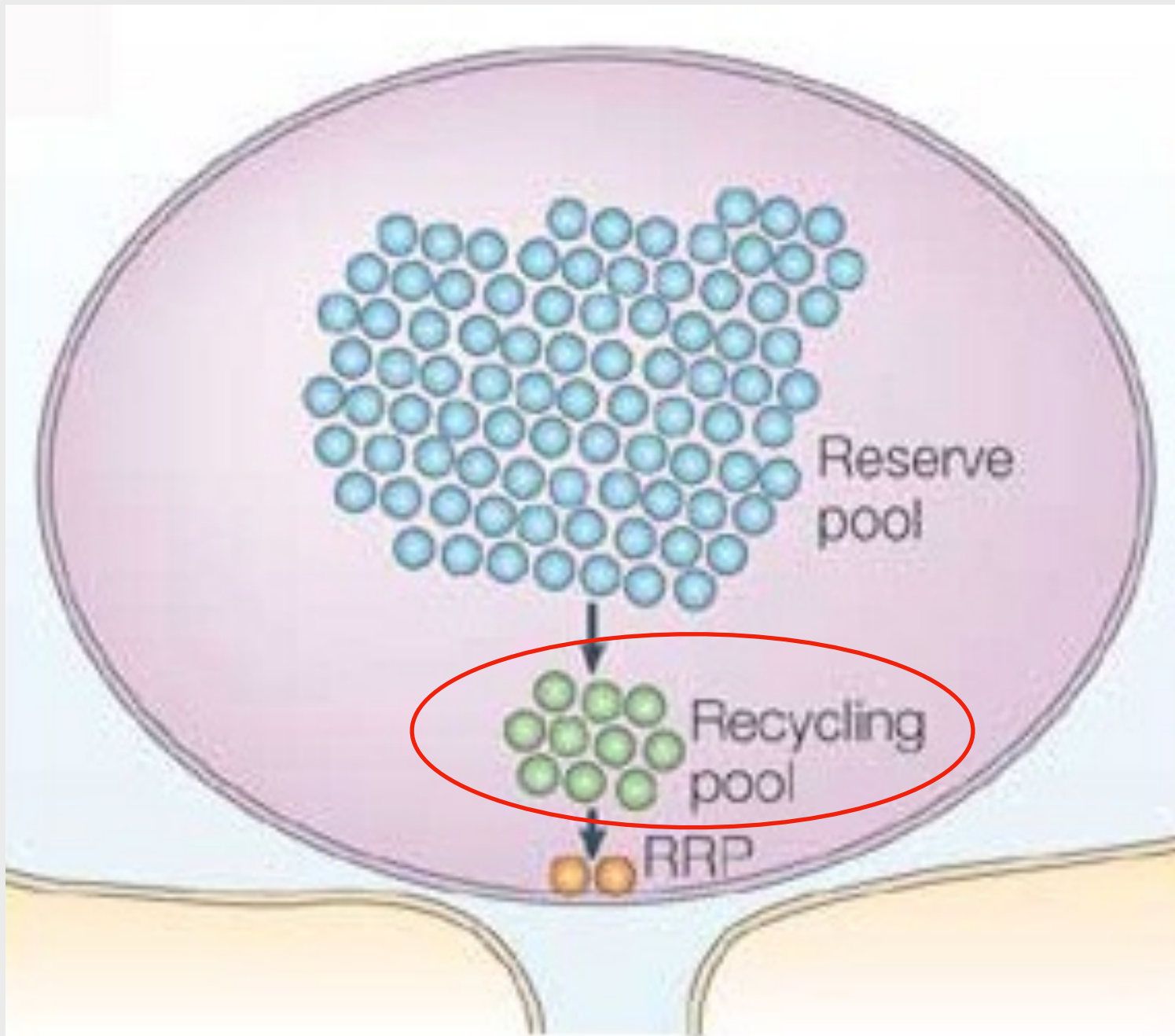


Readily Releasable Pool :

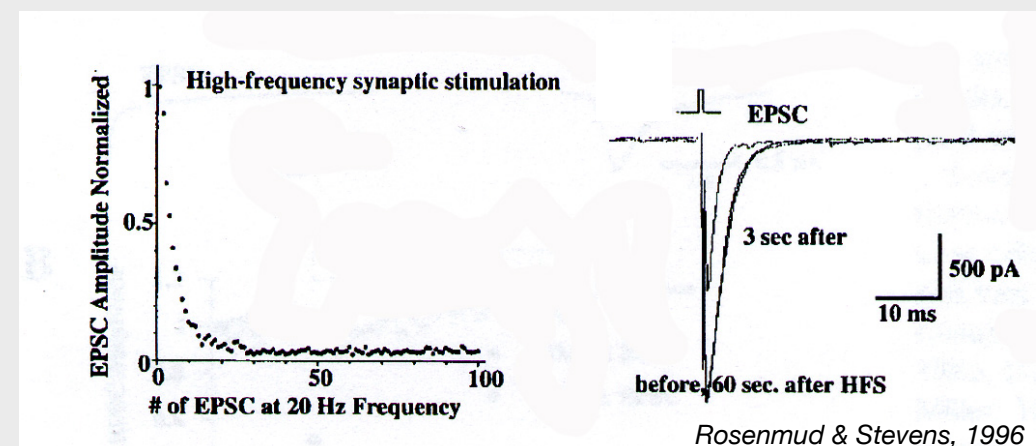
- readily available
- quickly emptied

the synaptic vesicle cycle

3 synaptic vesicle pools

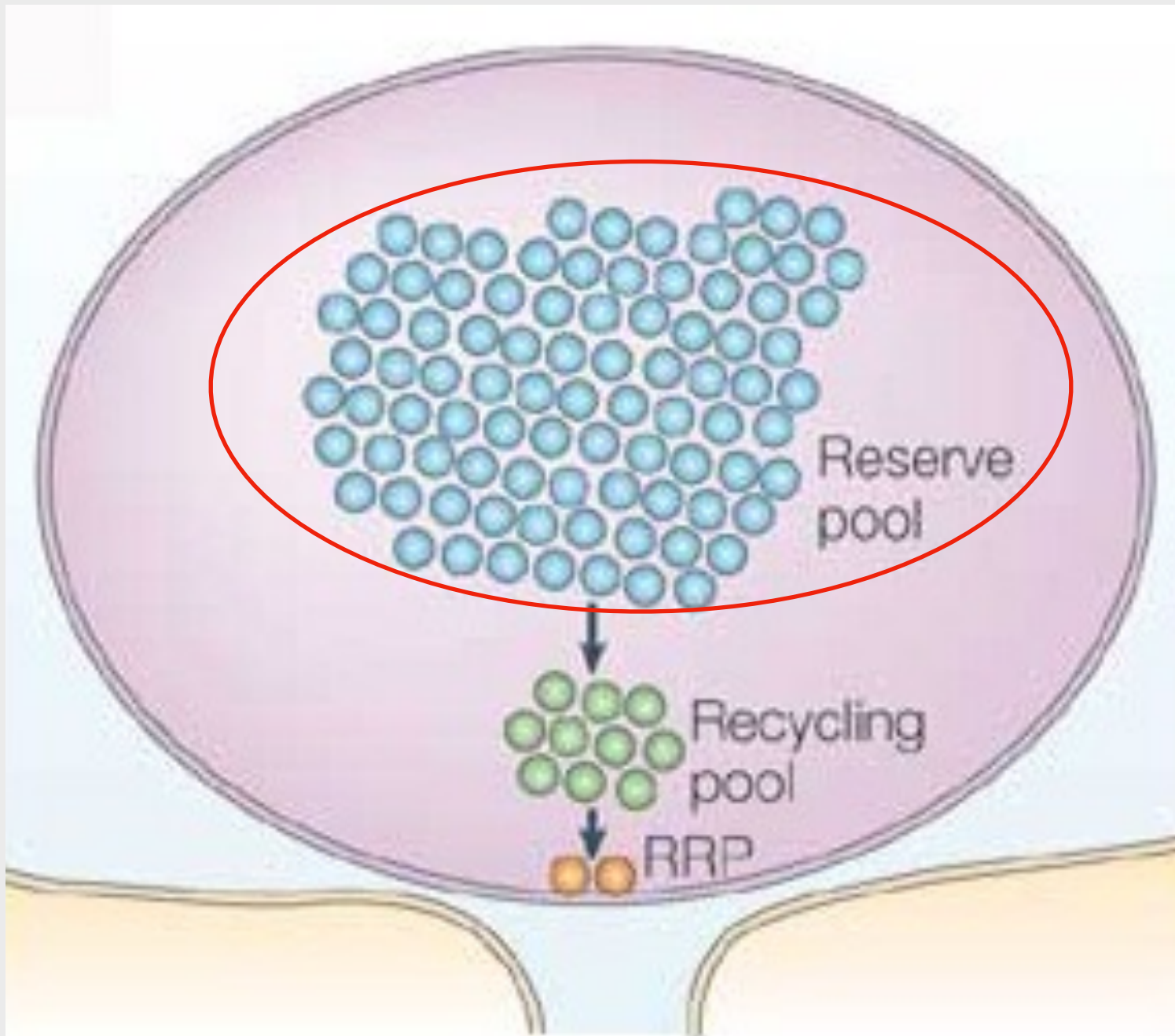


- quickly refills the RRP
- emptied after repeated physiological stimulation



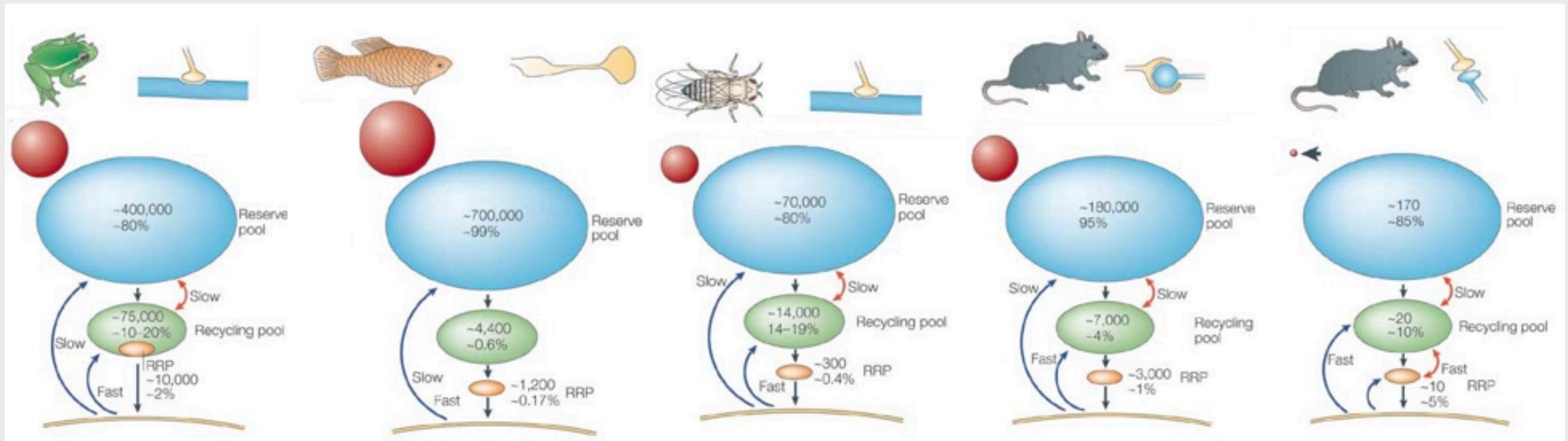
the synaptic vesicle cycle

3 synaptic vesicle pools



- slowly refills the recycling pool
- emptied after repeated strong (unphysiological) stimulation

the synaptic vesicle cycle



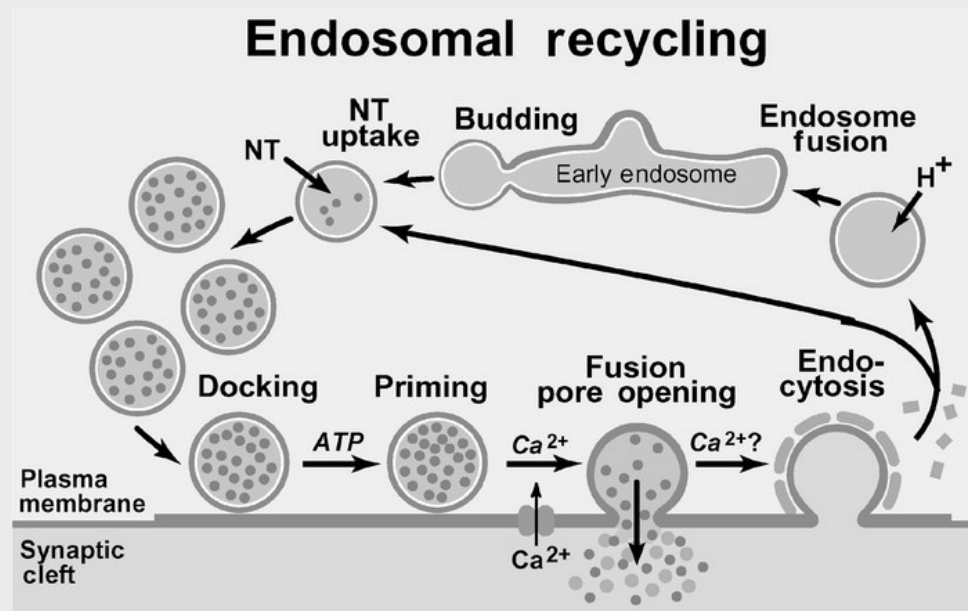
Rizzoli and Betz, 2005

Table 1 | **Characteristics of the vesicle pools**

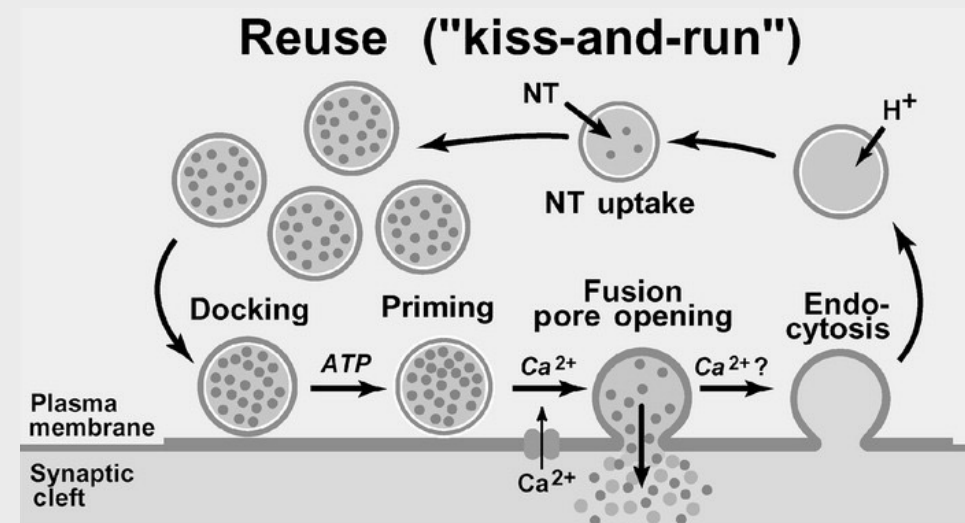
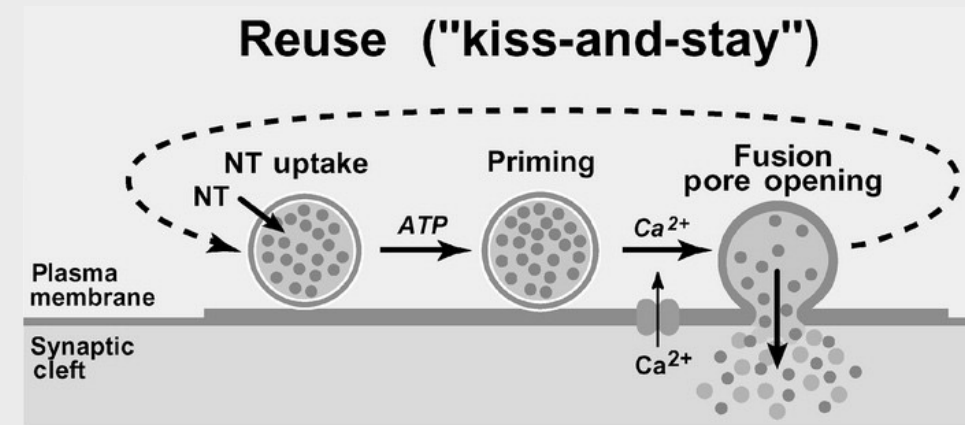
Pool	Readily releasable pool (RRP)	Recycling pool	Reserve pool
Size (% of all vesicles)	~1-2%	~10-20%	~80-90%
Location	Docked	Scattered	Scattered (bulk of vesicle cluster)
Released within	<1 second	A few seconds	Tens of seconds, minutes
Recycling	Fast (seconds)	Fast (seconds)	Slow (minutes)
Mixing with other pools	Fast mixing with recycling vesicles	Slow mixing with reserve	Slow mixing with other vesicles
Mobility in resting terminals	None — docked	High	Low (high in bipolar cells)

the synaptic vesicle cycle

complete vesicle fusion

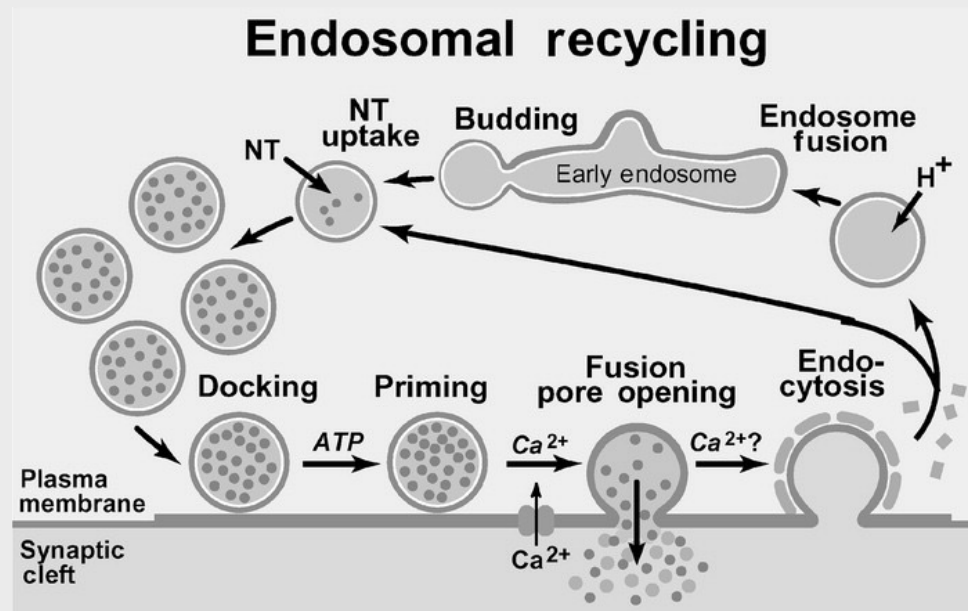


opening of a « fusion » pore



the synaptic vesicle cycle

complete vesicle fusion

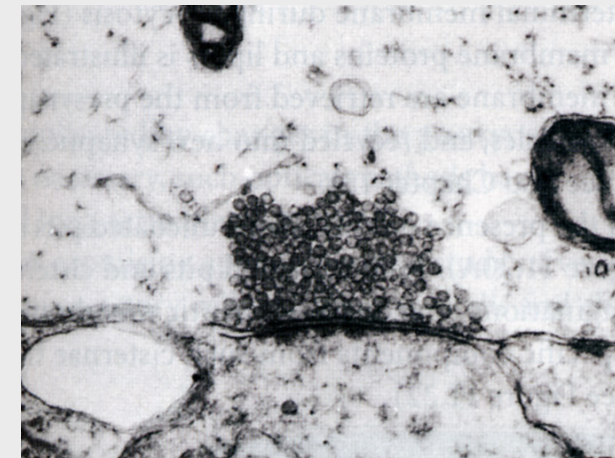


recycling duration :

- frog neuromuscular junction : 30s - 2min
- c. elegans neur. junct. : 8 - 18s
- drosophila neur. junct. : 14s
- goldfish retinal cells : 2 - 20s
- calix of Held : 1 - 23s
- hippocampal synapses : 7 - 30s

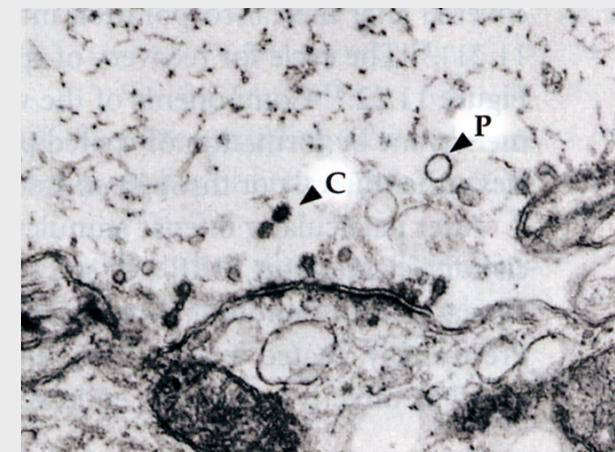
Lamprey giant axon

control

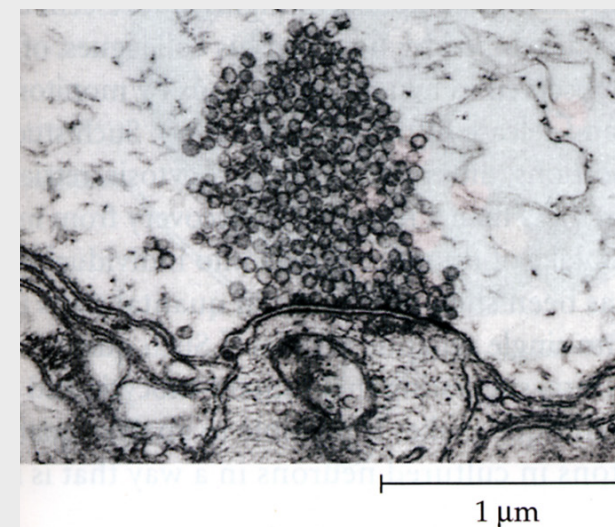


after 20Hz stimulation for 15 min

C : coated vesicles; P : pleomorphic



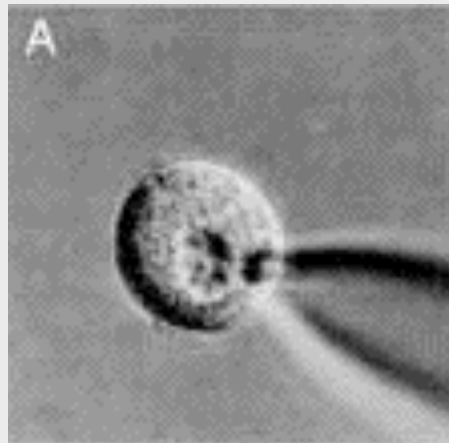
1h after stimulation



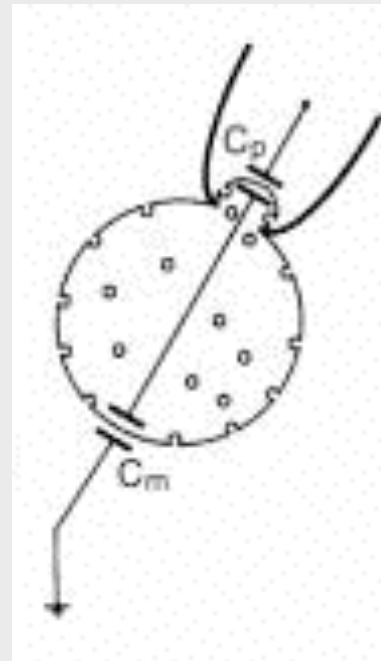
the synaptic vesicle cycle

opening of a « fusion » pore

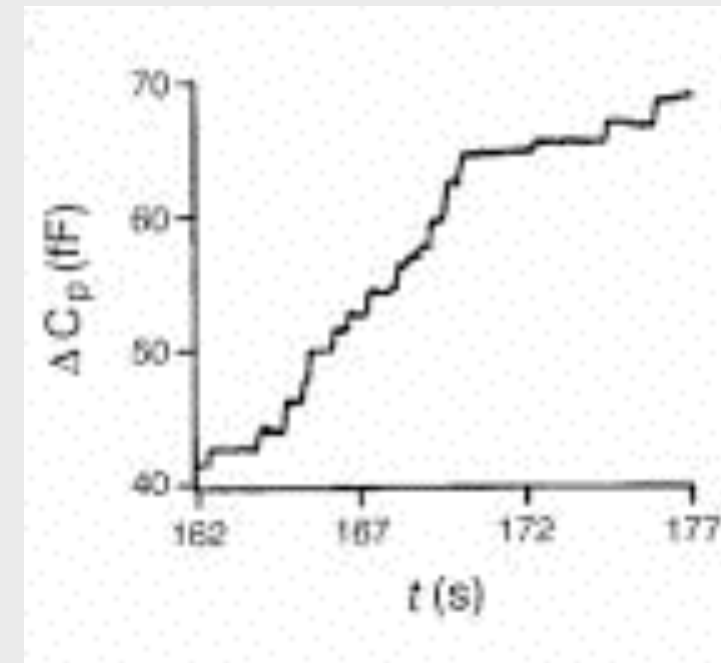
mastocytes



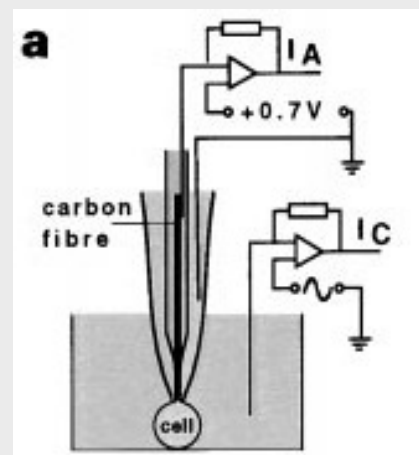
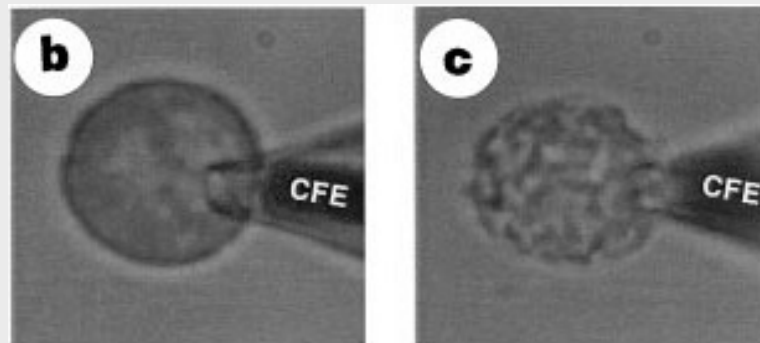
∅ of a single vesicle > 100nm



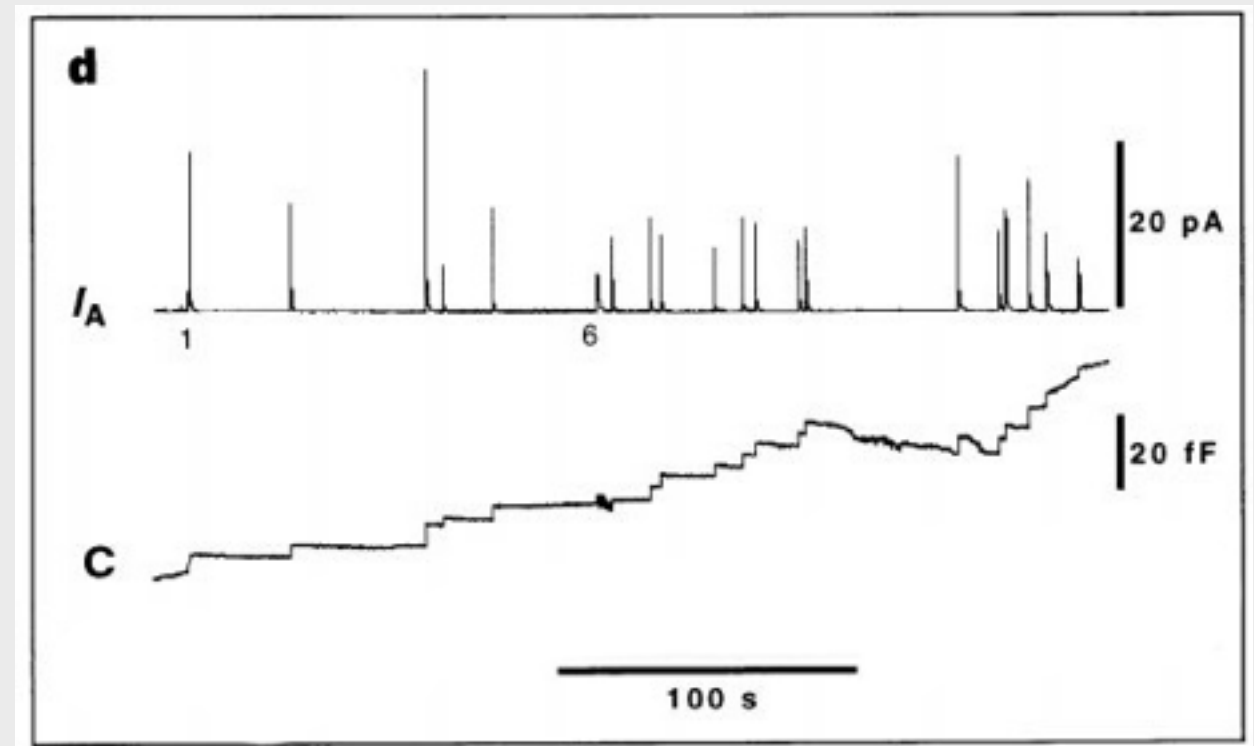
capacitance measurement



chromaffin cell



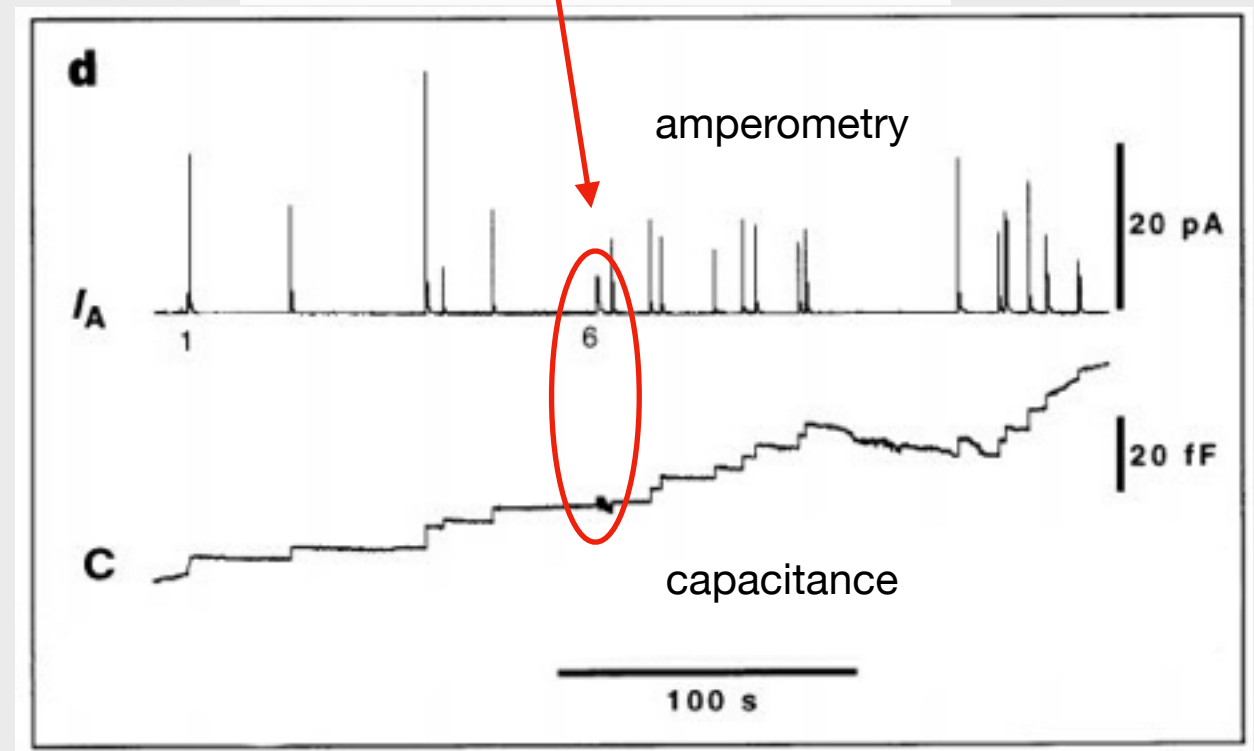
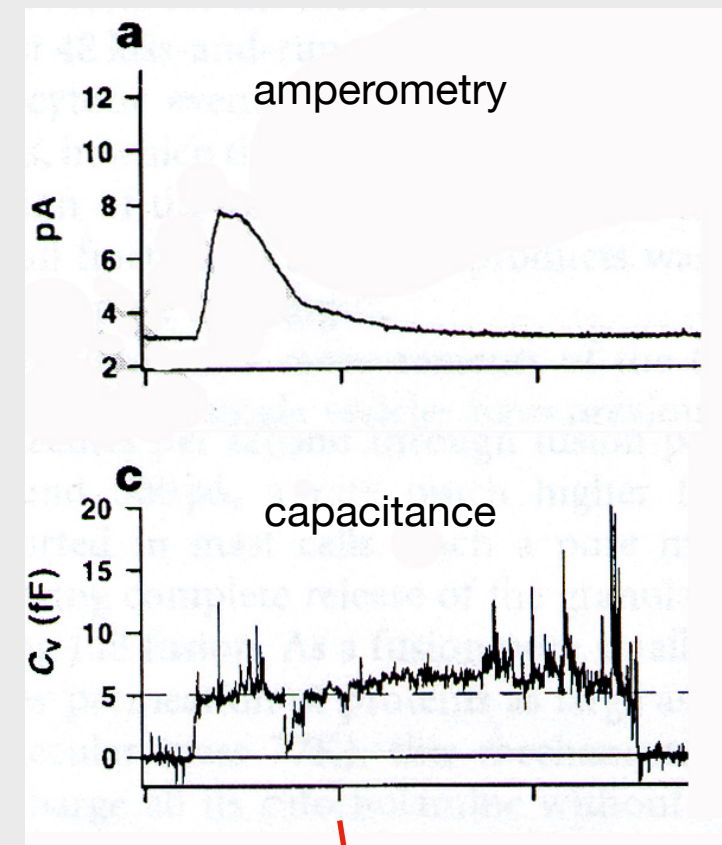
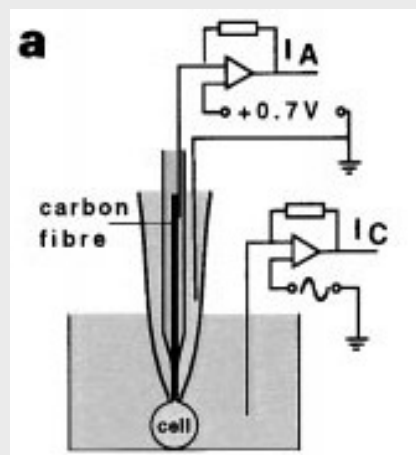
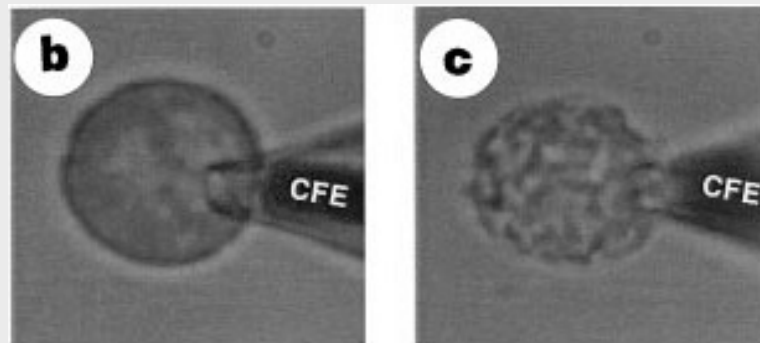
Amperometry : oxidation or reduction of catecholamines or serotonin generating a current in the carbon fibre



the synaptic vesicle cycle

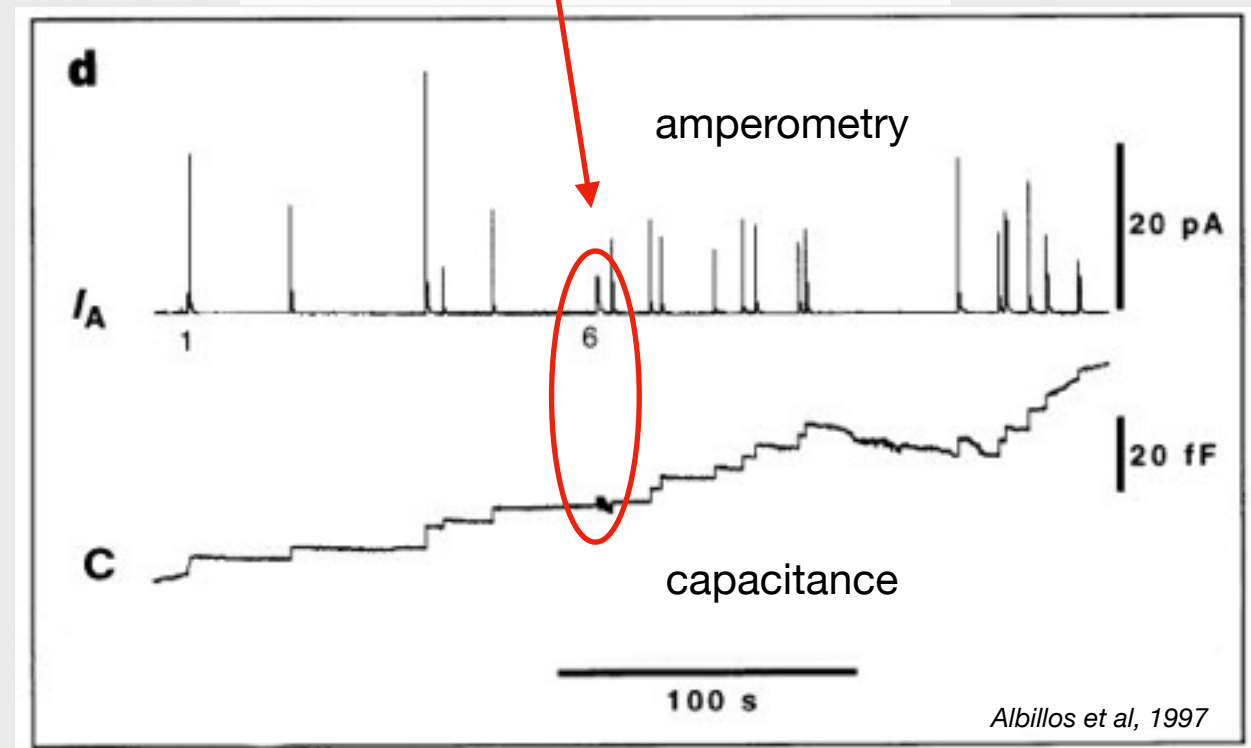
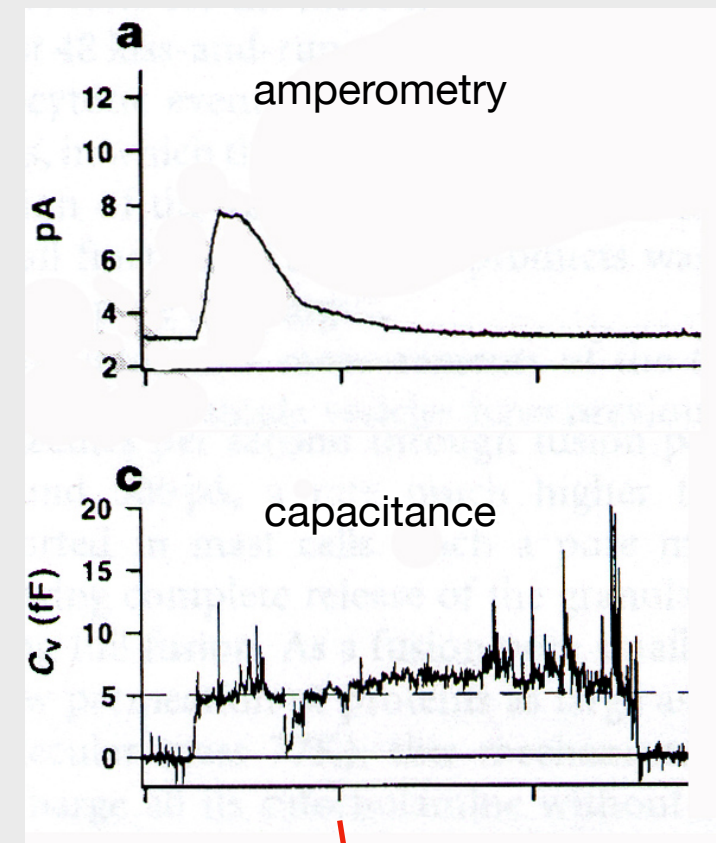
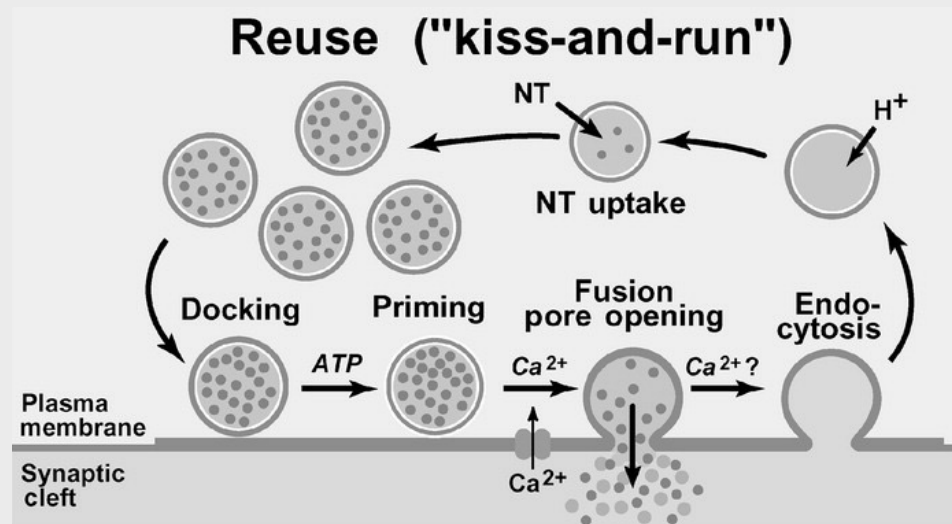
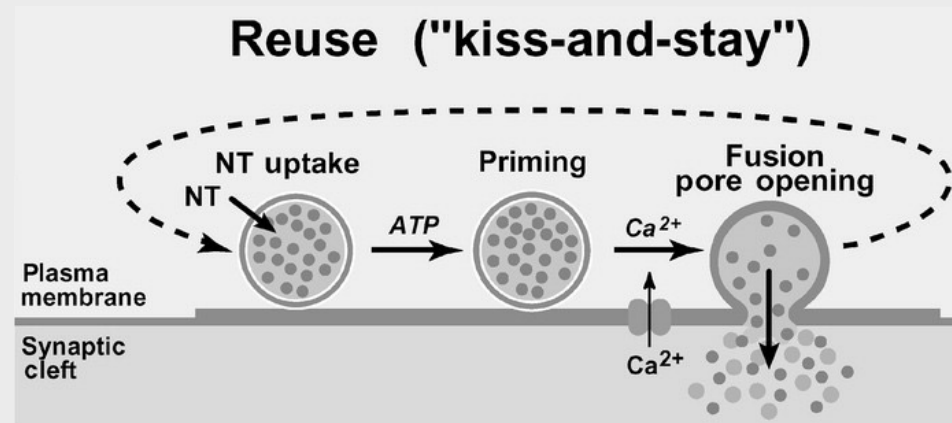
opening of a « fusion » pore

chromaffin cell



the synaptic vesicle cycle

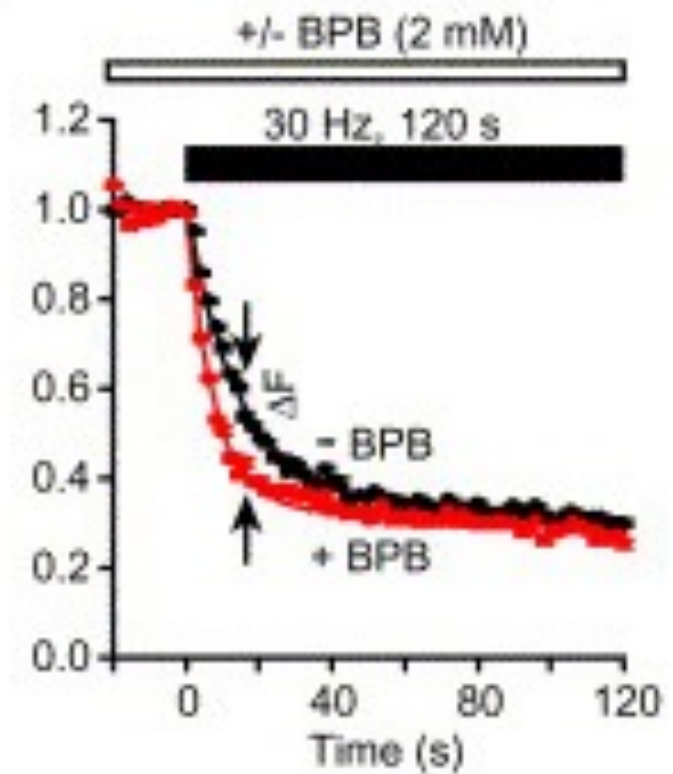
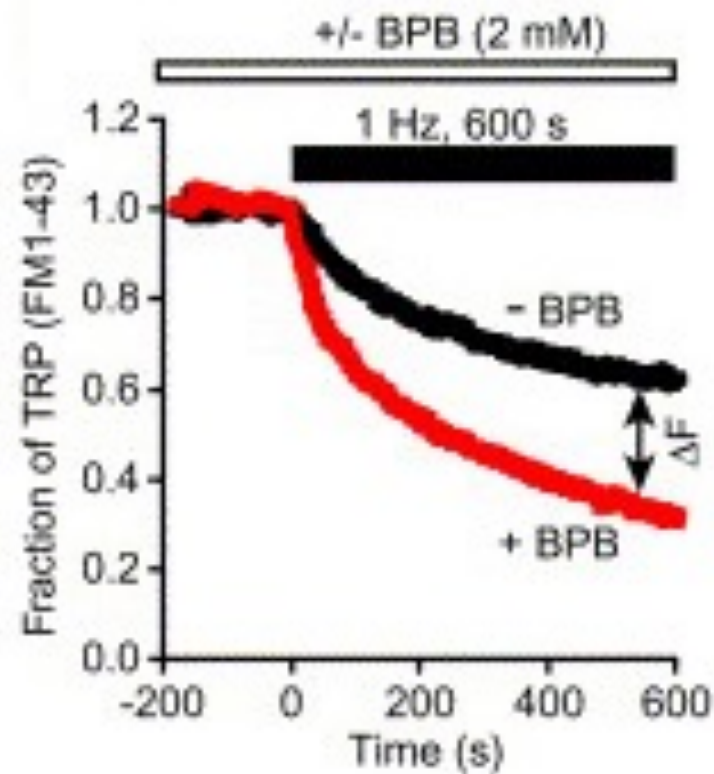
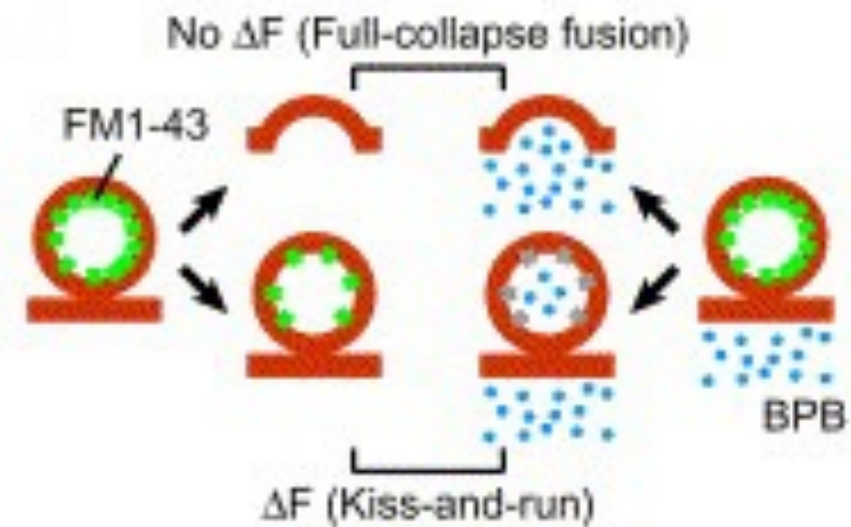
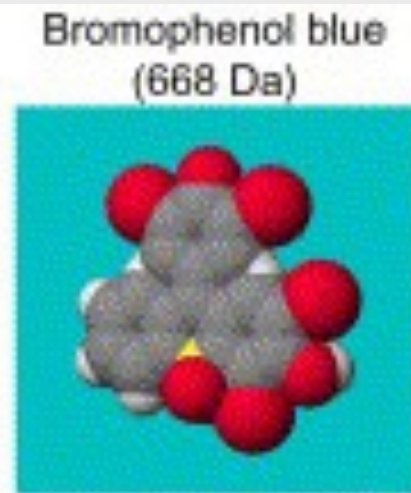
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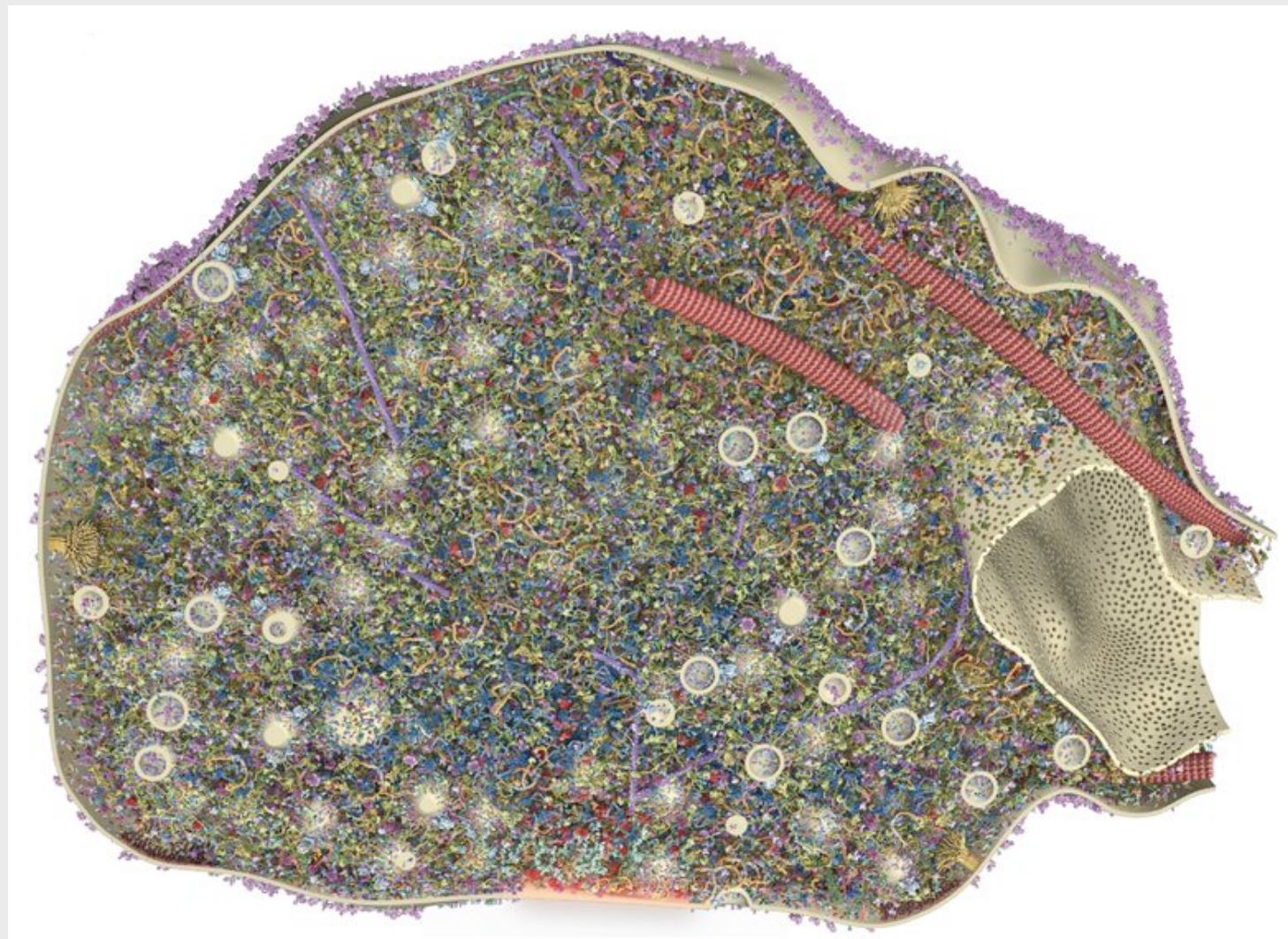
the synaptic vesicle cycle

opening of a « fusion » pore

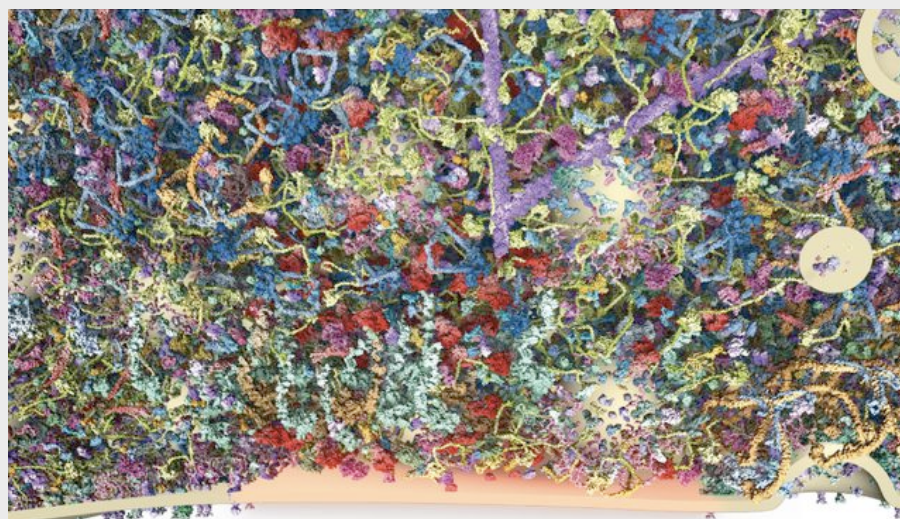
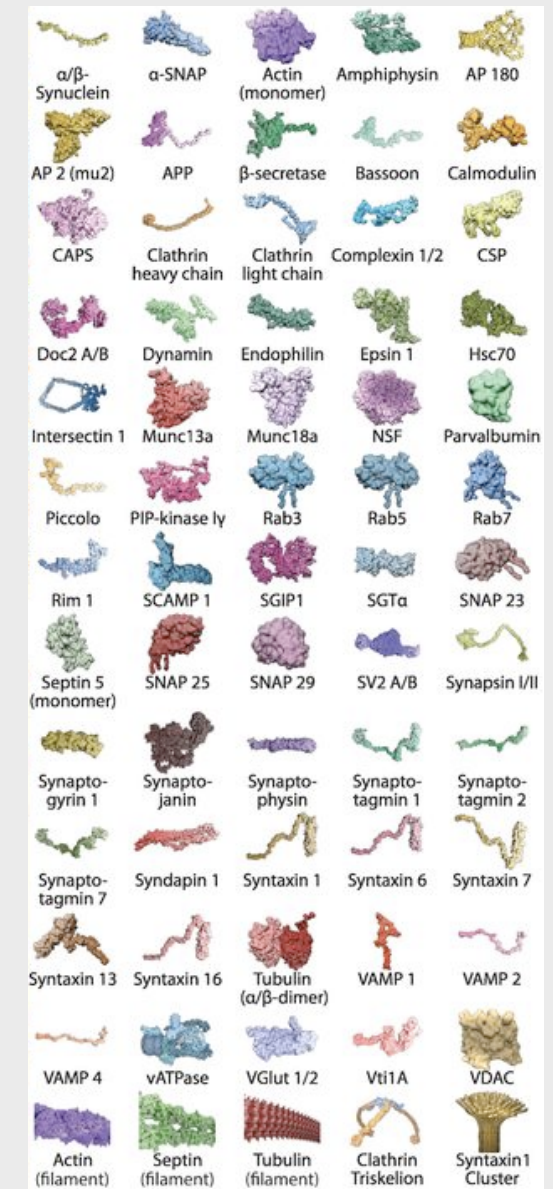
- BPB small molecule similar in size to glutamate
- does not pass through the membrane
- quench fluorescent dyes



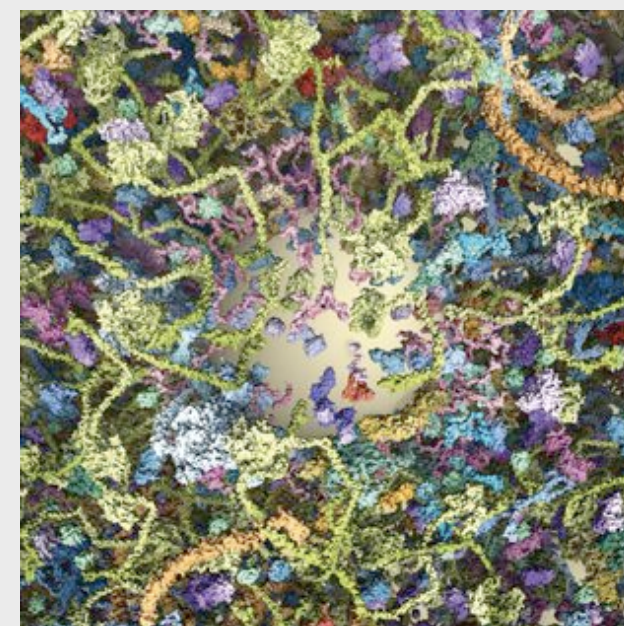
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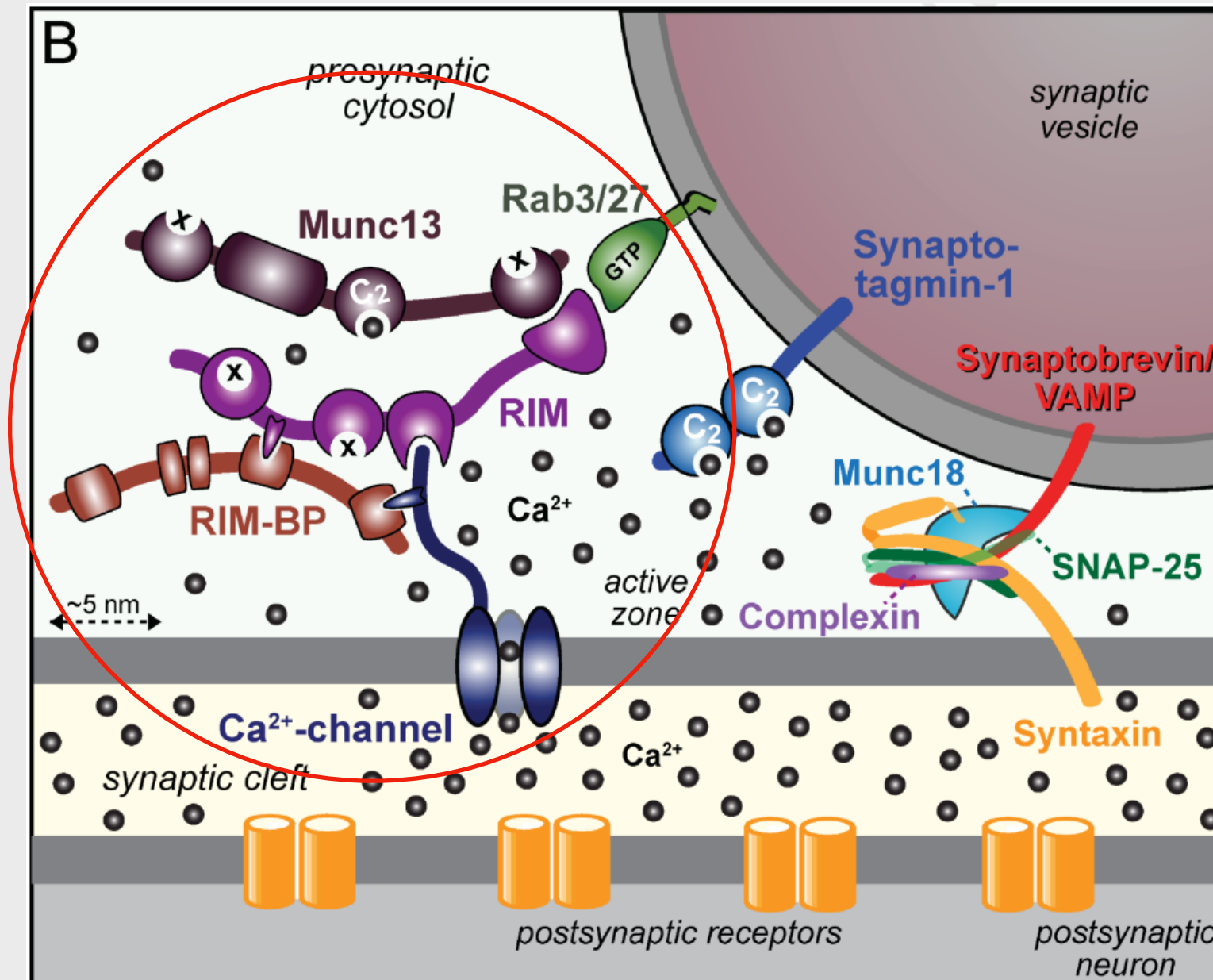
active zone area



one vesicle within the vesicle cluster

vesicle diameter = 42nm

key elements in vesicle exocytosis



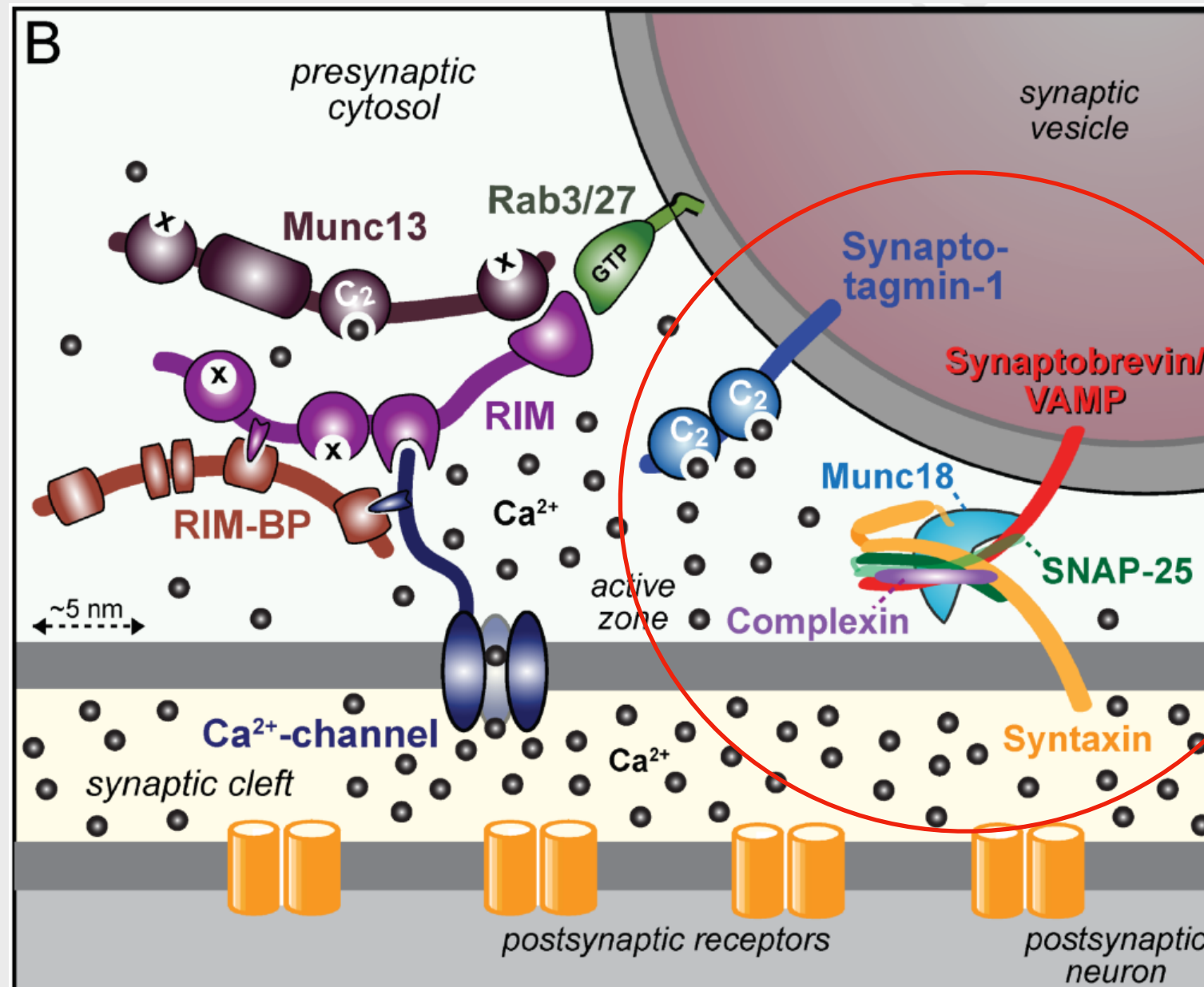
SNARE : SNAP receptor
SNAP : soluble NSF
attachment protein (fusion
with membrane of the Golgi)

Südhof 2013

active zone protein complex : RIM, Munc13, RIM-BP, and a Ca^{2+} -channel

- RIM binding to vesicular rab proteins (Rab3 and Rab27 isoforms) mediates vesicle docking
- RIM binding to the central priming factor Munc13 activates vesicle priming
- RIM binding to the Ca^{2+} channel recruits the channels within 100 nm of the docked vesicles

key elements in vesicle exocytosis



SNARE : SNAP receptor
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Südhof 2013

the core fusion machine composed of the SNARE/SM protein complex: synaptobrevin/VAMP, syntaxin-1, and SNAP-25 (SNARE). Munc18-1 (SM protein).

the Ca²⁺-sensor synaptotagmin-1 functions using complexin (bound to the SNARE complex) as an assistant.

key elements in vesicle exocytosis

